A conditional construction of a hereditarily atomic domain without ACCP

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Abstract

We construct a tower $\mathbb{Q}(\sqrt{65}) = F_0 \subset F_1 \subset \cdots$ of totally real number fields, each quadratic over the previous one, whose ring of integers D is an atomic almost Dedekind domain that does not satisfy the ascending chain condition on principal ideals (ACCP). The domain D has exactly two non-invertible maximal ideals. A conjecture of Coykendall, Gotti and Hasenauer is equivalent to the statement that every hereditarily atomic domain, that is, every domain all of whose subrings are atomic, satisfies ACCP. We prove that the ring of integers of a tower built in this way is hereditarily atomic, and therefore contradicts the conjecture, provided every sufficiently late step F_n/F_{n-1} satisfies two arithmetic conditions. The first, a Selmer condition (S), is the injectivity of an explicit $\mathbb{F}_2$-linear map defined by Hilbert symbols at the two dyadic primes. The second, an odd unit condition (O), asks, for each odd prime p ramified in F_n, that one explicit vector lie in the $\mathbb{Z}_p$-span of the images of certain relative units in a product of local unit quotients at the ramified places above p. The proof controls factorizations in an arbitrary subring through a copy of $\mathbb{Z}[1/2]/\mathbb{Z}$ in the Picard group of D, through the residue fields of D, and through approximation by a single global unit. Each condition concerns one quadratic step. We give a finite sufficient test for (O) and a reformulation of (S) through an auxiliary quadratic extension. Whether some tower satisfies both conditions at every late step is left open.

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1 Introduction

An integral domain is *atomic* if every nonzero nonunit is a finite product of irreducible elements. It satisfies the *ascending chain condition on principal ideals* (ACCP) if every ascending chain of principal ideals is eventually constant. Every domain that satisfies ACCP is atomic [1, p. 2], and Grams showed that the converse fails [7]. Throughout, subrings contain the identity.

Coykendall, Gotti and Hasenauer call a domain *hereditarily atomic* if every subring is atomic [2, Definition 2.1], and *hereditarily ACCP* if every subring satisfies ACCP [2, Section 6]. They conjecture that the two properties coincide [2, Conjecture 6.1]. The conjecture is equivalent to the statement that every domain without ACCP contains a subring that is not atomic [2, Proposition 6.2]. Equivalently, every hereditarily atomic domain satisfies ACCP. For commutative cancellative monoids the corresponding statement is a theorem of Gotti and Li: every hereditarily atomic monoid satisfies ACCP [6, Theorem 4.2]. For domains, [2, Theorem 6.4] proves that the two properties coincide under each of seven additional conditions. In characteristic zero, each of these conditions either requires an element that is transcendental over $\mathbb{Q}$ or forces the domain itself to be non-atomic.

This paper looks for a counterexample among rings of algebraic integers, where no element is transcendental. Let K be an algebraic extension of $\mathbb{Q}$ and let D be its ring of integers. Two known facts shape the search. First, the field K is hereditarily atomic exactly when D is a Dedekind domain [2, Theorem 4.4(1)]. The question therefore concerns D, not K. Second, every element of D is integral over $\mathbb{Z}$, so every subring $S \subseteq D$ satisfies $S^\times = S \cap D^\times$ (the proof of Lemma 3.4 applies). By a result of Grams (see [2, Proposition 5.1]), if D satisfied ACCP then so would every subring of D. A counterexample of this form must therefore fail ACCP itself; in particular D cannot be a Dedekind domain.

When K is the union of a tower of number fields in which the ramification index stays bounded along every sequence of primes V_n of F_n with $V_{n+1} \mid V_n$, D is an *almost Dedekind domain*: its localization at every maximal ideal is a discrete valuation ring [5, §36] (see Section 2.3 and Lemma 3.2). Divisibility in D is then read off from valuations, as in a Dedekind domain. Two features distinguish D from a Dedekind domain. A maximal ideal of D need not be finitely generated. The Picard group $\mathrm{Pic}(D)$, the group of invertible ideals modulo principal ideals, is the direct limit of the ideal class groups of the fields in the tower, and it can be large. An almost Dedekind domain need not be

atomic (see the discussion in [2, Section 2]). Our construction uses both features to make D atomic without ACCP. Since such a D is not Dedekind, some overring of D is not atomic [2, Theorem 3.6]. This overring contains D properly, since D itself is atomic, so it is not a subring of D, and this does not settle the question.

1.1 Main results

Our first result is unconditional.

Theorem A. *There is a tower $F_0 = \mathbb{Q}(\sqrt{65}) \subset F_1 \subset F_2 \subset \cdots$ of totally real number fields with $[F_n : F_{n-1}] = 2$ for every $n \geq 1$ such that the ring of integers D of $K = \bigcup_n F_n$ has the following properties.*

(a) *D is an almost Dedekind domain.*

(b) *D is atomic.*

(c) *D does not satisfy ACCP.*

(d) *Exactly two maximal ideals of D are not invertible. Their residue fields are $\mathbb{F}_{47}$ and $\mathbb{F}_7$.*

Section 4 constructs the tower (Theorem 4.11). Properties (a)–(d) follow from Lemma 3.2, Theorem 3.7 and the residue fields recorded in (T1). In F_0 the primes 47 and 7 split. At every later step one prime above 47 and one prime above 7 split again, and along every other sequence of primes V_n of F_n with $V_{n+1} \mid V_n$ the primes are eventually inert in each step. Following the primes that keep splitting gives two chains of primes of residue degree one: primes A_n of F_n above 47 and primes B_n of F_n above 7, with

$$A_{n-1}\mathcal{O}_{F_n} = A_n P_n, \qquad B_{n-1}\mathcal{O}_{F_n} = B_n Q_n.$$

The two non-invertible maximal ideals $\mathfrak{m}_A$ and $\mathfrak{m}_B$ of D are the limits of the chains (A_n) and (B_n). The primes P_n and Q_n split off from the chains at level n give invertible maximal ideals of D. The failure of ACCP is explicit: there are elements $y_n \in \mathcal{O}_{F_n}$ with $(y_n) = A_n B_n$, starting with $y_0 = -16 + 3\sqrt{65}$, such that $y_0 D \subsetneq y_1 D \subsetneq y_2 D \subsetneq \cdots$.

Each step of this tower is a *scaled step* (Definition 4.8). Informally, $F_n = F_{n-1}(y_n)$, where y_n is a root of a polynomial $X^2 - tX + y_{n-1}$, so that y_n has relative norm y_{n-1}. The coefficient $t \in \mathcal{O}_{F_{n-1}}$ is given by an explicit formula (4) in terms of y_{n-1}, an auxiliary element c with $(c) = A_{n-1}R$ for an auxiliary prime R, an integer $H \geq 5$ with the scale $h = 2^H$, and a power $u = \epsilon_H^k$, with k odd, of a unit ϵ_H of F_0 chosen so that $u - 1$ has 2-adic valuation exactly H at both dyadic primes. The name refers to the scale h. A second element $\beta_n = c(u-1)/h + u y_n$ also has relative norm y_{n-1}, and $(\beta_n) = A_n Q_n$. Consequently the element

$$z_n = \frac{y_n \beta_n}{y_{n-1}}$$

has relative norm one and $(z_n) = A_n P_n^{-1}$. Section 3 isolates six properties (T1)–(T6) of a tower with such elements, and the tower of Theorem A has all of them. Our second result is conditional.

Theorem B. *Let $F_0 \subset F_1 \subset \cdots$, with elements $y_n, \beta_n \in \mathcal{O}_{F_n}$, be a tower that satisfies the hypotheses* (T1)–(T6) *of Section 3 and in which every step $F_{n-1} \subset F_n$ is a scaled step in the sense of Definition 4.8. Let D be the ring of integers of $K = \bigcup_n F_n$. Suppose that, for every sufficiently large n, the step F_n/F_{n-1} satisfies both the Selmer condition* (S) *of Section 7 and the odd unit condition* (O) *of Section 8. Then D is hereditarily atomic and does not satisfy ACCP.*

Theorem B is proved at the end of Section 8. Both conditions concern a single step. Write $F = F_{n-1}$ and $L = F_n = F(\sqrt{\delta})$, where $\delta = t^2 - 4y_{n-1}$, and let T be the set of all finite primes of F that ramify in L. The field F has exactly two *dyadic* primes, that is, primes above 2.

The Selmer condition. Let $\Delta(F, \delta)$ be the group of square classes $[x] \in F^\times/F^{\times 2}$ of totally positive elements x such that $v_\mathfrak{p}(x)$ is even at every prime $\mathfrak{p}$ outside T and the two dyadic primes, and such that at each $\mathfrak{p} \in T$ the element x is a square or δ times a square in the completion $F_\mathfrak{p}$. It is a finite $\mathbb{F}_2$-vector space that contains $[\delta]$. The Hilbert symbols $(x, -\delta)_v$ at the two dyadic places v (Section 2.3) define an $\mathbb{F}_2$-linear map λ_δ from $\mathrm{Sel}_\delta = \Delta(F, \delta)/\langle[\delta]\rangle$ to $\mathbb{F}_2^2$. Condition (S) is that λ_δ is injective. Hilbert reciprocity places the image of λ_δ in the line of vectors with coordinate sum zero, and Sel_δ is never zero (24). Thus (S) says that Sel_δ has exactly two elements and that its nonzero class $[x]$ satisfies $(x, -\delta)_v = -1$ at both dyadic places. The formal definition is (S).

The odd unit condition. Let p be an odd prime that ramifies in F_n; by (T1), $p \geq 5$. At each place v of F_n above p with ramification index $e(v/p) = 2$, let W_v, the *coefficient ring* of $(F_n)_v$, be the ring of integers of the largest unramified subfield of the completion $(F_n)_v$. The quotient $C_v = \mathcal{O}^\times_{(F_n)_v}/W_v^\times$ is a free $\mathbb{Z}_p$-module of finite rank (Lemma 8.1). Let $M_{n,p}$ be the product of the groups C_v over all such places v. The *relative units*, the units of F_n whose norm to F_{n-1} is 1, map to $M_{n,p}$, and so does z_n, which is a unit at these places. Condition (O) at p asks that the image of z_n lie in the $\mathbb{Z}_p$-span of the images of those relative units whose residues at the two dyadic primes of F_n have trivial p-primary component. Two features of (O) matter. It is a condition in the whole product $M_{n,p}$: the image of z_n must be approximated at all places above p simultaneously, not at each place separately. And it asks for membership in a $\mathbb{Z}_p$-span: knowing that this span has full rank, that is, finite index in the submodule of $M_{n,p}$ on which the nontrivial automorphism of F_n/F_{n-1} acts by -1, does not suffice. The formal definition is (O), which imposes (O) at every odd prime ramified in F_n.

1.2 How the proof works

The proof of Theorem B controls factorizations in an arbitrary subring $S \subseteq D$ with three tools: the Picard group of D, the residue fields of D, and approximation of the elements z_n by global units.

The Picard group: bounded factorizations and the failure of ACCP. Divisibility in D is decided by valuations (Lemma 3.2). Since all maximal ideals of D other than $\mathfrak{m}_A$ and $\mathfrak{m}_B$ are invertible, $\mathrm{Pic}(D)$ decides which products of them are principal. The classes of the split-off primes generate a copy of $\mathbb{Z}[1/2]/\mathbb{Z}$ in $\mathrm{Pic}(D)$, in which $[P_jD]$ corresponds to 2^{-j-1} and $[Q_jD]$ to -2^{-j-1} (Lemma 3.3). This uses (T3): no nontrivial ideal class of F_{n-1} becomes principal in F_n. Let κ and n_1 be integers with $\kappa \sum_{j>n_1} 2^{-j-1} < 1$. A proper nonzero ideal $\prod_{j>n_1} P_j^{e_j}$ with $0 \leq e_j \leq \kappa$ has class represented by a number strictly between 0 and 1, so it is not principal. Hence an element x with $v_{\mathfrak{m}_A}(x) = 0$ or $v_{\mathfrak{m}_B}(x) = 0$, called *one-sided*, has bounded factorization lengths in every subring of D (Proposition 3.6). Mixing the two chains cancels the classes: y_{n-1}/y_n generates P_nQ_n, which gives the strict chain $y_0D \subsetneq y_1D \subsetneq \cdots$. An element with both valuations positive splits in D into two one-sided factors, so D is atomic (Theorem 3.7). These arguments occupy Section 3.

The residue fields: which orders a subring can contain. Each step has a prime that ramifies in it and is the only ramified prime above its rational prime (an *isolated ramified prime*, Definition 3.1 and hypothesis (T5)). This forces every proper subfield of K to be a number field, so a subring S with fraction field different from K is a bounded factorization domain (Proposition 5.2). Suppose instead that S has an element with unbounded factorization lengths. The long factorizations produce elements of S supported on late primes P_j, Q_j. At a maximal ideal V of odd residue characteristic different from $\mathfrak{m}_A, \mathfrak{m}_B$, the two-primary components of the residues of y_n have order 2^{n+o_V} for large n, with residue offsets o_V that distinguish the maximal ideals over the same rational

prime (T6). These orders force S to map onto every odd residue field of D, and to separate the maximal ideals of D over each odd prime (Proposition 5.9). Teichmüller lifting, which lifts each nonzero residue to a root of unity of order prime to p in the completion (Section 2.3), then determines the completions of the orders $S \cap F_n$: at odd primes unramified throughout the tower they are maximal (Proposition 5.11); above a ramified prime, for all sufficiently large n, they are maximal at the places of ramification index one, and at a place V of ramification index two they are fixed by one nonnegative integer s_V independent of n (Proposition 5.12). Section 5 carries this out.

Unit approximation: moving a factorization into the subring. Let $x \in S$ have unbounded factorization lengths and positive valuation at both $\mathfrak{m}_A$ and $\mathfrak{m}_B$. We split x in D into two one-sided factors. A factor with a given divisor is determined only up to a unit of D, and it lies in S only if it lies in the order $S \cap F_N$ for some N. An order differs from the maximal order at only finitely many primes, so this is a finite list of local conditions. By the previous paragraph, for large N there are none at the odd places of ramification index one, so the conditions sit at the dyadic places and at the odd places of ramification index two. The factor differs from an element of S by a product of powers of elements z_n (Lemma 6.3). So it suffices that every late z_n be approximable by global units simultaneously at all these places. Condition (JT_A) of Section 6 states this approximation in a compact group $\mathcal{U}_N$ that is the product of the unit group of $\mathcal{O}_{F_N} \otimes \mathbb{Z}_2$ and the quotients C_v at the odd places of ramification index two. Under this condition one global unit moves both factors into $S \cap F_N$ (Lemma 6.4), and each factor is bounded in S because it is one-sided. Hence S is atomic (Theorem 6.1).

Class field theory: from approximation to (S) and (O). At the dyadic places, class field theory identifies the obstruction to approximating an element by global units with the Artin symbol of its ideal, that is, the image of the ideal under the Artin map of class field theory (Section 2.3). This symbol lies in the Galois group of the maximal abelian pro-2 extension unramified outside 2 and split at the real places (Lemma 7.4). Under (S), the relevant quotient of this group has exponent two (Proposition 7.7), and Kummer theory, which describes quadratic extensions by square classes (Section 2.3), identifies its quadratic characters with the classes of Sel_δ (Lemma 7.5). The Artin symbol of $(z_n) = A_n P_n^{-1}$ then vanishes, because every class of Sel_δ is a square locally at A_{n-1}; this gives Theorem 7.2 in Section 7. At the odd places of ramification index two, the quotients C_v are pro-p, and approximation becomes exactly condition (O) (Lemma 8.4); for $p \geq 5$ the p-adic logarithm identifies each C_v with the coefficient ring W_v, which gives ranks and a finite test. Proposition 8.9 combines the two, and Section 8 completes the proof of Theorem B. Let $U^0_{n,p}$ be the $\mathbb{Z}_p$-span in the definition of (O), and $M^-_{n,p}$ the (-1)-eigenspace of $M_{n,p}$ described in Section 1.1. Equality $U^0_{n,p} = M^-_{n,p}$ implies (O) at p, and Nakayama's lemma for compact $\mathbb{Z}_p$-modules reduces it to a rank computation modulo p. This gives a finite sufficient test (Proposition 8.5).

1.3 What is proved and what remains

Theorem A is unconditional. Theorem B requires (S) and (O) at every sufficiently late step of one tower. Each condition is a statement about a single quadratic step F_n/F_{n-1} and its unit group. Condition (S) is a question of linear algebra over $\mathbb{F}_2$ on a finite group of square classes. For (O), Proposition 8.5 gives a sufficient test at each prime p by a finite computation with finitely many relative units. Appendix B reformulates (S) through the auxiliary quadratic extension $F(\sqrt{\delta_+})$, where δ_+ is an explicit factor of the discriminant δ, and Appendix A shows that the construction can prescribe, at finitely many odd primes in each step, how closely z_n is approximated by elements of the coefficient rings (Proposition A.3). We do not construct an infinite tower in which every late step satisfies both (S) and (O). Section 9 states this remaining problem as an existence statement

for one-step extensions of finite towers (Hypothesis 9.2), shows that it yields a hereditarily atomic domain without ACCP (Proposition 9.3), and explains why prescribed local precision does not by itself give (O).

1.4 Organization

Section 2 fixes notation and collects the background from factorization theory and algebraic number theory, with references. Section 3 states the hypotheses (T1)–(T6) and proves the factorization results that use only them. Section 4 constructs the tower of Theorem A. Section 5 determines the orders that a subring with unbounded factorization lengths must contain, and Section 6 proves hereditary atomicity from unit approximation. Sections 7 and 8 reduce unit approximation to (S) and (O) and prove Theorem B. Section 9 formulates the remaining existence problem. Appendix A proves the local precision results, Appendix B reformulates (S), and Appendix C records the dependencies among the main results.

2 Preliminaries

2.1 Notation and conventions

All rings are commutative domains with identity, and subrings contain that identity. Throughout, p and ℓ denote rational primes, and p is odd unless stated otherwise. The letter $\mathfrak{m}$ always denotes a maximal ideal, and generic primes of a number field are written $\mathfrak{p}$ and $\mathfrak{q}$. A generic number field is written $\mathcal{E}$. Places are written v and w. For a place v of a number field $\mathcal{E}$ we write $\mathcal{E}_v$ for the completion, $\mathcal{O}_{\mathcal{E}_v}$ for its ring of integers, k_v for its residue field, f_v for the residue degree and $e(v/p)$ for the ramification index over p. At a finite place, v also denotes the normalized valuation, and we write $v_{\mathfrak{p}}$ when the place is given by a prime $\mathfrak{p}$; on $\mathbb{Q}_2$ and its unramified extensions, v_2 is normalized by $v_2(2) = 1$. For a finite abelian group, or an element of one, the subscript (2) denotes the two-primary part. Ideal class groups are written $\mathrm{Cl}(\mathcal{E})$, ray class groups $\mathrm{Cl}_{\mathfrak{f}}(\mathcal{E})$ for a modulus $\mathfrak{f}$, and Pic denotes the Picard group. For $x \in \mathcal{E}^\times$ we write $[x]$ for its class in $\mathcal{E}^\times/\mathcal{E}^{\times 2}$, called its *square class*, and $[x]_{\mathfrak{p}}$ for its class in $\mathcal{E}_{\mathfrak{p}}^\times/\mathcal{E}_{\mathfrak{p}}^{\times 2}$.

The tower. The fields are $F_0 = \mathbb{Q}(\sqrt{65}) \subset F_1 \subset \cdots$ with $[F_n : F_{n-1}] = 2$. We write σ_n for the nontrivial automorphism of F_n/F_{n-1}, $E_n = \mathcal{O}_{F_n}^\times$ for the unit group, and E_n^1 for the group of *relative units*, the kernel of the norm map $E_n \to E_{n-1}$. We put $K = \bigcup_n F_n$ and $D = \mathcal{O}_K$. The letter n always denotes a level of the tower; m, N and N' denote further levels. The elements y_n, β_n, z_n and z_n' are those of hypothesis (T2), and the residue offsets o_V are those of (T6) (Section 3).

Primes. Hypothesis (T1) fixes two chains of primes (A_n) and (B_n): A_n and B_n are primes of F_n with residue fields $\mathbb{F}_{47}$ and $\mathbb{F}_7$, and

$$A_{n-1}\mathcal{O}_{F_n} = A_nP_n, \qquad B_{n-1}\mathcal{O}_{F_n} = B_nQ_n.$$

Thus A_{n-1} and B_{n-1} split in F_n, and P_n and Q_n are the primes split off from the chains (A_n) and (B_n) at level n. A prime P_j or Q_j of F_j, its unique extension to each later field, and its extension to D are denoted by the same letter. The maximal ideals of D determined by the two chains are $\mathfrak{m}_A$ and $\mathfrak{m}_B$; they are the two non-invertible maximal ideals of D (Theorem 3.7). A maximal ideal of D is usually denoted V, with $V_n = V \cap \mathcal{O}_{F_n}$, normalized valuation v_V and residue field $k(V) = D/V$; likewise $k(\mathfrak{p})$ denotes the residue field of a prime $\mathfrak{p}$ of a number field, for instance $k(V_n) = \mathcal{O}_{F_n}/V_n$. The tower primes A_n, B_n, P_j, Q_j, the auxiliary prime R and the detecting primes G_i are written in italic capitals.

One quadratic step. When we study a single step we write $F = F_{n-1}$, $L = F_n$, $\sigma = \sigma_n$, $A = A_{n-1}$, $B = B_{n-1}$, $y = y_{n-1}$, and $A\mathcal{O}_L = A'P$, $B\mathcal{O}_L = B'Q$. The parameters of a scaled step are an integer $H \geq 5$, the scale $h = 2^H$, a unit $\epsilon = \epsilon_H$ of F_0, an exponent k, the unit $u = \epsilon^k$ and an element c of F. They determine the coefficient $t \in \mathcal{O}_F$ of (4), and y_n is a root of $X^2 - tX + y$. The discriminant of the step is

$$\delta = t^2 - 4y = \frac{\delta_-\delta_+}{(chu)^2}, \qquad \delta_- = c^2 - h^2 y, \qquad \delta_+ = c^2(u-1)^2 - h^2 y(u+1)^2,$$

so that $L = F(\sqrt{\delta})$. The auxiliary prime R satisfies $(c) = AR$, and the detecting primes G_i (Definition 4.5) are primes of F, ramified in L, whose residue characters separate the square classes of virtual units (Section 2.3). The set of dyadic primes of F is $\mathcal{D}$, and T is the set of all finite primes of F ramified in L.

Groups and maps. The letter U is used for unit groups: $U_L = \prod_{v|2}(1 + 2\mathcal{O}_{L_v})$ is the group of dyadic principal units of L, $\mathcal{U}_N$ is the compact group of (20) in which unit approximation is measured, and $U^0_{n,p} \subseteq U^{\mathrm{all}}_{n,p}$ are the $\mathbb{Z}_p$-modules of (41). The group of square classes of virtual units of a number field $\mathcal{E}$ is $\mathcal{V}(\mathcal{E})$ (Section 2.3). The Selmer group $\Delta(F, \delta)$, its quotient Sel_δ and the map λ_δ are defined in Section 7; the modules $M_{n,p}$ and the maps $\mu_{n,p}$ and $\mathrm{res}_{n,p}$ are defined in Section 8. Hilbert symbols $(x, b)_v$ take values in $\{\pm 1\}$, and $[x, b]_v \in \mathbb{F}_2$ is defined by $(-1)^{[x,b]_v} = (x, b)_v$. At a place v of F_n of residue characteristic p, $W_{n,v}$ is the coefficient ring of $(F_n)_v$ (Section 2.3), always written with the level first; W_v abbreviates it when the level is fixed.

Letters such as x, b, e, g, m, r, s, ξ, γ, ψ, I, the indeterminate X, and the letter χ for a quadratic character, are also used as local variables; each such use is declared where it occurs. Apart from these, every symbol keeps one meaning throughout the paper.

2.2 Factorization background

Let S be a domain. An *irreducible element*, or *atom*, of S is a nonzero nonunit that is not a product of two nonunits. The domain S is *atomic* if every nonzero nonunit is a finite product of atoms. It satisfies the *ascending chain condition on principal ideals* (ACCP) if every chain $x_1S \subseteq x_2S \subseteq \cdots$ of principal ideals is eventually constant. It is a *bounded factorization domain* (BFD) if it is atomic and, for each nonzero nonunit x, the number of atoms in a factorization of x into atoms is bounded [1, p. 1]. By [1, Theorem 2.4], S is a BFD if and only if, for each nonzero nonunit x, the lengths r of the expressions $x = x_1 \cdots x_r$ with all x_i nonunits of S are bounded. This is the form used in Section 3, where an element with such a bound is called *bounded* in S. There are implications

$$\text{BFD} \implies \text{ACCP} \implies \text{atomic}$$

[1, p. 2 and Section 2]. Neither implication reverses: a monoid domain of Anderson, Anderson and Zafrullah satisfies ACCP but is not a BFD [1, Example 2.1], and Grams constructed an atomic domain without ACCP [7]. For a systematic treatment of factorization theory we refer to [4].

Hereditary atomicity is defined in Section 1. We use one descent property of ACCP. Let S be a subring of a domain that satisfies ACCP, and suppose that every element of S that is a unit of the larger domain is a unit of S. Then S satisfies ACCP, by a result of Grams (see [2, Proposition 5.1]). If the larger domain is the ring of integers of an algebraic extension of $\mathbb{Q}$, this unit condition holds automatically, because every element is integral over $\mathbb{Z}$ (the proof of Lemma 3.4 applies). For a subring S of D it reads $S^\times = S \cap D^\times$, and it shows that a nonunit of S is a nonunit of D, so an element that is bounded in D is bounded in every subring of D that contains it, with the same bound.

An *almost Dedekind domain* is a domain whose localization at every maximal ideal is a discrete valuation ring [5, §36], [3, Section 2.2]. An almost Dedekind domain has dimension at most one and is a Prüfer domain [5, p. 434], that is, every nonzero finitely generated ideal is invertible [2, Section 2]. It is a Dedekind domain exactly when it is Noetherian [2, Section 2]. An invertible ideal is finitely generated. Hence a maximal ideal of an almost Dedekind domain is invertible if and only if it is finitely generated. As in Section 2.1, v_V is the normalized valuation at a maximal ideal V; divisibility is then decided by these valuations (Lemma 3.2). For $x \neq 0$ in an almost Dedekind domain S, the *divisor* of x is the family $(v_V(x))_V$ indexed by the maximal ideals V of S, and its *support* is the set of those V with $v_V(x) > 0$. The Picard group $\mathrm{Pic}(S)$ of a domain S is the group of invertible fractional ideals modulo nonzero principal fractional ideals. One-sided elements (Definition 3.5), which carry the boundedness argument, are defined in Section 3.

2.3 Number-theoretic background

This subsection collects the facts from algebraic number theory that the paper uses. We assume familiarity with Dedekind domains, the splitting of primes in extensions, units and ideal class groups. Each item below says which factorization question it serves and gives a precise reference; we use Milne's notes [8, 9] throughout.

Towers of number fields. The facts in this paragraph hold for every tower of number fields; we state them for the tower $F_0 \subset F_1 \subset \cdots$, with $K = \bigcup_n F_n$ and $D = \mathcal{O}_K$. Then $D = \bigcup_n \mathcal{O}_{F_n}$, and the maximal ideals V of D correspond bijectively to the sequences $(V_n)_{n\geq 0}$ of primes V_n of F_n with $V_{n+1} \mid V_n$, via $V_n = V \cap \mathcal{O}_{F_n}$. In a step F_n/F_{n-1} of degree two, a prime $\mathfrak{p}$ of F_{n-1} with ramification indices e_i and residue degrees f_i in F_n satisfies $\sum_i e_i f_i = 2$ [8, Theorem 3.34]. So $\mathfrak{p}$ either splits into two primes, or is inert, or ramifies. If F_n/F_{n-1} is Galois and w is a place of F_n above a place v of F_{n-1}, then $(F_n)_w/(F_{n-1})_v$ is Galois with group the decomposition group of w [8, Proposition 8.10]; in particular $(F_n)_w = (F_{n-1})_v$ when v splits. The localization D_V is the union of the localizations of the rings $\mathcal{O}_{F_n}$ at V_n. If the ramification index $e(V_n/p)$ is the same for all large n, a uniformizer at one such level remains a uniformizer at all later levels, and D_V is a discrete valuation ring (Lemma 3.2).

Every invertible ideal of D is finitely generated, hence is extended from an ideal of some $\mathcal{O}_{F_n}$. An ideal of $\mathcal{O}_{F_n}$ that becomes principal in D already becomes principal in some $\mathcal{O}_{F_m}$, since the generator and the coefficients witnessing the equality lie in a finite level (proof of Lemma 3.3). Hence $\mathrm{Pic}(D)$ is the direct limit of the class groups $\mathrm{Cl}(F_n)$ under the maps induced by extension of ideals. These maps need not be injective; an ideal class that becomes principal in a larger field is said to *capitulate*. Hypothesis (T3) excludes capitulation in every step. For factorization this is the decisive point: the valuations of an element determine its divisor, and $\mathrm{Pic}(D)$ decides which divisors are principal.

Units and square classes. The unit group $\mathcal{O}_\mathcal{E}^\times$ of a number field $\mathcal{E}$ with r_1 real and r_2 complex places is finitely generated of rank $r_1 + r_2 - 1$ [8, Theorem 5.1]. If $\mathcal{E}$ is totally real, its torsion units are ± 1, so $\mathcal{O}_\mathcal{E}^\times/\mathcal{O}_\mathcal{E}^{\times 2}$ is an $\mathbb{F}_2$-vector space of dimension $[\mathcal{E} : \mathbb{Q}]$. An element $x \in \mathcal{E}^\times$ is a *virtual unit* if $v_\mathfrak{p}(x)$ is even for every prime $\mathfrak{p}$, that is, if $x\mathcal{O}_\mathcal{E} = I^2$ for a fractional ideal I. We write

$$\mathcal{V}(\mathcal{E}) = \{[x] \in \mathcal{E}^\times/\mathcal{E}^{\times 2} : v_\mathfrak{p}(x) \equiv 0 \pmod 2 \text{ for every prime } \mathfrak{p}\}$$

for the group of square classes of virtual units. Sending $[x]$ to the class of I gives an exact sequence

$$0 \longrightarrow \mathcal{O}_\mathcal{E}^\times/\mathcal{O}_\mathcal{E}^{\times 2} \longrightarrow \mathcal{V}(\mathcal{E}) \longrightarrow \mathrm{Cl}(\mathcal{E})[2] \longrightarrow 0,$$

so $\dim_{\mathbb{F}_2} \mathcal{V}(\mathcal{E}) = [\mathcal{E} : \mathbb{Q}] + \dim_{\mathbb{F}_2} \mathrm{Cl}(\mathcal{E})[2]$ when $\mathcal{E}$ is totally real. Units are the only freedom in changing a factor of an element without changing its divisor, so the images of global units in local unit groups are the decisive objects in Sections 6 to 8. The detecting primes of Section 4 are chosen so that every nonzero class of $\mathcal{V}(F_{n-1})$ has a nonsquare residue at one of them, and Proposition 4.7 uses them to obtain (T3) and (T4).

Local fields at an odd prime. Let $\mathcal{K}$ be a finite extension of $\mathbb{Q}_p$ with ring of integers $\mathcal{O}$, maximal ideal $\mathfrak{m}$ and residue field $k_{\mathcal{K}}$ of order q. The ring $\mathcal{O}$ contains exactly q roots of $X^q - X$, and reduction maps them bijectively and multiplicatively onto $k_{\mathcal{K}}$; they are the *Teichmüller representatives* [8, Remark 7.36]. This holds for every prime p, and Section 7 uses it at $p = 2$. The field $\mathcal{K}$ contains a largest unramified extension $\mathcal{K}_0$ of $\mathbb{Q}_p$, obtained by adjoining the roots of unity in $\mathcal{K}$ of order prime to p [8, Proposition 7.50 and Corollary 7.51]. Its ring of integers W is the *coefficient ring* of $\mathcal{K}$; it contains the Teichmüller representatives. Suppose that $\mathcal{K}/\mathcal{K}_0$ has degree two, so that it is totally ramified. Then $\mathcal{O} = W \oplus \pi W$ for every uniformizer π [8, Proposition 7.55 and Remark 7.56]. Since p is odd, completing the square in the minimal polynomial of a uniformizer gives a uniformizer π with $\pi^2 \in pW^\times$; we call such a π a *square uniformizer*. Newton's lemma lifts approximate roots of polynomials to roots [8, Theorem 7.32].

The p-adic exponential and logarithm are controlled by the following bound. Let v be the normalized valuation of $\mathcal{K}$ and $e = v(p)$ its absolute ramification index, and put $\mathfrak{a} = \{x \in \mathcal{K} : v(x) > e/(p-1)\}$. For $i \geq 1$ one has $v_p(i!) \leq (i-1)/(p-1)$ [8, Example 7.29], and $v_p(i) \leq (i-1)/(p-1)$ because $p^j \geq 1 + j(p-1)$. Hence, for $x \in \mathfrak{a}$, the terms $x^i/i!$ and x^i/i have valuation at least $v(x) + (i-1)\big(v(x) - e/(p-1)\big)$, which exceeds $v(x)$ for $i \geq 2$ and tends to infinity. So the series $\exp(x)$ and $\log(1+x)$ converge on $\mathfrak{a}$, and $v(\exp(x) - 1) = v(\log(1+x)) = v(x)$. The same bounds show that, for $x, x' \in \mathfrak{a}$ and $\mu = \min\{v(x), v(x')\}$, each term of total degree d in the expansions of the composed series $\exp(\log(1+X))$, $\log(\exp(X))$ and $\log((1+X)(1+Y))$ at (x, x') has valuation at least $\mu + (d-1)\big(\mu - e/(p-1)\big)$. Therefore the formal identities $\exp(\log(1+X)) = 1 + X$, $\log(\exp(X)) = X$ and $\log((1+X)(1+Y)) = \log(1+X) + \log(1+Y)$ may be evaluated on $\mathfrak{a}$, and log is an isomorphism from the multiplicative group $1 + \mathfrak{a}$ onto the additive group $\mathfrak{a}$, with inverse exp (compare the proof of [9, Chapter III, Lemma 2.4]). In particular, if $e < p - 1$, then $\mathfrak{a} = \mathfrak{m}$, and log maps $1 + \mathfrak{m}^r$ onto $\mathfrak{m}^r$ for every $r \geq 1$.

These facts describe the orders a subring can contain. If $x \in \mathcal{O}$ has residue $\bar{x} \in k_{\mathcal{K}}$, the powers x^{q^r} converge to the Teichmüller representative of $\bar{x}$. So a subring whose image in a residue field is the whole field contains, after completion, all Teichmüller representatives and hence the whole coefficient ring. This is how Section 5 reduces membership in an order to one nonnegative integer s_V at each ramified place (Proposition 5.12). For $p \geq 5$ and $[\mathcal{K} : \mathcal{K}_0] \leq 2$, so that $e \leq 2 < p - 1$, the logarithm makes the quotients $\mathcal{O}^\times / W^\times$ into free $\mathbb{Z}_p$-modules, on which condition (O) is linear (Lemma 8.1). By (T1), only primes $p \geq 5$ occur in (O); Section 8 explains why $p = 3$ is excluded.

Hilbert symbols. Let v be a place of a number field $\mathcal{E}$. For $x, b \in \mathcal{E}_v^\times$ the Hilbert symbol $(x, b)_v \in \{\pm 1\}$ equals 1 exactly when b is a norm from $\mathcal{E}_v(\sqrt{x})$, and it is bimultiplicative. At a finite place this is [9, Chapter III, Theorem 4.4(a),(d)]. At a real place it is elementary (compare [9, Chapter III, Remark 4.8]): b is a norm from $\mathbb{R}(\sqrt{x})$ exactly when $x > 0$ or $b > 0$, so $(x, b)_v = -1$ exactly when $x < 0$ and $b < 0$. In particular, at a real place $(x, b)_v = 1$ whenever $x > 0$ or $b > 0$ there.

Let v be a finite place of odd residue characteristic and π a uniformizer at v. For $x \in \mathcal{E}_v^\times$ with $v(x)$ even, write $\chi_v(x) \in \mathbb{F}_2$ for 0 if the residue of the unit $x\pi^{-v(x)}$ is a square and 1 otherwise; we call χ_v the *residue character* at v. Another uniformizer changes $\pi^{-v(x)}$ by an even power of a

unit, so $\chi_v(x)$ does not depend on π. The nonzero squares form a subgroup of index two in $k_v^\times$, so χ_v is a homomorphism on the group of elements of even valuation, and it vanishes on squares. By Newton's lemma [8, Theorem 7.32], $\chi_v(x) = 0$ exactly when x is a square in $\mathcal{E}_v$. For $x, b \in \mathcal{E}_v^\times$ with $v(x)$ even, we have $[x, b]_v = v(b)\chi_v(x)$, that is, $(x, b)_v = (-1)^{v(b)\chi_v(x)}$. For $x, b \in \mathcal{E}^\times$ with x a unit at v this is [9, Chapter VIII, 5.8]; it extends to local x and b, since both sides depend only on the square classes of x and b in $\mathcal{E}_v^\times$, which contain elements of $\mathcal{E}^\times$. In general write $x = \pi^{2k}w$ with w a unit; bimultiplicativity gives $(x, b)_v = (\pi^k, b)_v^2(w, b)_v = (w, b)_v$, and $\chi_v(x) = \chi_v(w)$. In particular $(x, b)_v = 1$ when $v(x)$ and $v(b)$ are both even, for instance when x and b are both units at v. For $x, b \in \mathcal{E}^\times$ the product formula $\prod_v (x, b)_v = 1$ holds over all places of $\mathcal{E}$ [9, Chapter V, Example 5.4]. In a quadratic step $L = F(\sqrt{\delta})$, the symbols decide whether an element of F is locally a norm from L. The product formula is the global relation among them that we use. It places the values of the map λ_δ of Section 7 on the line of vectors with coordinate sum zero, and Section 7 also uses it to show that a local condition at A_{n-1} holds automatically.

Ray class groups and primes in ray classes. A modulus $\mathfrak{f}$ of a number field $\mathcal{E}$ is a nonzero ideal $\mathfrak{f}_0$ of $\mathcal{O}_\mathcal{E}$ together with a set of real places. The ray class group $\mathrm{Cl}_\mathfrak{f}(\mathcal{E})$ is the group of fractional ideals prime to $\mathfrak{f}_0$ modulo the principal ideals (x) with $x \equiv 1 \pmod{\mathfrak{f}_0}$ and x positive at the real places of $\mathfrak{f}$. It is finite, and it sits in an exact sequence

$$0 \longrightarrow \mathcal{O}_\mathcal{E}^\times / \mathcal{O}_{\mathcal{E},\mathfrak{f},1}^\times \longrightarrow \prod_{v \mid \mathfrak{f} \text{ real}} \{\pm 1\} \times (\mathcal{O}_\mathcal{E}/\mathfrak{f}_0)^\times \longrightarrow \mathrm{Cl}_\mathfrak{f}(\mathcal{E}) \longrightarrow \mathrm{Cl}(\mathcal{E}) \longrightarrow 0,$$

where $\mathcal{O}_{\mathcal{E},\mathfrak{f},1}^\times$ is the group of units congruent to 1 modulo $\mathfrak{f}_0$ and positive at the real places of $\mathfrak{f}$ [9, Chapter V, Theorem 1.7]. By class field theory there is a finite abelian extension of $\mathcal{E}$, the ray class field, whose Galois group is identified with $\mathrm{Cl}_\mathfrak{f}(\mathcal{E})$ by the Artin map, which sends a prime $\mathfrak{p}$ not dividing $\mathfrak{f}_0$ to its Frobenius element [9, Chapter V, Theorems 3.5 and 3.6, Corollary 3.7 and Example 3.9]. The Chebotarev density theorem [9, Chapter VIII, Theorem 7.4], together with the fact that the primes of absolute residue degree at least two have Dirichlet density zero [9, Chapter VI, Proposition 4.5 and Corollary 4.6], shows that every class of $\mathrm{Cl}_\mathfrak{f}(\mathcal{E})$ contains infinitely many primes of absolute residue degree one. In Section 4, an element c with $(c) = A_{n-1}R$ satisfying finitely many congruence and sign conditions requires the auxiliary prime R to lie in a prescribed class of a ray class group, and Chebotarev's theorem supplies such primes. In Section 7 the moduli $2^m\mathcal{O}_F$, for all m, give the Galois group in which the obstruction to dyadic unit approximation lies (Lemma 7.4).

Quadratic Kummer theory. Let $\mathcal{E}$ be a number field, and fix an algebraic closure. Sending a finite abelian extension of $\mathcal{E}$ of exponent two inside this closure to the group of elements of $\mathcal{E}^\times$ that are squares in the extension is a bijection onto the subgroups of $\mathcal{E}^\times$ that contain $\mathcal{E}^{\times 2}$ with finite index. The extension corresponding to such a subgroup is generated by the square roots of its elements, and its degree equals that index [9, Chapter VII, Appendix A, Theorem A.3]. In particular, the quadratic extensions of $\mathcal{E}$ correspond to the nonzero square classes. If $x \in \mathcal{E}^\times$ is a unit at a finite place v of odd residue characteristic, then $\mathcal{E}(\sqrt{x})$ is unramified at v [9, Chapter VII, Appendix A, Proposition A.5]. If $v(x)$ is odd, then $\mathcal{E}(\sqrt{x})$ is ramified at v. Hence $\mathcal{E}(\sqrt{x})$ is ramified at such v exactly when $v(x)$ is odd: if $v(x)$ is even, choose $a \in \mathcal{E}^\times$ with $v(a) = v(x)/2$; then $\mathcal{E}(\sqrt{x}) = \mathcal{E}(\sqrt{x/a^2})$ and x/a^2 is a unit at v. Locally, with a a power of a uniformizer, the same argument and Newton's lemma show that $\mathcal{E}_v(\sqrt{x})$ is the unramified quadratic extension of $\mathcal{E}_v$ exactly when x is a nonsquare unit times an even power of a uniformizer, that is, when $v(x)$ is even and $\chi_v(x) = 1$. At a real place, $\mathcal{E}(\sqrt{x})$ is split exactly when x is positive there. In Section 7, the

group that obstructs dyadic unit approximation is a Galois group, and Kummer theory identifies the quadratic characters of the relevant quotient of this group with the classes of Sel_δ (Lemma 7.5). This turns an approximation problem in a profinite group into linear algebra over $\mathbb{F}_2$.

Curves over finite fields. For a smooth projective geometrically connected curve of genus g over $\mathbb{F}_q$, the number of its $\mathbb{F}_q$-rational points differs from $q+1$ by at most $2g\sqrt{q}$ [10, Section 1]. In Section 4 this bound supplies values of c that keep a given odd prime inert in the next step (Lemma 4.2) and, once a residue field is large, values that give the residue of y_n a prescribed two-primary component (Lemma 4.4). This is what allows the construction to prescribe the orders of the two-primary components of residues, and hence the residue offsets of (T6).

3 Almost Dedekind towers and factorization

This section isolates the properties of the tower that the factorization arguments use, and it proves everything about the ring D itself that Theorem A asserts. Section 3.1 states six hypotheses (T1)–(T6) on a tower of quadratic extensions and says where each one is used. Only (T1)–(T3) are needed in the present section. Hypothesis (T1) describes how primes split in the tower. It makes D an almost Dedekind domain in which divisibility is decided by valuations, as in a Dedekind domain (Lemma 3.2). Hypotheses (T2) and (T3) supply principal ideals and ideal classes that embed a copy of $\mathbb{Z}[1/2]/\mathbb{Z}$ in $\mathrm{Pic}(D)$ (Lemma 3.3). Comparing classes in this subgroup shows that a nonzero nonunit whose valuation vanishes at one of the two maximal ideals $\mathfrak{m}_A, \mathfrak{m}_B$ has bounded factorization lengths in every subring of D that contains it (Proposition 3.6). Section 6 applies this to the two factors into which an element of a subring is split. Finally, Theorem 3.7 shows that D is atomic, does not satisfy ACCP, and has exactly two non-invertible maximal ideals. Hypotheses (T4)–(T6) are used only from Section 5 on, to control the subrings of D.

3.1 The structural hypotheses

Let $F_0 = \mathbb{Q}(\sqrt{65}) \subset F_1 \subset \cdots$ be a tower of totally real number fields with $[F_n : F_{n-1}] = 2$, and let σ_n, E_n, K and $D = \mathcal{O}_K$ be as in Section 2.1. We write $\operatorname{Max} D$ for the set of maximal ideals of D. A *branch* is a sequence $(V_n)_{n \geq m}$ of primes V_n of F_n with $V_{n+1} \mid V_n$. The branches with $m = 0$ correspond bijectively to the maximal ideals V of D, via $V_n = V \cap \mathcal{O}_{F_n}$ (Section 2.3), and we use the same letter for a maximal ideal and its branch. Primes P_j and Q_j follow the convention of Section 2.1.

Hypothesis (T5) uses the following notion. If a step F_n/F_{n-1} has an isolated ramified prime, defined below, then every subfield of F_n contains F_{n-1} or is contained in it (Lemma 5.1). This is what forces every proper subfield of K to be a number field.

Definition 3.1. Let L/F be a quadratic extension of number fields. A prime G of F is an *isolated ramified prime* of L/F if its residue characteristic ℓ is unramified in F, $F_G = \mathbb{Q}_\ell$, G ramifies in L, and every other prime of F over ℓ is unramified in L/F.

We impose the following hypotheses. Each concerns the fields of the tower and their rings of integers, not the subrings of D.

(T1) There are primes A_n, B_n of F_n with $(F_n)_{A_n} = \mathbb{Q}_{47}$ and $(F_n)_{B_n} = \mathbb{Q}_7$ such that

$$A_{n-1}\mathcal{O}_{F_n} = A_n P_n, \qquad B_{n-1}\mathcal{O}_{F_n} = B_n Q_n,$$

where $A_0 = (47, \sqrt{65} - 21)$ and $B_0 = (7, \sqrt{65} - 3)$. Thus A_{n-1} and B_{n-1} split in F_n. We call (A_n) and (B_n) the two *chains*, and we call P_n and Q_n, the second primes of F_n over A_{n-1} and B_{n-1}, the primes *split off* from the chains. We write $\mathfrak{m}_A, \mathfrak{m}_B$ for the maximal ideals of D determined by the chains (A_n) and (B_n). They have residue fields $\mathbb{F}_{47}$ and $\mathbb{F}_7$. Moreover:

- every P_n and every Q_n is unramified and inert in each subsequent step;
- the two dyadic primes of F_0 are unramified and inert in every step;
- for each odd rational prime $p \notin \{47, 7\}$ there is a level after which all primes over p are unramified and inert in every step;
- every branch over 47 or 7 other than (A_n) and (B_n) is eventually unramified and inert, that is, unramified and inert in every step from some level on;
- absolute ramification indices at odd primes are at most two, and 3, 7 and 47 are unramified in every F_n.

(T2) There are $y_0 = -16 + 3\sqrt{65}$ and $y_n, \beta_n \in \mathcal{O}_{F_n}$ for $n \geq 1$ satisfying

$$(y_n) = A_n B_n, \quad (\beta_n) = A_n Q_n, \quad N_{F_n/F_{n-1}} y_n = N_{F_n/F_{n-1}} \beta_n = y_{n-1}.$$

Define

$$z_n = \frac{y_n \beta_n}{y_{n-1}}, \qquad z_n' = \frac{y_n}{\beta_n}. \tag{1}$$

Then $(z_n) = A_n P_n^{-1}$, $(z_n') = B_n Q_n^{-1}$, and both relative norms are one.

(T3) For each $n \geq 1$, extension of ideals induces an injective homomorphism $\mathrm{Cl}(F_{n-1}) \to \mathrm{Cl}(F_n)$, and $[A_0]$ has order two in $\mathrm{Cl}(F_0)$.

(T4) The unit norms satisfy $N_{F_n/F_{n-1}}(E_n) \subseteq E_{n-1}^2$ for every n.

(T5) Every step F_n/F_{n-1} has an isolated ramified prime G (Definition 3.1).

(T6) If $V \in \operatorname{Max} D \setminus \{\mathfrak{m}_A, \mathfrak{m}_B\}$ has odd residue characteristic, then there is an integer o_V, the *residue offset* of V, such that, for all sufficiently large n, the two-primary component of the residue of y_n at V has order 2^{n+o_V}. Distinct such V over the same rational prime have distinct residue offsets.

The assertions of (T2) about z_n and z_n' follow from the displayed identities. Indeed $(y_{n-1}) = A_{n-1}B_{n-1}$ (for $n = 1$ because $y_0 \in A_0 \cap B_0$ and $N_{F_0/\mathbb{Q}}(y_0) = -329 = -7 \cdot 47$), so $y_{n-1}\mathcal{O}_{F_n} = A_n P_n B_n Q_n$ by (T1), which gives the two principal ideals, and the relative norm of z_n is $y_{n-1} \cdot y_{n-1}/y_{n-1}^2 = 1$, and that of z_n' is $y_{n-1}/y_{n-1} = 1$.

In (T6), the residue field D/V is an algebraic extension of a finite field, so every element of its multiplicative group has finite order. The *two-primary component* of such an element is its component of two-power order, which is therefore well defined. Hypothesis (T6) is a statement about the residues of y_n at the finite levels, required for large n separately for each V; no starting level uniform in V is required.

The role of each hypothesis.

- (T1) determines the maximal ideals of D and makes D almost Dedekind (Lemma 3.2). Its local clauses (two unramified dyadic primes, ramification index at most two at odd primes, 3 unramified) restrict the local fields that occur in Sections 5–8.

- (T2) supplies the elements used throughout. The y_n give the strictly increasing chain $y_0D \subsetneq y_1D \subsetneq \cdots$, and the conjugates $\sigma_n(\beta_n)$ split an element into two one-sided factors (Theorem 3.7). The z_n and z'_n give the class identities of Lemma 3.3, and they are the elements that Section 6 approximates by global units.
- (T3) excludes capitulation of ideal classes. It makes the map ι of Lemma 3.3 injective, and this injectivity drives Proposition 3.6.
- (T4) keeps the two-primary components of the residues of global units at a fixed odd maximal ideal $V \neq \mathfrak{m}_A, \mathfrak{m}_B$ inside a finite group (Lemma 5.7). This is used in Proposition 5.9.
- (T5) forces every proper subfield of K to be a number field (Lemma 5.1 and Proposition 5.2).
- (T6) lets a subring with unbounded factorization lengths recover every odd residue field of D and separate the maximal ideals of D over each odd prime (Proposition 5.9).

Section 4 constructs a tower that satisfies all six hypotheses (Theorem 4.11).

3.2 Valuations

The first lemma shows that, for divisibility, D behaves like a Dedekind domain: each element has a valuation at every maximal ideal, and these valuations decide divisibility and units. The difference from the Dedekind case is recorded in (2): the valuation of an element at $\mathfrak{m}_A$ reappears at infinitely many of the maximal ideals P_j, and similarly for $\mathfrak{m}_B$ and the Q_j. The facts on splitting and completions used in the proof are recalled in Section 2.3.

Lemma 3.2. *Under* (T1), *D is almost Dedekind. Denote the normalized valuation at $V \in \operatorname{Max} D$ by v_V. For every $0 \neq x \in D$,*

$$v_{P_j}(x) = v_{\mathfrak{m}_A}(x), \qquad v_{Q_j}(x) = v_{\mathfrak{m}_B}(x) \quad (j \gg 0), \tag{2}$$

and x has positive valuation at only finitely many maximal ideals outside $\{\mathfrak{m}_A, \mathfrak{m}_B, P_j, Q_j : j \geq 1\}$. Each P_j, Q_j is invertible. An element of D is a unit of D exactly when all its valuations are zero, and $x' \mid_D x$ exactly when $v_V(x') \leq v_V(x)$ at every $V \in \operatorname{Max} D$. More generally, an element $x \in K^\times$ lies in D exactly when $v_V(x) \geq 0$ at every $V \in \operatorname{Max} D$, and lies in $D^\times$ exactly when all its valuations are zero.

Proof. We show that each localization is a discrete valuation ring, then compare valuations along the two chains, then count the remaining maximal ideals.

Step 1: D is almost Dedekind. Let $V \in \operatorname{Max} D$ have residue characteristic p. The local ring D_V is the increasing union of the local rings of the $\mathcal{O}_{F_n}$ at V_n. The ramification index $e(V_n/p)$ is nondecreasing in n. By (T1) it is at most two if p is odd, and it is constant if $p = 2$, because the dyadic primes of F_0 are unramified in every step. Hence it is constant from some level on. From that level on, a uniformizer at one level remains a uniformizer at every later level. Every nonzero element of D_V lies in one of these later local rings, where it is a power of this uniformizer times a unit. Thus D_V is a valuation domain with value group $\mathbb{Z}$, that is, a discrete valuation ring.

Step 2: the chains. Place x in F_m. By (T1) and Section 2.3, the completions along the chains (A_n) and (B_n) are all $\mathbb{Q}_{47}$ and $\mathbb{Q}_7$, so each step along a chain is unramified of local degree one. The prime P_j with $j > m$ lies over A_{j-1}, has completion $\mathbb{Q}_{47}$, and is unramified in every later step. Therefore $v_{P_j}(x) = v_{A_{j-1}}(x) = v_{A_m}(x) = v_{\mathfrak{m}_A}(x)$ for every $j > m$. The same argument for (B_n) gives $v_{Q_j}(x) = v_{\mathfrak{m}_B}(x)$ for $j > m$. This proves (2).

Step 3: the remaining maximal ideals. If $x \in V$ and V lies over the rational prime p, then $N_{F_m/\mathbb{Q}}(x) \in V \cap \mathbb{Z} = p\mathbb{Z}$. So only the finitely many rational primes dividing $N_{F_m/\mathbb{Q}}(x)$ can occur. For a rational prime other than 47 and 7, (T1) supplies a common level after which no prime over it splits, so finitely many maximal ideals of D lie over it. Over 47 and 7, consider the primes of F_0 conjugate to A_0 and B_0. The set of infinite branches through one of them is the inverse limit of the finite sets of primes above it at the successive levels, and is therefore compact. Each such branch is eventually inert by (T1), so some finite-level prime on it has exactly one infinite continuation. The set of branches through that prime is then an open singleton. A compact discrete space is finite, so these branches are finitely many. The branches through A_0 are (A_n) and, for each $j \geq 1$, the branch of P_j, because P_j is inert in every later step; the same holds for B_0 with the Q_j. Hence all remaining maximal ideals over 47 and 7 are $\mathfrak{m}_A$, $\mathfrak{m}_B$ and the P_j, Q_j. This proves the finiteness assertion.

Step 4: invertibility of P_j and Q_j. The ideal $P_j\mathcal{O}_{F_n}$ remains prime for every $n \geq j$, so P_jD is maximal. It is invertible because extension of ideals preserves the inverse of the invertible ideal $P_j \subset \mathcal{O}_{F_j}$. The same argument applies to Q_j.

Step 5: units and divisibility. Inside K we have $D = \bigcap_{V \in \operatorname{Max} D} D_V$. Hence $x/x' \in D$ exactly when $v_V(x) - v_V(x') \geq 0$ for every V, and $x^{-1} \in D$ exactly when $v_V(x) \leq 0$, that is $v_V(x) = 0$, for every V. The same equality $D = \bigcap_V D_V$ gives the criteria for elements $x \in K^\times$: $x \in D$ exactly when every $v_V(x) \geq 0$, and $x, x^{-1} \in D$ exactly when every $v_V(x) = 0$. □

Support and divisor of elements of D are as in Section 2.2.

3.3 The Picard group

Since D is almost Dedekind, the valuations of an element determine its principal ideal, and the question of which ideals are principal is answered by $\operatorname{Pic}(D)$. By Section 2.3, $\operatorname{Pic}(D)$ is the direct limit of the class groups $\operatorname{Cl}(F_n)$, and (T3) makes every map in this direct system injective. The next lemma locates the classes of the primes of the two chains in a copy of $\mathbb{Z}[1/2]/\mathbb{Z}$.

The mechanism is visible already at the first steps. The ideal A_0 has class of order two, which corresponds to $\frac{1}{2}$. In F_1 it splits as A_1P_1, and the element z_1 of (T2) shows that A_1 and P_1 have the same class. So the class of A_0 is twice the class of A_1, which therefore corresponds to $\frac{1}{4}$. Continuing, $[A_nD]$ corresponds to 2^{-n-1}. This matches the divisors. For $m > n$, iterating (T1) gives $A_n\mathcal{O}_{F_m} = A_mP_{n+1}\cdots P_m$, and correspondingly $2^{-n-1} = 2^{-m-1} + \sum_{j=n+1}^{m} 2^{-j-1}$. The primes of the chain (B_n) and the Q_j carry the negative classes, because y_n and z'_n are principal.

Lemma 3.3 (The classes of A_n, B_n, P_j, Q_j in $\operatorname{Pic}(D)$)**.** *Under* (T1)–(T3)*, there is an injective homomorphism $\iota : \mathbb{Z}[1/2]/\mathbb{Z} \longrightarrow \operatorname{Pic}(D)$ such that*

$$\begin{aligned} [A_nD] &= \iota(2^{-n-1} + \mathbb{Z}), \quad [B_nD] = -[A_nD], \\ [P_jD] &= \iota(2^{-j-1} + \mathbb{Z}), \quad [Q_jD] = -[P_jD]. \end{aligned} \tag{3}$$

The lemma says nothing about the rest of $\operatorname{Pic}(D)$, and nothing more is needed.

Proof. We compute the classes at each finite level, show that no nonzero class of a finite level vanishes in D, and then pass to the limit.

Step 1: classes at level n. The principal ideal $(z_n) = A_nP_n^{-1}$ gives $[P_n] = [A_n]$ in $\operatorname{Cl}(F_n)$. Consequently extension sends $[A_{n-1}]$ to $[A_nP_n] = 2[A_n]$. By (T3), $[A_0]$ has order two, and if $[A_{n-1}]$ has order 2^n, then injectivity of extension shows that $2[A_n]$ has order 2^n, so $[A_n]$ has order 2^{n+1}. By induction, $[A_n]$ has order 2^{n+1} for every n. The principal ideal $(y_n) = A_nB_n$ gives $[B_n] = -[A_n]$, and $(z'_n) = B_nQ_n^{-1}$ gives $[Q_n] = [B_n] = -[P_n]$.

Step 2: no capitulation in D. A class of a finite level that becomes principal in D already becomes principal in a finite level. Indeed, an equality $ID = uD$ involves finitely many generators of I and of I^{-1}, the element u, and finitely many coefficients witnessing both inclusions $ID \subseteq uD$ and $uD \subseteq ID$. All these data lie in a common finite level. Thus (T3) prevents a nonzero class of a finite level from vanishing in $\mathrm{Pic}(D)$, and the class of A_nD in $\mathrm{Pic}(D)$ has order 2^{n+1}.

Step 3: the limit. By Step 1, $2[A_nD] = [A_{n-1}D]$ in $\mathrm{Pic}(D)$. So the cyclic subgroups generated by the classes $[A_nD]$ are compatible, and $\iota(a2^{-n-1} + \mathbb{Z}) = a[A_nD]$ is a well-defined homomorphism on $\mathbb{Z}[1/2]/\mathbb{Z}$, the union of its cyclic subgroups generated by $2^{-n-1}+\mathbb{Z}$. It is injective because $[A_nD]$ and $2^{-n-1} + \mathbb{Z}$ have the same order 2^{n+1}. The identities (3) follow from Step 1. □

3.4 Bounded factorizations of one-sided elements

Let S be a domain and x a nonzero nonunit of S. We consider expressions $x = x_1 \cdots x_r$ with arbitrary nonunits x_i of S, not assumed irreducible. We say that x is *bounded* in S, or has *bounded factorization lengths* in S, if the lengths r of these expressions are bounded. A longest such expression is a product of atoms, since a factor that is not an atom could be split into two nonunits to give a longer expression. So a bounded element has a factorization into atoms, and a domain in which every nonzero nonunit is bounded is a BFD (Section 2.2). This convention lets us prove boundedness before atomicity is known.

The next lemma lets us compare factorizations in a subring S with valuations in D: a nonunit of S is a nonunit of D.

Lemma 3.4 (Units in subrings)**.** *For every subring $S \subseteq D$, one has $S^\times = S \cap D^\times$.*

Proof. Every element of D is integral over $\mathbb{Z} \subseteq S$. Let $s \in S$ have an inverse in D, and let $s^{-k} + a_{k-1}s^{-(k-1)} + \cdots + a_0 = 0$ with $a_i \in \mathbb{Z}$. Multiplying by s^{k-1} expresses s^{-1} as $-(a_{k-1} + a_{k-2}s + \cdots + a_0 s^{k-1}) \in S$. □

Definition 3.5. A nonzero $x \in D$ is *one-sided* if $v_{\mathfrak{m}_A}(x) = 0$ or $v_{\mathfrak{m}_B}(x) = 0$, that is, if $x \notin \mathfrak{m}_A \cap \mathfrak{m}_B$.

The proof below shows that, in a factorization of a one-sided element x, every factor has positive valuation at $\mathfrak{m}_A$ or $\mathfrak{m}_B$ or at one of finitely many maximal ideals determined by x. An element of $\mathfrak{m}_A \cap \mathfrak{m}_B$ behaves differently: the classes of P_n and Q_n cancel, and y_{n-1}/y_n has divisor P_nQ_n. So $y_0 = y_n \prod_{k=1}^{n}(y_{k-1}/y_k)$ is a product of $n+1$ nonunits of D for every n, and y_0 is not bounded in D.

Proposition 3.6. *Assume* (T1)–(T3). *Let $S \subseteq D$ be a subring. Each one-sided nonzero nonunit of S has bounded factorization lengths in S, with a bound that can be chosen independently of S.*

Proof. We bound the number of factors that can meet $\mathfrak{m}_A$, and show that every other factor must have positive valuation at one of finitely many maximal ideals fixed in advance.

Step 1: the cutoff. Suppose first that $(v_{\mathfrak{m}_A}(x), v_{\mathfrak{m}_B}(x)) = (\kappa_A, 0)$ with $\kappa_A > 0$. By (2), we may choose n_1 such that $v_{P_j}(x) = \kappa_A$ and $v_{Q_j}(x) = 0$ for $j > n_1$, and such that $\kappa_A \sum_{j>n_1} 2^{-j-1} < 1$. Let $\mathcal{H}$ be the set of maximal ideals at which x has positive valuation, other than $\mathfrak{m}_A$ and the P_j with $j > n_1$. It is finite by Lemma 3.2.

Step 2: the factors. Consider a factorization $x = x_1 \cdots x_r$ in nonunits of S. Each factor is a nonunit of D by Lemma 3.4, and divides x in D. Since valuations are additive, at most κ_A factors have positive valuation at $\mathfrak{m}_A$. A remaining factor has valuation zero at $\mathfrak{m}_A$ and at $\mathfrak{m}_B$. By Lemma 3.2 it therefore has finite support, and this support is nonempty because the factor is a nonunit of D. Suppose that this support were contained in $\{P_j : j > n_1\}$. Then the principal ideal

of the factor would be $\prod_{j>n_1} P_j^{e_j} D$, with $0 \le e_j \le \kappa_A$ because the factor divides x, and with not all e_j zero. By Lemma 3.3 its class would be

$$\iota\left(\sum_{j>n_1} e_j 2^{-j-1} + \mathbb{Z}\right),$$

where the finite sum lies strictly between zero and one. This class is nonzero because ι is injective, which contradicts principality. Hence every remaining factor has positive valuation at a member of $\mathcal{H}$.

Step 3: the bound. Again by additivity of valuations,

$$r \le \kappa_A + \sum_{V \in \mathcal{H}} v_V(x).$$

The case $(v_{\mathfrak{m}_A}(x), v_{\mathfrak{m}_B}(x)) = (0, \kappa_B)$ with $\kappa_B > 0$ is symmetric. If $v_{\mathfrak{m}_A}(x) = v_{\mathfrak{m}_B}(x) = 0$, the whole support of x is finite, each factor has positive valuation somewhere in it, and $r \le \sum_V v_V(x)$. These bounds involve only x and D, not S. □

3.5 The ring of integers of the tower

We now prove the properties of D stated in Theorem A. The conjugates $\sigma_n(\beta_n)$ of the elements β_n of (T2) split an element with both valuations positive into two one-sided factors, the elements y_n give an ascending chain of principal ideals, and finite generation distinguishes $\mathfrak{m}_A$ and $\mathfrak{m}_B$ from the other maximal ideals.

Theorem 3.7. *Assume* (T1)–(T3). *The almost Dedekind domain D is atomic, does not satisfy ACCP, and has exactly two non-invertible maximal ideals, namely $\mathfrak{m}_A$ and $\mathfrak{m}_B$.*

Proof. We prove the three assertions in turn.

Step 1: atomicity. Let x be a nonzero nonunit of D. If x is one-sided, Proposition 3.6 with $S = D$ shows that x is bounded in D, so a longest expression of x is a factorization into atoms. It remains to treat x with $\kappa_A = v_{\mathfrak{m}_A}(x)$ and $\kappa_B = v_{\mathfrak{m}_B}(x)$ both positive. By (2), choose n_1 such that $v_{P_j}(x) = \kappa_A$ and $v_{Q_j}(x) = \kappa_B$ for $j > n_1$. The automorphism σ_n interchanges the two primes A_n, P_n over A_{n-1} and the two primes B_n, Q_n over B_{n-1}, so for each n the element $\sigma_n(\beta_n)$ has ideal $P_n B_n$. Choose κ_B distinct indices $i_1, \dots, i_{\kappa_B} > n_1$, and put $x' = \prod_k \sigma_{i_k}(\beta_{i_k})$. Then $(v_{\mathfrak{m}_A}(x'), v_{\mathfrak{m}_B}(x')) = (0, \kappa_B)$, the valuations of x' at the P_j are at most one, those at the Q_j are at most κ_B, and all its support outside $\mathfrak{m}_A, \mathfrak{m}_B$ consists of primes P_j, Q_j with $j > n_1$. By Lemma 3.2, $x' \mid_D x$. Both x' and x/x' are one-sided nonunits of D, since $v_{\mathfrak{m}_B}(x') = \kappa_B > 0$ and $v_{\mathfrak{m}_A}(x/x') = \kappa_A > 0$. Proposition 3.6 gives atomic factorizations of both, and their product is an atomic factorization of x.

Step 2: failure of ACCP. Since $(y_{n-1}) = A_{n-1}B_{n-1}$, (T1) gives $y_{n-1}\mathcal{O}_{F_n} = A_n P_n B_n Q_n$, so the quotient y_{n-1}/y_n has ideal $P_n Q_n$ in $\mathcal{O}_{F_n}$. It is therefore an integral nonunit of D, and

$$y_0 D \subsetneq y_1 D \subsetneq y_2 D \subsetneq \cdots$$

is a strictly increasing chain of principal ideals.

Step 3: the non-invertible maximal ideals. Let $V \notin \{\mathfrak{m}_A, \mathfrak{m}_B\}$. By prime avoidance, choose $x \in V \setminus (\mathfrak{m}_A \cup \mathfrak{m}_B)$. Its support is finite by Lemma 3.2. Adjoin to x an element of D that is a uniformizer at V (a numerator of a uniformizer of D_V) and, for every other maximal ideal in the support of x, an element of V outside that maximal ideal. These finitely many elements generate V,

since the ideal they generate is contained in V and agrees with V after localization at every maximal ideal. A finitely generated nonzero ideal of an almost Dedekind domain is invertible (Section 2.2), so V is invertible. In contrast, any finite set of elements of $\mathfrak{m}_A$ lies in every sufficiently late P_j, by (2), and so cannot generate $\mathfrak{m}_A$. Likewise $\mathfrak{m}_B$ is not finitely generated. Since every invertible ideal is finitely generated, $\mathfrak{m}_A$ and $\mathfrak{m}_B$ are precisely the non-invertible maximal ideals. □

Lemma 3.2 and Theorem 3.7, together with the residue fields recorded in (T1), prove parts (a)–(d) of Theorem A for every tower satisfying (T1)–(T3); part (a) uses only (T1). Theorem A then follows from the existence of such a tower, which Theorem 4.11 provides.

4 Construction of the tower

This section constructs the tower of Theorem A. Theorem 4.11 produces an infinite tower $F_0 = \mathbb{Q}(\sqrt{65}) \subset F_1 \subset \cdots$ that satisfies the hypotheses (T1)–(T6) of Section 3 and in which every step is a scaled step (Definition 4.8). With Lemma 3.2 and Theorem 3.7, this proves Theorem A. Scaled steps are also the steps to which Theorem B applies.

What one step must achieve. The tower is built one quadratic step at a time. Suppose that $F = F_{n-1}$ has been constructed, together with primes $A = A_{n-1}$ and $B = B_{n-1}$ and an element $y = y_{n-1}$ with $(y) = AB$. The next field is $L = F(\alpha)$, where α is a root of a polynomial $X^2 - tX + y$ with $t \in \mathcal{O}_F$, and α becomes y_n. Then $N_{L/F}\alpha = y$ automatically, and the construction consists in choosing t. The field L must be totally real, since the assumptions at the next level require it; the step arranges this by making the discriminant of $X^2 - tX + y$ totally positive. Beyond this, the step must achieve five things.

(1) *The chains continue* (hypotheses (T1) and (T2)). The primes A and B split in L; the ideal (α) is the product of one prime A' above A and one prime B' above B; and a second element β of norm y generates $A'Q$, where Q is the other prime of L above B. Together with the injectivity in (2), these identities give the class of A' order 2^{n+1} in $\mathrm{Cl}(L)$.

(2) *No capitulation and square unit norms* (hypotheses (T3) and (T4)). No nontrivial ideal class of F becomes principal in L, and the norm of every unit of L is a square in F. Both follow from making finitely many chosen primes of F, the detecting primes of Definition 4.5, ramify in L. The residue characters at these primes detect every nonzero class of the group $\mathcal{V}(F)$ of square classes of virtual units (Section 2.3).

(3) *All other branches are eventually inert* (hypothesis (T1)). At each step, finitely many primes of F carry an inert prescription: L is required to be unramified and inert there. New ramification is allowed only over rational primes that are unramified in F, which keeps the absolute ramification indices at odd primes at most two.

(4) *Residue offsets* (hypothesis (T6)). At finitely many odd primes, the step prescribes whether the prime splits or is inert, and it prescribes the two-primary component of the residue of α. Along a branch that stays inert, the relation $N_{F_n/F_{n-1}} y_n = y_{n-1}$ doubles the order of this component at each step, once the residue fields have order congruent to 1 modulo 4 and the component is nontrivial. Prescribing the component at finitely many steps therefore fixes the residue offsets o_V and makes them distinct over each rational prime.

(5) *An isolated ramified prime* (hypothesis (T5)). One of the detecting primes is ramified in L, and every other prime of F over the same rational prime is unramified in L.

Why the scaled family. Apart from the existence of β, each requirement is a local condition on L at finitely many primes of F. The scaled family (4) of Section 4.2 meets the requirement on β by an identity, and it turns every local condition into a congruence condition on a single element $c \in F$. For $\beta = a + u\alpha$ with $a \in F$ and $u \in F^\times$ one has $N_{L/F}\beta = a^2 + aut + u^2 y$, so $N_{L/F}\beta = y$ exactly when $t = (y(1-u^2) - a^2)/(au)$. The family takes $a = c(u-1)/h$, where $h = 2^H$ is the scale and u is a power of a fixed unit of F_0 with $v_2(u-1) = H$ at both dyadic primes (5). This makes a a unit at the dyadic primes, which allows them to stay inert. The discriminant of $X^2 - tX + y$ is then $\delta_-\delta_+/(chu)^2$, where δ_- and δ_+ are explicit polynomials in c, u and $h^2 y$. At a prime V of F at which this discriminant is nonzero, the splitting of V in L and the residues of α therefore depend only on the classes of c and u modulo a sufficiently high power of V. Finally, c is chosen with $(c) = AR$ for an auxiliary prime R. The factor A places α and β at the same prime of L above A, and R is chosen with $u \equiv -1 \pmod{R}$, which makes t integral. Such a prime R, compatible with all congruence conditions on c, exists by the Chebotarev density theorem applied to a ray class field (Section 2.3).

Organization. Section 4.1 computes the arithmetic of the base field $F_0 = \mathbb{Q}(\sqrt{65})$ (Lemma 4.1). Section 4.2 states the assumptions under which one step is made and defines the scaled family. Sections 4.3 and 4.4 solve the local problems at a single odd prime: Lemma 4.2 realizes an inert prescription, and Lemma 4.4 realizes a prescribed two-primary component of the residue of α once the residue field is large. Both lemmas count points on curves over finite fields. The rest of the section assembles the step and the tower. Lemma 4.6 finds primes with prescribed Frobenius in a finite abelian extension at which a fixed unit has residue order prime to given odd primes; Definition 4.5 defines detecting primes; Proposition 4.7 combines all local conditions into one quadratic step; and Definition 4.8 defines scaled steps by the properties of that step used in later sections. A finite procedure (Section 4.9) then separates the residue offsets, and Definition 4.10 and Theorem 4.11 assemble the infinite tower.

4.1 The base field

The tower starts from the real quadratic field $F_0 = \mathbb{Q}(\sqrt{65})$. Its class group has order two and is generated by the class of A_0, which starts the chain of classes $[A_n]$ of order 2^{n+1}. The unit $\epsilon_0 = 8 + \sqrt{65}$ supplies the unit ϵ_H of the scaled family. The following lemma collects the facts about F_0 that the construction uses. Each can be checked by hand. Put

$$\omega = \frac{1+\sqrt{65}}{2}, \qquad \epsilon_0 = 8 + \sqrt{65}, \qquad y_0 = -16 + 3\sqrt{65},$$

and recall from (T1) that $A_0 = (47, \sqrt{65} - 21)$ and $B_0 = (7, \sqrt{65} - 3)$.

Lemma 4.1 (The base field). (a) *$\mathcal{O}_{F_0} = \mathbb{Z}[\omega]$, where $\omega^2 - \omega - 16 = 0$, and the discriminant of F_0 is 65. The prime 2 splits as $2\mathcal{O}_{F_0} = \mathfrak{d}_0\mathfrak{e}_0$ with*

$$\mathfrak{d}_0 = (2, \omega - 1), \qquad \mathfrak{e}_0 = (2, \omega),$$

the prime 3 is inert, and A_0 and B_0 are primes of residue degree one above 47 and 7. In particular $(F_0)_{\mathfrak{d}_0} = (F_0)_{\mathfrak{e}_0} = \mathbb{Q}_2$, $(F_0)_{A_0} = \mathbb{Q}_{47}$ and $(F_0)_{B_0} = \mathbb{Q}_7$.

(b) *$(\omega + 4) = \mathfrak{e}_0^2$, and $\mathrm{Cl}(F_0) \simeq \mathbb{Z}/2\mathbb{Z}$, generated by each of the classes $[\mathfrak{e}_0]$, $[A_0]$ and $[B_0]$.*

(c) *$(y_0) = A_0 B_0$.*

(d) ϵ_0 *is a unit of norm* -1*, and* $\epsilon_0^2 - 1 = 16\epsilon_0$*. Its residue at* A_0 *is* 29*, of order* 46 *in* $\mathbb{F}_{47}^\times$*, and its residue at* B_0 *is* 4*, of order* 3 *in* $\mathbb{F}_7^\times$*.*

(e) $\epsilon_0 \equiv 1 \pmod 8$ *at* $\mathfrak{d}_0$ *and* $\epsilon_0 \equiv -1 \pmod 8$ *at* $\mathfrak{e}_0$*. Hence* ϵ_0^{23} *is a square in* $(F_0)_{\mathfrak{d}_0}$*, lies in the square class of* -1 *in* $(F_0)_{\mathfrak{e}_0}$*, and has nonsquare residue at* A_0*.*

Proof. We verify the statements in order; all computations are with explicit integers.

Step 1: part (a). Since 65 is squarefree and $65 \equiv 1 \pmod 4$, the ring of integers is $\mathbb{Z}[\omega]$ and the discriminant is 65 [8, Example 2.41]. One checks $\omega^2 - \omega - 16 = 0$ directly. By the Dedekind–Kummer theorem [8, Theorem 3.41], the factorization of a rational prime p is read off from the factorization of $X^2 - X - 16$ modulo p. Modulo 2 this polynomial is $X(X-1)$, so $2\mathcal{O}_{F_0} = (2,\omega)(2,\omega-1)$ with both factors of residue degree one. Modulo 3 it is $X^2 - X - 1$, which takes the values $2, 2, 1$ at $0, 1, 2$; so 3 is inert. Since $21^2 = 65 + 8 \cdot 47$ and $3^2 = 65 - 8 \cdot 7$, the number 65 is a square modulo 47 and modulo 7, so 47 and 7 split [8, Example 3.44]. Since $X^2 - X - 16 \equiv (X-11)(X-37) \pmod{47}$ and $X^2 - X - 16 \equiv (X-2)(X-6) \pmod 7$, Theorem 3.41 of [8] shows that $A_0 = (47, \omega - 11)$ and $B_0 = (7, \omega - 2)$ are primes of residue degree one; note that $\sqrt{65} - 21 = 2(\omega - 11)$ and $\sqrt{65} - 3 = 2(\omega - 2)$. A prime of residue degree one over p in a quadratic field that splits p has completion $\mathbb{Q}_p$, which gives the last assertion.

Step 2: part (b). The Minkowski bound of F_0 is $\frac{2!}{2^2}\sqrt{65} = \sqrt{65}/2 < 5$ [8, Theorem 4.3], so every ideal class contains an integral ideal of norm at most 4. Such an ideal is a product of primes above 2 and 3. The ideal (3) is prime and principal, and $\mathfrak{d}_0\mathfrak{e}_0 = (2)$, so $\mathrm{Cl}(F_0)$ is generated by $[\mathfrak{e}_0] = -[\mathfrak{d}_0]$. The conjugate of ω is $1 - \omega$, so $N(\omega + 4) = (\omega + 4)(5 - \omega) = -\omega^2 + \omega + 20 = 4$. Moreover $\omega + 4 \in (2, \omega) = \mathfrak{e}_0$, and $\omega + 4 \equiv \omega \equiv 1 \pmod{\mathfrak{d}_0}$. Hence $(\omega + 4) = \mathfrak{e}_0^2$, and $2[\mathfrak{e}_0] = 0$. It remains to show that $\mathfrak{e}_0$, A_0 and B_0 are not principal. A generator of an ideal of norm p has the form $(a + b\sqrt{65})/2$ with $a, b \in \mathbb{Z}$ and norm $(a^2 - 65b^2)/4 = \pm p$, so $a^2 - 65b^2 = \pm 4p$. For $p = 2$, 47 and 7 one has $4p \equiv 3 \pmod 5$, so $a^2 \equiv \pm 3 \pmod 5$. The squares modulo 5 are $0, 1, 4$, which excludes this. Thus $\mathrm{Cl}(F_0) \simeq \mathbb{Z}/2\mathbb{Z}$, and each of the three classes generates it.

Step 3: part (c). Modulo A_0 one has $\sqrt{65} \equiv 21$, so $y_0 \equiv -16 + 63 = 47 \equiv 0$. Modulo B_0 one has $\sqrt{65} \equiv 3$, so $y_0 \equiv -16 + 9 = -7 \equiv 0$. Thus A_0B_0 divides (y_0). Since $N(y_0) = 256 - 9 \cdot 65 = -329 = -47 \cdot 7$, the ideal (y_0) has norm 329, and $(y_0) = A_0B_0$.

Step 4: part (d). $N(\epsilon_0) = 64 - 65 = -1$ and $\epsilon_0^2 = 129 + 16\sqrt{65} = 1 + 16\epsilon_0$. Modulo A_0 the residue of ϵ_0 is $8 + 21 = 29$. Here $29^2 = 841 \equiv 42 \not\equiv 1 \pmod{47}$, and $29^{23} \equiv -1 \pmod{47}$ by Euler's criterion, because 29 is not a square modulo 47: by quadratic reciprocity $\left(\frac{29}{47}\right) = \left(\frac{47}{29}\right) = \left(\frac{18}{29}\right) = \left(\frac{2}{29}\right) = -1$, since $29 \equiv 5 \pmod 8$. So the order of 29 divides 46 but not 2 or 23, and it equals 46. Modulo B_0 the residue of ϵ_0 is $8 + 3 = 11 \equiv 4$, and $4^3 = 64 \equiv 1 \pmod 7$ with $4 \neq 1$, so the order is 3.

Step 5: part (e). Note that $\epsilon_0 = 2\omega + 7$ and that $\omega(\omega - 1) = 16$. At $\mathfrak{d}_0 = (2, \omega - 1)$ the element ω is a unit, so $v_2(\omega - 1) = 4$, $\omega \equiv 1 \pmod{16}$ and $\epsilon_0 \equiv 9 \equiv 1 \pmod 8$. At $\mathfrak{e}_0 = (2, \omega)$ the element $\omega - 1$ is a unit, so $v_2(\omega) = 4$ and $\epsilon_0 \equiv 7 \equiv -1 \pmod 8$. An element $x \in \mathbb{Z}_2$ with $x \equiv 1 \pmod 8$ is a square, by Newton's lemma [8, Theorem 7.32] applied to $X^2 - x$ at $X = 1$: there $|1 - x|_2 \leq 2^{-3} < 2^{-2} = |2|_2^2$. Since 23 is odd, ϵ_0^{23} is congruent to 1 modulo 8 at $\mathfrak{d}_0$, hence a square there, and $-\epsilon_0^{23}$ is congruent to 1 modulo 8 at $\mathfrak{e}_0$, so ϵ_0^{23} lies in the square class of -1 there. Finally the residue of ϵ_0^{23} at A_0 is $29^{23} \equiv -1$, which is not a square modulo 47 because $47 \equiv 3 \pmod 4$. □

In particular, $F = F_0$ with $A = A_0$, $B = B_0$ and $y = y_0$ satisfies the standing assumptions of Section 4.2 with $n = 1$.

4.2 The scaled family

This subsection fixes the setting of one step and defines the family of quadratic extensions from which each step is chosen.

Standing assumptions. Let $n \geq 1$, and write $F = F_{n-1}$, $A = A_{n-1}$, $B = B_{n-1}$ and $y = y_{n-1}$. We assume:

- F is totally real;
- F has exactly two dyadic primes $\mathfrak{d}$ and $\mathfrak{e}$, above $\mathfrak{d}_0$ and $\mathfrak{e}_0$ respectively, and both are absolutely unramified;
- the completions of F at A and B are $F_A = \mathbb{Q}_{47}$ and $F_B = \mathbb{Q}_7$;
- A lies over A_0 and B lies over B_0;
- $(y) = AB$, and the class $[A]$ has order 2^n in $\mathrm{Cl}(F)$.

Fix an integer $H \geq 5$ and set

$$\begin{gathered} h = 2^H, \qquad \epsilon = \epsilon_H = \epsilon_0^{46\cdot 2^{H-4}} = (\epsilon_0^{23})^{2^{H-3}}, \qquad u = \epsilon^k, \\ a = c(u-1)/h, \qquad \delta_- = c^2 - h^2 y, \qquad \delta_+ = c^2(u-1)^2 - h^2 y(u+1)^2, \\ t = \frac{y(1-u^2)-a^2}{au} = -\frac{h^2 y + c^2}{ch} + \frac{\delta_-}{chu}, \qquad \delta = t^2 - 4y = \frac{\delta_-\delta_+}{(chu)^2}, \\ L = F(\alpha), \quad \alpha^2 - t\alpha + y = 0, \qquad \beta = a + u\alpha. \end{gathered} \tag{4}$$

Here $c \in F^\times$ and the positive integer k are parameters. The field L is the candidate for F_n, with $y_n = \alpha$ and $\beta_n = \beta$. We will choose $(c) = AR$, with R an *auxiliary prime* of absolute residue degree one selected in the proof of Proposition 4.7, c totally positive, and k in an infinite arithmetic progression of positive integers.

The formulas in (4) are explained as follows. If α' is the other root of $X^2 - tX + y$, then $(a + u\alpha)(a + u\alpha') = a^2 + aut + u^2 y$. So the first expression for t is exactly the condition that β have norm y. Substituting $a = c(u-1)/h$ into it gives the second expression for t, and expanding $t^2 - 4y$ gives the factorization of δ. Thus $L = F(\sqrt{\delta})$, and δ has the same square class as $\delta_-\delta_+$.

The unit u at the dyadic primes, at A and at B. We use the normalization $v_2(2) = 1$ (Section 2.1). At both dyadic primes,

$$v_2(\epsilon^k - 1) = H, \qquad (\epsilon^k - 1)/2^H \equiv 1 \pmod 2 \quad (k \text{ odd}). \tag{5}$$

This follows by repeated squaring of ϵ_0^{46}, starting from the identity $\epsilon_0^2 - 1 = 16\epsilon_0$ of Lemma 4.1(d). That identity gives $v_2(\epsilon_0^2 - 1) = 4$ at $\mathfrak{d}_0$ and $\mathfrak{e}_0$. Since $\epsilon_0^{46} - 1 = (\epsilon_0^2 - 1)(1 + \epsilon_0^2 + \cdots + \epsilon_0^{44})$ and the second factor is a sum of 23 terms congruent to 1 modulo 2, it is a unit, and $v_2(\epsilon_0^{46} - 1) = 4$. If $v_2(x-1) = s \geq 2$, then $v_2(x+1) = 1$ and $v_2(x^2 - 1) = s + 1$. Applying this $H - 4$ times gives $v_2(\epsilon - 1) = H$. The same factorization argument, with k odd in place of 23, gives $v_2(\epsilon^k - 1) = H$. Then $(\epsilon^k - 1)/2^H$ is a unit of $(F_0)_{\mathfrak{d}_0} = \mathbb{Q}_2$, respectively $(F_0)_{\mathfrak{e}_0} = \mathbb{Q}_2$, so it is 1 modulo 2. These formulas persist in unramified extensions, in particular at the dyadic primes of F.

Since $F_A = \mathbb{Q}_{47}$ and $F_B = \mathbb{Q}_7$, the residue fields at A and B are $\mathbb{F}_{47}$ and $\mathbb{F}_7$. Hence $\epsilon \equiv 1 \pmod A$, because 46 divides the exponent of ϵ. By Lemma 4.1(d), the residue of ϵ_0 at B is a root of $X^2 - 16X - 1$ modulo 7, that is, 4 or 5, of order three or six in $\mathbb{F}_7^\times$. The exponent of ϵ is even and prime to 3, so the residue of ϵ at B has order three. These facts use only the residue fields at

A and B. Since A and B lie over A_0 and B_0, the residues of ϵ_0 at A and B are 29 and 4, as at A_0 and B_0; Step 5 of the proof of Proposition 4.7 uses that 29 is not a square modulo 47. Thus $k \equiv 1 \pmod 6$ ensures that a is a unit at the dyadic primes and at B, while $v_A(a) \geq 2$. Indeed, k odd gives $v_2(u-1) = H = v_2(h)$; $k \equiv 1 \pmod 3$ gives $u \equiv \epsilon \not\equiv 1 \pmod B$; and $v_A(u-1) \geq 1$. Here c is a unit at the dyadic primes and at B, and $v_A(c) = 1$, as arranged in Proposition 4.7.

4.3 Inert prescriptions

Hypothesis (T1) requires every branch other than (A_n) and (B_n) to be eventually unramified and inert. At each step the construction therefore imposes the following condition at finitely many primes. An *inert prescription* at a prime V of F is the requirement that V be unramified and inert in L; the finite set of primes carrying an inert prescription at a step is its *inert set*. At an odd prime V, an inert prescription holds exactly when $F_V(\sqrt{\delta})$ is the unramified quadratic extension of F_V, that is, when δ is a nonsquare unit times an even power of a uniformizer (Section 2.3, quadratic Kummer theory). This is a condition on the square class of $\delta_-\delta_+$ in F_V.

The Weil bound. To meet this condition we must show that a polynomial in the residue of c takes a nonsquare value at some point of the residue field $\mathbb{F}_q$. The residue prescriptions of Section 4.4 need more. There the residue of α must have a prescribed two-primary component, and the residues with this component form one coset of the subgroup of odd order in $\mathbb{F}_q^\times$. We need many elements of this coset at which a certain discriminant is a nonzero square, so that one of them survives a few exclusions. In each case we count points on a double cover: of the projective line, $Z^2 = \Psi(X)$, for Lemma 4.2 and Lemma 4.4(1), and of a conic for Lemma 4.4(2). For a smooth projective geometrically connected curve C of genus g over $\mathbb{F}_q$, the Weil bound of Section 2.3 states

$$\bigl|\#C(\mathbb{F}_q) - (q+1)\bigr| \leq 2g\sqrt{q}.$$

We apply it to curves of the following form. Let q be odd, let χ be the quadratic character of $\mathbb{F}_q^\times$, extended by $\chi(0) = 0$ (this χ takes values in $\{0, \pm 1\}$, unlike the residue characters χ_v of Section 2.3), and let $\Psi \in \mathbb{F}_q[X]$ be squarefree of even degree $2g+2$. The affine curve $Z^2 = \Psi(X)$ has a smooth projective geometrically connected model C of genus g. Indeed, C is glued from $Z^2 = \Psi(X)$ and $Z'^2 = X'^{2g+2}\Psi(1/X')$ along $X' = 1/X$, $Z' = Z/X^{g+1}$. Both charts are smooth because Ψ and its reversal $X'^{2g+2}\Psi(1/X')$ are squarefree, and C is geometrically irreducible because Ψ is not a square in $\overline{\mathbb{F}}_q[X]$. The double cover $X\colon C \to \mathbb{P}^1$ is branched exactly at the $2g+2$ roots of Ψ, so the Riemann–Hurwitz formula gives $2g_C - 2 = -4 + (2g+2)$ for the genus g_C of C, that is, $g_C = g$. The points at infinity are $X' = 0$, $Z' = \pm\sqrt{\mathrm{lc}(\Psi)}$, where $\mathrm{lc}(\Psi)$ is the leading coefficient. Above each $x \in \mathbb{F}_q$ it has exactly $1 + \chi(\Psi(x))$ rational points, and it has two rational points at infinity when the leading coefficient of Ψ is a square in $\mathbb{F}_q^\times$. In that case

$$\#C(\mathbb{F}_q) = q + 2 + \sum_{x \in \mathbb{F}_q} \chi(\Psi(x)), \qquad \text{so} \qquad \sum_{x \in \mathbb{F}_q} \chi(\Psi(x)) \geq -1 - 2g\sqrt{q}.$$

Thus, if $\sqrt{q}$ is large compared with g, the sum of the values $\chi(\Psi(x))$ is small compared with q, and Ψ takes nonzero square values at many points of $\mathbb{F}_q$.

Lemma 4.2 (Inert prescription at an odd prime)**.** *Let $\mathcal{K}$ be a local field of odd residue characteristic and let y, h, u_0 be units of $\mathcal{K}$. Some unit $c \in \mathcal{K}$ makes $\mathcal{K}(\sqrt{\delta})$ the unramified quadratic extension of $\mathcal{K}$, where δ is the discriminant in (4) formed with $u = u_0$. This property persists in sufficiently small neighbourhoods of c, u_0.*

Proof. Since $\delta = \delta_-\delta_+/(chu_0)^2$, it suffices to choose c so that $\delta_-\delta_+$ is a nonsquare unit times an even power of a uniformizer π. We write $\mathbb{F}_q$ for the residue field and a bar for residues.

Case 1: y is a square in $\mathcal{K}$. Take c near $h\sqrt{y}$ so that $v(\delta_-) = 2$. For instance, $c = h\sqrt{y} + \pi^2 x$ with x a unit gives $\delta_- = \pi^2 x(2h\sqrt{y} + \pi^2 x)$, whose leading unit has residue $2\bar{h}\sqrt{\bar{y}}\,\bar{x}$. Since the derivative $2c$ of δ_- with respect to c is a unit, this leading unit can be chosen freely by the choice of $\bar{x}$. At $c \equiv h\sqrt{y}$ the other factor δ_+ has residue $\bar{h}^2\bar{y}(\bar{u}_0 - 1)^2 - \bar{h}^2\bar{y}(\bar{u}_0 + 1)^2 = -4\bar{h}^2\bar{y}\bar{u}_0 \neq 0$. Choosing $\bar{x}$ so that the leading unit of $\delta_-\delta_+$ is a nonsquare makes δ have nonsquare unit part and even valuation.

Case 2: $\bar{y}$ is a nonsquare in $\mathbb{F}_q$. Write $c = h\xi$. Then $\delta_-\delta_+ = h^4(\xi^2 - y)\big(\xi^2(u_0-1)^2 - y(u_0+1)^2\big)$, so after rescaling c by h the relevant square class is that of the value at $\bar{\xi}$ of

$$\Psi(X) = (X^2 - \bar{y})\big(X^2(\bar{u}_0 - 1)^2 - \bar{y}(\bar{u}_0 + 1)^2\big).$$

If $\Psi(\bar{\xi})$ is a nonsquare in $\mathbb{F}_q^\times$ for some $\bar{\xi} \in \mathbb{F}_q^\times$, then any lift ξ of $\bar{\xi}$ makes $\delta_-\delta_+$ a nonsquare unit. We show that such a $\bar{\xi}$ exists.

For $\bar{u}_0 \neq \pm 1$, Ψ is a squarefree quartic with no roots in $\mathbb{F}_q$: its roots satisfy $X^2 = \bar{y}$ or $X^2 = \bar{y}\big((\bar{u}_0+1)/(\bar{u}_0-1)\big)^2$, and $\bar{y}$ is a nonsquare; the two values of X^2 differ because $\bar{u}_0 \neq 0$. The curve $Z^2 = \Psi(X)$ has genus one. It has two rational points at infinity, since the leading coefficient $(\bar{u}_0 - 1)^2$ is a square, and two above 0, since $\Psi(0) = \bar{y}^2(\bar{u}_0 + 1)^2$ is a nonzero square. If every nonzero argument gave a square, the curve would have $2q+2$ rational points. This contradicts the genus-one bound $q + 1 + 2\sqrt{q} < 2q + 2$.

If $\bar{u}_0 = 1$, then $\Psi(X) = -4\bar{y}(X^2 - \bar{y})$. The identity $\sum_{x\in\mathbb{F}_q} \chi(x^2 - \bar{y}) = -1$ gives $\sum_{x\in\mathbb{F}_q} \chi(\Psi(x)) = -\chi(-\bar{y})$, whereas $\Psi(0) = 4\bar{y}^2$ is a nonzero square. So the sum over the $q-1$ nonzero arguments, none of which is a root, is $-\chi(-\bar{y}) - 1 \leq 0 < q - 1$.

If $\bar{u}_0 = -1$, then $\Psi(X) = 4X^2(X^2 - \bar{y})$. At nonzero arguments remove the square factor $4X^2$; the sum of $\chi(x^2 - \bar{y})$ over $x \in \mathbb{F}_q^\times$ is $-1 - \chi(-\bar{y}) \leq 0 < q - 1$.

In each case some nonzero argument has a nonsquare value. Lift it to c/h. Finally, the square class of a nonzero element of $\mathcal{K}$ is locally constant, and δ depends continuously on c and u_0; this proves persistence. □

4.4 Residue prescriptions

Hypothesis (T6) concerns the two-primary components of the residues of the elements y_n. Let V be an odd prime of F other than A and B, so that y is a unit at V, and suppose that c, h and u are units at V. The residue of α at a prime of L above V is a root of the reduction of $X^2 - tX + y$. So prescribing it requires control of the trace t, and this trace is not arbitrary. Write $\mathbb{F}_q$ for the residue field of V, and write u, y and ξ also for the residues of u, y and c/h. For fixed $u, y \in \mathbb{F}_q^\times$, the possible trace at $\xi \in \mathbb{F}_q^\times$ is

$$t_u(\xi) = -\frac{u-1}{u}\xi - \frac{y(u+1)}{u\xi}, \tag{6}$$

by the second expression for t in (4). As in Section 2.1, $x_{(2)}$ denotes the two-primary component of an element x of a finite multiplicative group.

Definition 4.3. Let V be an odd prime of F other than A and B, with residue field $k(V)$. A *split residue prescription* at V is the requirement that V split in L and that the residue of α at one of the two primes above V have a prescribed two-primary component $b \in k(V)^\times_{(2)}$. An *inert residue prescription* at V is the requirement that V be inert in L and that the residue of α in the quadratic extension of $k(V)$ have a prescribed two-primary component 1 or -1. Both are called *residue prescriptions.*

The next lemma realizes residue prescriptions over large residue fields. In part (1) the root x must lie in the coset of the odd-order subgroup of $\mathbb{F}_q^\times$ determined by b. This coset has $(q-1)/(q-1)_2$ elements, and the size condition makes it large compared with the genus of the curve that the Weil bound is applied to.

Lemma 4.4 (Residue prescriptions over large residue fields)**.** *Let $\mathbb{F}_q$ have odd characteristic, and let $u, y \in \mathbb{F}_q^\times$.*

1. *Put $(q-1)_2 = 2^{v_2(q-1)}$. If $(q-1)/(q-1)_2 > 4\sqrt{q}+12$, then for every $b \in \mathbb{F}_{q\,(2)}^\times$ some $\xi \in \mathbb{F}_q^\times$ makes $X^2 - t_u(\xi)X + y$ have two distinct roots $x, y/x \in \mathbb{F}_q^\times$, nonzero trace, and $x_{(2)} = b$.*
2. *If $q \equiv 1 \pmod 4$, $q \geq 81$, and $y_{(2)} = 1$, then, for either sign, some $\xi \in \mathbb{F}_q^\times$ makes that polynomial irreducible, with nonzero trace and root $\theta \in \mathbb{F}_{q^2}$ satisfying $\theta_{(2)} = \pm 1$.*

Proof. In both parts we first choose the root (x in part (1), θ in part (2)) with the prescribed two-primary component and then solve for ξ; the Weil bound shows that a suitable root exists.

Step 1: which traces occur. Multiplying $t = t_u(\xi)$ by $u\xi$ shows that a given trace t is attained at $\xi \in \mathbb{F}_q^\times$ exactly when

$$(u-1)\xi^2 + ut\xi + y(u+1) = 0. \tag{7}$$

For $u \neq \pm 1$, this is a quadratic equation in ξ, and its discriminant in the variable ξ is

$$\operatorname{disc}_\xi = u^2t^2 - 4(u-1)(u+1)y = u^2(t^2-4y) + 4y.$$

A nonzero square value of disc_ξ suffices: then (7) has two solutions in $\mathbb{F}_q$, and both are nonzero because their product $y(u+1)/(u-1)$ is nonzero. For $u = 1$ or -1, (6) reads $t_u(\xi) = -2y/\xi$ or $t_u(\xi) = -2\xi$, so every nonzero trace is attained directly.

Step 2: part (1). Take $x = bY^{(q-1)_2}$ with $Y \in \mathbb{F}_q^\times$ and $t = x + y/x$. Since $\mathbb{F}_q^\times$ is cyclic of order $q-1$, the map $Y \mapsto Y^{(q-1)_2}$ maps $\mathbb{F}_q^\times$ onto the subgroup of odd order, so this parametrizes exactly the elements with two-primary component b. The polynomial $X^2 - tX + y$ has the roots x and y/x; they are distinct when $x^2 \neq y$, and the trace is nonzero when $x^2 \neq -y$. Since $t^2 - 4y = (x^2-y)^2/x^2$, one has

$$x^2 \operatorname{disc}_\xi = \Psi(x), \qquad \Psi(X) = u^2(X^2-y)^2 + 4yX^2.$$

Suppose $u \neq \pm 1$. As a quadratic polynomial in X^2, Ψ has discriminant $16y^2(1-u^2) \neq 0$ and constant term $u^2y^2 \neq 0$. So Ψ is squarefree, has nonzero constant term and has four distinct nonzero roots. Since $(q-1)_2$ is prime to the characteristic, the polynomial $\Psi_b(Y) = \Psi(bY^{(q-1)_2})$ is squarefree of degree $4(q-1)_2$, with leading coefficient the square u^2b^4. Its double cover $Z^2 = \Psi_b(Y)$ has genus $2(q-1)_2 - 1$ and two rational points at infinity. The curve bound gives

$$\sum_{Y \in \mathbb{F}_q} \chi\big(\Psi_b(Y)\big) \geq -1 - (4(q-1)_2 - 2)\sqrt{q}.$$

The number of $Y \in \mathbb{F}_q$ at which $\Psi_b(Y)$ is a nonzero square is therefore at least $\big(q - 4(q-1)_2 - 1 - (4(q-1)_2 - 2)\sqrt{q}\big)/2$, because Ψ_b has at most $4(q-1)_2$ zeros. Discard $Y = 0$, and the at most $4(q-1)_2$ arguments for which $x^2 = y$ or $x^2 = -y$ (each equation has at most two solutions x, each with at most $(q-1)_2$ preimages Y). A lower bound for the remaining arguments with nonzero square value is

$$\frac{q - 3 - 12(q-1)_2 - (4(q-1)_2 - 2)\sqrt{q}}{2} > 0,$$

where positivity follows from $q-1 > (q-1)_2(4\sqrt{q}+12)$ and $q>1$. At such an argument, $\mathrm{disc}_\xi = \Psi(x)/x^2$ is a nonzero square, so Step 1 gives ξ. This proves (1). If $u=\pm 1$, the coset has more than four elements, so excluding $x^2=\pm y$ suffices.

Step 3: part (2). The odd-order subgroup of $\mathbb{F}_q^\times$ contains y and a unique b with $b^2=y$, because squaring is bijective on a group of odd order. Let $\mathcal{C}$ be the norm-one conic for $\mathbb{F}_{q^2}/\mathbb{F}_q$; its rational points are the $\zeta \in \mathbb{F}_{q^2}$ with $\zeta^{q+1}=1$, and there are $q+1$ of them. Put $\theta = \pm b\zeta^2$. Because $v_2(q+1)=1$, the element ζ^2 has odd order, so these elements have norm $b^2=y$ and two-primary component ± 1. Since $\zeta^q=\zeta^{-1}$, the conjugate of θ is $\pm b\zeta^{-2}$. Their trace is $t=\pm b(\zeta^2+\zeta^{-2})$, so $t^2-4y = y(\zeta^2-\zeta^{-2})^2$ and

$$\mathrm{disc}_\xi = u^2 y\big((\zeta^2-\zeta^{-2})^2 + 4/u^2\big).$$

The prefactor $u^2y=(ub)^2$ is a square in $\mathbb{F}_q$. Let $u\neq\pm1$. Over $\overline{\mathbb{F}}_q$, the conic $\mathcal{C}$ is a projective line with coordinate ζ. Its two points at infinity, $\zeta=0$ and $\zeta=\infty$, are defined over $\mathbb{F}_{q^2}$ and interchanged by the Frobenius of $\mathbb{F}_{q^2}/\mathbb{F}_q$, so neither is $\mathbb{F}_q$-rational. The function in parentheses on $\mathcal{C}$ has eight simple geometric zeros: their equation is $\zeta^8+(4/u^2-2)\zeta^4+1=0$, which has distinct roots because, as a quadratic in ζ^4, its discriminant is $16(1-u^2)/u^4\neq 0$ and its constant term is 1. Its two geometric poles are $\zeta=0$ and $\zeta=\infty$, of order four, and neither is $\mathbb{F}_q$-rational. The double cover of $\mathcal{C}$ defined by the square root of this function is branched exactly at the eight zeros. Multiplying the function by the square ζ^4 shows that this double cover is the smooth projective model of $Z^2=\zeta^8+(4/u^2-2)\zeta^4+1$, a curve of the type above with $2g+2=8$, so it has genus three. Above a rational point of $\mathcal{C}$ at which the function takes the value x there are exactly $1+\chi(x)$ rational points of the double cover, and no rational point lies above a pole. Since $\#\mathcal{C}(\mathbb{F}_q)=q+1$, the Weil bound shows that the number of conic points giving a nonzero square is at least

$$\frac{q+1-6\sqrt{q}-8}{2}.$$

Exclude the points with $\zeta^4=1$ or $\zeta^4=-1$. Such a point satisfies $\zeta^8=1$, and since $\zeta^{q+1}=1$ and $\gcd(8,q+1)=2$, it satisfies $\zeta^2=1$; so there are only two, namely $\zeta=\pm1$. This leaves at least $(q-11-6\sqrt{q})/2>0$ points, using $q\geq 81$. These exclusions say exactly that $\theta\notin\mathbb{F}_q$ and $t\neq 0$. At a remaining point, Step 1 gives ξ, and $X^2-t_u(\xi)X+y$ has the roots θ and its conjugate, so it is irreducible.

For $u=\pm1$, every nonzero trace is attained. The elements of $\mathbb{F}_{q^2}^\times$ of norm y and two-primary component ±1 are the $(q+1)/2$ elements $\pm b\zeta^2$, and at most one of them, namely $\pm b$, fails the two exclusions. This proves (2). □

Growth of residue fields. Every finite list of odd residue fields reaches both size conditions after sufficiently many inert quadratic extensions. Indeed, for $q_j=q_0^{2^j}$ and $j\geq 1$,

$$2^{v_2(q_j-1)} = 2^{v_2(q_0^2-1)+j-1}, \qquad \frac{q_j-1}{2^{v_2(q_j-1)}\sqrt{q_j}} \longrightarrow \infty.$$

After at least one inert step also $q\equiv 1 \pmod 4$. Once both conditions and $q\equiv 1 \pmod 4$ hold, they persist together under further inert or split steps.

Combining the local conditions. At distinct primes these prescriptions are independent finite congruences on c, once the classes of u modulo fixed powers of these primes are fixed. Indeed, by Lemma 4.2 an inert prescription holds for all c in a neighbourhood of a solution, and a residue prescription depends only on the residues of c/h, u and y at the prime; the Chinese remainder theorem combines conditions at distinct primes. In Proposition 4.7 the classes of u are fixed by $u_0=\epsilon$ and retained by the choice of k. The local extensions are then the intended unramified ones.

Under a residue prescription, the reduction of $X^2 - tX + y$ at V is separable, with two roots in the residue field for a split prescription and none for an inert one; so V is unramified in L and splits or is inert as prescribed. At a prime of the inert set, the discriminant square class of Lemma 4.2 makes V unramified and inert in L.

4.5 Virtual units and detecting primes

Hypotheses (T3) and (T4) require that a step L/F make no nontrivial ideal class of F principal and that the norms of units of L be squares of units of F. Both follow from one property: every virtual unit of F that is a norm from L is a square in F (Step 9 of the proof of Proposition 4.7). A virtual unit that is a norm from L has a square residue, after an even power of a uniformizer is removed, at every prime of odd residue characteristic that ramifies in L. It therefore suffices that L ramify at primes whose quadratic residue characters separate the square classes of virtual units. This subsection gives such primes a name.

Recall from Section 2.3 that $x \in F^\times$ is a virtual unit if $v_\mathfrak{p}(x)$ is even at every prime $\mathfrak{p}$ of F, and that the square classes of virtual units form the group $\mathcal{V}(F)$. The exact sequence there, from units modulo squares to $\mathcal{V}(F)$ to $\mathrm{Cl}(F)[2]$, shows that $\mathcal{V}(F)$ is a finite-dimensional $\mathbb{F}_2$-vector space.

Let G be a prime of F of odd residue characteristic. The residue character χ_G of Section 2.3 is defined on the elements $x \in F^\times$ with $v_G(x)$ even, is a homomorphism, and vanishes on squares. Since every virtual unit has even valuation at G, χ_G induces a homomorphism $\mathcal{V}(F) \to \mathbb{F}_2$.

Definition 4.5. Let L/F be a quadratic extension of number fields, let $y \in F^\times$ with $[y] \notin \mathcal{V}(F)$, and let $r = \dim_{\mathbb{F}_2} \mathcal{V}(F)$. Primes $G_1, \dots, G_r$ of F are *detecting primes* for (F, y) in L if:

(a) each G_i has absolute residue degree one and is an isolated ramified prime of L/F (Definition 3.1); the G_i have distinct odd rational characteristics; and y is a unit and a square modulo each G_i;

(b) there is a basis $[x_1], \dots, [x_r]$ of $\mathcal{V}(F)$ with each x_j a unit at every G_i and $\chi_{G_i}(x_j) = 1$ if $i = j$, $\chi_{G_i}(x_j) = 0$ if $i \neq j$.

Condition (b) says that the characters χ_{G_i}, restricted to $\mathcal{V}(F)$, form the dual basis of $[x_1], \dots, [x_r]$. Later arguments use only its consequence that the map $\mathcal{V}(F) \to \mathbb{F}_2^r$, $[x] \mapsto (\chi_{G_i}(x))_i$, is injective.

4.6 The auxiliary prime: ray classes and unit orders

In a scaled step the element c generates AR, where R is the auxiliary prime, and c must be totally positive and satisfy finitely many congruences modulo powers of fixed primes. These requirements become one condition on R. Suppose that $c_{\mathrm{ref}} \in \mathcal{O}_F$ is totally positive, satisfies the congruences, and has $v_A(c_{\mathrm{ref}}) = 1$, and write $(c_{\mathrm{ref}}) = AJ$. Let $\mathfrak{f}$ be a modulus whose finite part is a high power of the product of the fixed primes and which contains every real place. If a prime R lies in the class of J in the ray class group $\mathrm{Cl}_\mathfrak{f}(F)$, then $R = J \cdot (w)$ with w totally positive and $w \equiv 1$ modulo the finite part of $\mathfrak{f}$. The element $c = c_{\mathrm{ref}} w$ then generates AR, is totally positive and satisfies the same congruences. Thus we need a prime in a prescribed ray class. Class field theory identifies $\mathrm{Cl}_\mathfrak{f}(F)$ with the Galois group of the ray class field of $\mathfrak{f}$, and a prime lies in a given class exactly when its Frobenius element in that field is the corresponding Galois element (Section 2.3; [9, Chapter V, Theorems 3.5 and 3.6]). The Chebotarev density theorem [9, Chapter VIII, Theorem 7.4] then supplies infinitely many such primes.

The auxiliary prime must satisfy one further condition. The exponent k in $u = \epsilon^k$ will satisfy $k \equiv 1 \pmod{\mathcal{M}}$ for a fixed integer $\mathcal{M}$, in order to keep the congruence classes of u, and also $u \equiv -1 \pmod{R}$, which makes the trace t integral at R. Both are possible when the order of ϵ modulo R is $2s$ with s odd and prime to $\mathcal{M}$. The two-adic valuation of this order is fixed by a Hilbert symbol computation (Step 5 of the proof of Proposition 4.7). The next lemma controls the odd part. For an odd prime ℓ, the condition that the order of ϵ modulo R be prime to ℓ is a condition on the Frobenius element of R in a field obtained by adjoining roots of unity and a root of ϵ. Kummer theory [9, Chapter VII, Appendix A, Theorem A.3] computes the Galois group of this field, and the proof shows that it is independent enough of the ray class field for Chebotarev's theorem to impose both conditions at once.

Lemma 4.6. *Let F be a totally real number field, let $\epsilon \in \mathcal{O}_F^\times$ have infinite order, let $\mathcal{R}/F$ be a finite abelian extension, and let $\gamma \in \mathrm{Gal}(\mathcal{R}/F)$. Let $\mathcal{P}$ be a finite set of odd rational primes. Then infinitely many primes R of F of absolute residue degree one are unramified in $\mathcal{R}$, have Frobenius element γ in $\mathcal{R}/F$, and have the property that the order of the residue of ϵ modulo R is prime to every $\ell \in \mathcal{P}$. These primes R can be chosen outside any given finite set of primes.*

Proof. We fix the ℓ-adic valuation of $NR - 1$ by a Frobenius condition in a cyclotomic extension, require ϵ to be an ℓ^{s_ℓ}th power modulo R by a Frobenius condition in a Kummer extension, and apply Chebotarev's theorem to the compositum of these extensions with $\mathcal{R}$. Excluding a finite set of rational characteristics excludes only finitely many primes of F.

Step 1: the cyclotomic condition. Choose a prime R_0 of F of absolute residue degree one, unramified in $\mathcal{R}$, with Frobenius γ, prime to every $\ell \in \mathcal{P}$ and outside the excluded set, and put $s_\ell = v_\ell(NR_0 - 1)$. Such primes exist by Chebotarev's theorem, since primes of absolute residue degree at least two have Dirichlet density zero [9, Chapter VI, Proposition 4.5 and Corollary 4.6]. Let $F_\mu = F(\mu_{\ell^{s_\ell+1}} : \ell \in \mathcal{P})$ and let γ' be the Frobenius element of R_0 in F_μ/F. It agrees with γ on $F_\mu \cap \mathcal{R}$, since both are Frobenius elements of R_0. For a prime R of absolute residue degree one not dividing ℓ, the Frobenius element of R acts on $\mu_{\ell^{s_\ell+1}}$ by $\zeta \mapsto \zeta^{NR}$. Hence γ' fixes $\mu_{\ell^{s_\ell}}$, and every such R with Frobenius γ' in F_μ/F satisfies $NR \equiv NR_0 \pmod{\ell^{s_\ell+1}}$, so that $v_\ell(NR-1) = s_\ell$. This applies to the primes eventually selected.

Step 2: roots of ϵ. Let $\ell \in \mathcal{P}$ with $s_\ell > 0$. Since $\mathcal{O}_F^\times$ is finitely generated and ϵ has infinite order, we can write $\epsilon = \eta_\ell^{\ell^{a_\ell}}$ with $\eta_\ell \in \mathcal{O}_F^\times$ and a_ℓ maximal. Then η_ℓ is not an ℓth power in F, because an ℓth root of a unit is a unit. Fix an embedding of an algebraic closure of F into $\mathbb{C}$, so that complex conjugation acts on it. An element x of F that becomes an ℓth power in F_μ was already an ℓth power in F. Indeed, multiplying an ℓth root of x in F_μ by a root of unity in $\mu_\ell \subseteq F_\mu$ gives the real ℓth root of x in F_μ. This real root lies in the real subfield of the abelian extension F_μ/F, the fixed field of complex conjugation, which is Galois and totally real over F. Every conjugate of the root over F is a real root of the same binomial $X^\ell - x$ of odd degree, so uniqueness of the real root puts it in F. Consequently η_ℓ is not an ℓth power in F_μ. Let α_ℓ be the real root of $X^{\ell^{s_\ell}} - \eta_\ell$. By Kummer theory, $[F_\mu(\alpha_\ell) : F_\mu]$ is the order of the class of η_ℓ in $F_\mu^\times / F_\mu^{\times \ell^{s_\ell}}$ [9, Chapter VII, Appendix A, Theorem A.3]. This order is ℓ^{s_ℓ}: if $\eta_\ell^{\ell^{s_\ell-1}} = w^{\ell^{s_\ell}}$ with $w \in F_\mu$, then $\eta_\ell = \zeta w^\ell$ with $\zeta \in \mu_{\ell^{s_\ell-1}}$, and ζ is an ℓth power in $\mu_{\ell^{s_\ell}} \subseteq F_\mu$, so η_ℓ would be an ℓth power in F_μ.

Step 3: independence from $\mathcal{R}$. The extensions $F_\mu(\alpha_\ell)/F_\mu$ have coprime degrees. Their compositum F'_μ has Galois group over F_μ the product of the corresponding cyclic groups of odd order. The conjugates of α_ℓ over F are the elements $\zeta\alpha_\ell$ with $\zeta \in \mu_{\ell^{s_\ell}}$, so F'_μ/F is Galois, and complex conjugation restricts to an element $\tilde{c}$ of $\mathrm{Gal}(F'_\mu/F)$ that fixes every α_ℓ. If $g \in \mathrm{Gal}(F'_\mu/F_\mu)$ sends α_ℓ to $\zeta_g\alpha_\ell$, then $\tilde{c}g\tilde{c}^{-1}$ sends α_ℓ to $\zeta_g^{-1}\alpha_\ell$. So conjugation by $\tilde{c}$ inverts $\mathrm{Gal}(F'_\mu/F_\mu)$. Consequently

$$F'_\mu \cap F_\mu\mathcal{R} = F_\mu.$$

Indeed, the Galois group of this intersection over F_μ is a quotient of $\mathrm{Gal}(F'_\mu/F_\mu)$ on which conjugation by $\tilde{c}$ is inversion. Since $F_\mu\mathcal{R}/F$ is abelian, this conjugation is also trivial on it, and a group of odd order on which inversion is trivial is trivial.

Step 4: the Frobenius class. By the Kummer pairing, $g \mapsto (g\alpha_\ell/\alpha_\ell)_\ell$ is an isomorphism from $\mathrm{Gal}(F'_\mu/F_\mu)$ onto $\prod_\ell \mu_{\ell^{s_\ell}}$. Hence exactly one extension of γ' to F'_μ fixes every α_ℓ. By Step 3 and Step 1, this lift and γ on $\mathcal{R}$ come from a common element θ of $\mathrm{Gal}(F'_\mu\mathcal{R}/F)$. Every conjugate of θ has the same restriction to the abelian extension $F_\mu\mathcal{R}$, and it still fixes each α_ℓ: if ρ is a Galois element, then $\rho^{-1}\alpha_\ell = \zeta\alpha_\ell$ with $\zeta \in \mu_{\ell^{s_\ell}}$, and θ fixes ζ because γ' fixes $\mu_{\ell^{s_\ell}}$. Chebotarev's theorem now gives a set of positive density of primes of F, unramified in $F'_\mu\mathcal{R}$, whose Frobenius class is that of θ. Removing the primes of absolute residue degree at least two, the finitely many primes above $\mathcal{P}$ and the excluded primes leaves infinitely many primes R of absolute residue degree one.

Step 5: the order of ϵ. Let R be such a prime. Its Frobenius element in $\mathcal{R}/F$ is γ, and $v_\ell(NR-1) = s_\ell$ by Step 1. Let ℓ satisfy $s_\ell > 0$. The element α_ℓ is integral, since η_ℓ is a unit, and its Frobenius image is α_ℓ itself, so its residue $\bar{\alpha}_\ell$ at a prime above R satisfies $\bar{\alpha}_\ell^{NR} = \bar{\alpha}_\ell$. Hence $\bar{\alpha}_\ell$ lies in the residue field of R, and η_ℓ, and thus ϵ, is an ℓ^{s_ℓ}th power in that residue field. The multiplicative group of the residue field is cyclic of order $NR-1$, whose ℓ-part is ℓ^{s_ℓ}, so the order of ϵ is prime to ℓ. When $s_\ell = 0$, the order $NR-1$ of this group is prime to ℓ and the conclusion is immediate. □

4.7 One quadratic step

We now carry out a single step of the tower. Starting from a field $F = F_{n-1}$ with the data fixed before (4) (Section 4.2), Proposition 4.7 produces $L = F_n$ and the elements $y_n = \alpha$ and $\beta_n = \beta$ with the properties that (T2)–(T5) require at level n. It also respects any finite list of local prescriptions. These prescriptions are what the construction in Sections 4.9 and 4.10 uses to keep primes unramified, to split them, and to fix their residue offsets.

Proposition 4.7 (One quadratic step)**.** *Let F, A, B, y and $H \geq 5$ satisfy the assumptions stated before (4), and fix the following data.*

(a) A finite inert set of primes of F that contains both dyadic primes and contains neither A nor B.

(b) A finite set of forbidden characteristics*: rational primes that newly selected primes must avoid.*

(c) Finitely many split or inert residue prescriptions (Definition 4.3) of the kinds provided by Lemma 4.4, at odd primes of F other than A and B, with at most one prescription at each prime, and with the hypotheses of that lemma at each of them: if V carries a split residue prescription, its residue field $\mathbb{F}_q$ satisfies $(q-1)/(q-1)_2 > 4\sqrt{q}+12$ as in part (1); if V carries an inert residue prescription, then $q \equiv 1 \pmod 4$, $q \geq 81$, and the residue of y at V has trivial two-primary component, as in part (2). Primes with a split residue prescription lie outside the inert set; an inert residue prescription refines the inert prescription at that prime.

Let $\mathcal{X}$ be the finite set of rational primes consisting of 47 and 7, the rational characteristics of A and B; the rational characteristics of the primes of the inert set, among them 2; the rational characteristics of the primes carrying residue prescriptions; and the forbidden characteristics. Then there are c, R and an infinite arithmetic progression of positive integers k such that, for every such

k, (4) gives δ_-, δ_+ totally positive, $a, t \in \mathcal{O}_F$, and

$$N_{L/F}\alpha = N_{L/F}\beta = y, \qquad A\mathcal{O}_L = A'P, \quad B\mathcal{O}_L = B'Q,$$
$$(\alpha) = A'B', \qquad (\beta) = A'Q, \qquad \operatorname{ord}[A'] = 2^{n+1},$$
$$\mathrm{Cl}(F) \hookrightarrow \mathrm{Cl}(L), \qquad N_{L/F}(\mathcal{O}_L^\times) = \mathcal{O}_F^{\times 2}.$$

All prescribed inert and residue conditions hold. The choices can be made so that the residues of α at P and at Q are both nonsquares. The extension L/F has an isolated ramified prime (Definition 3.1) of absolute residue degree one whose rational characteristic is not in $\mathcal{X}$.

The characteristics 47 and 7 are placed in $\mathcal{X}$ so that the detecting primes and the auxiliary prime constructed in the proof lie over neither of them. In Step 2 of the proof, every other prime above the rational characteristic of a detecting prime is added to the inert set. Since that characteristic is outside $\mathcal{X}$, these primes are neither A nor B, which must split, and carry no earlier prescription.

Proof. We first fix the behaviour of c at finitely many primes (Steps 1–3), then realize these conditions by a generator c of AR, with the auxiliary prime R in a prescribed ray class and with a prescribed order of ϵ modulo R (Steps 4–6). Then we choose the exponent k (Step 7) and verify the conclusions (Steps 8 and 9). We use the following form of the trace from (4):

$$t = \frac{c^2(1-u) - h^2 y(1+u)}{chu} = \frac{c(1-u)}{hu} - \frac{hy(1+u)}{cu}. \tag{8}$$

Step 1: local conditions at the prescribed primes. Choose all local models with $u_0 = \epsilon$. At odd primes of the inert set without a residue prescription use Lemma 4.2. At a prime with a split residue prescription use Lemma 4.4(1), and for an inert residue prescription with component 1 or -1 use Lemma 4.4(2); their hypotheses hold by (c). For each of the two dyadic primes v, choose $\bar{c} \in k_v^\times$ with

$$\mathrm{Tr}_{k_v/\mathbb{F}_2}(\bar{y}/\bar{c}^2) = 1.$$

Then (5) gives $\bar{a} = \bar{c}$ and $\bar{t} = -\bar{c}$, so the reduction of $X^2 - tX + y$ at v is separable and irreducible, and v is unramified and inert in L. At A prescribe $v_A(c) = 1$, and at B prescribe a unit value. For $k \equiv 1 \pmod 6$ the residues of the trace at A and B are then

$$\bar{t}_A = -2\bar{h}\,\overline{y/c}, \qquad \bar{t}_B = -\frac{\bar{c}(\bar{\epsilon} - 1)}{\bar{h}\bar{\epsilon}}. \tag{9}$$

Indeed, in (8) at A the first summand has valuation at least two, because $u \equiv 1 \pmod A$, and the second is $-2h\,y/c$ modulo A. At B the second summand vanishes modulo B, and $\bar{u} = \bar{\epsilon}$ because $\bar{\epsilon}$ has order three. The leading coefficient of c at A determines $\overline{y/c}$, and the residue of c at B determines $\bar{t}_B$, since $\bar{\epsilon} \neq 1$. Choosing them independently makes these two nonzero residues nonsquares.

Step 2: detecting primes. We select detecting primes (Definition 4.5). The class $[y]$ is independent of $\mathcal{V}(F)$, because $v_A(y) = 1$ is odd. Choose representatives $x_1, \dots, x_r$ of a basis of $\mathcal{V}(F)$. Extend the dual basis of $\mathcal{V}(F)$ to $\mathcal{V}(F) \oplus \langle [y] \rangle$ by zero on $[y]$: let φ_i be the functional with $\varphi_i(x_j) = 1$ if $i = j$, $\varphi_i(x_j) = 0$ if $i \neq j$, and $\varphi_i(y) = 0$. Let $F_\mathcal{V} = F(\sqrt{x_1}, \dots, \sqrt{x_r}, \sqrt{y})$. Sending $g \in \mathrm{Gal}(F_\mathcal{V}/F)$ to the functional $[x] \mapsto g(\sqrt{x})/\sqrt{x} \in \{\pm 1\} \cong \mathbb{F}_2$ is injective, and both groups have order $|\mathcal{V}(F) \oplus \langle [y] \rangle|$ by Kummer theory [9, Chapter VII, Appendix A, Theorem A.3]. So every functional, in particular each φ_i, comes from a unique element of $\mathrm{Gal}(F_\mathcal{V}/F)$. At a prime $\mathfrak{p}$ of odd residue characteristic at which all x_j and y are units, the Frobenius element sends $\sqrt{x}$ to

$(-1)^{\chi_{\mathfrak{p}}(x)}\sqrt{x}$, by Euler's criterion. Chebotarev's theorem therefore gives, for each i, infinitely many odd primes of absolute residue degree one whose residue characters on $x_1, \dots, x_r, y$ are φ_i. Choose such primes G_i in distinct rational characteristics unramified in F and not in $\mathcal{X}$. Also avoid the characteristics of all primes prescribed earlier, including $G_1, \dots, G_{i-1}$, and the supports of $\epsilon \pm 1$ and of $x_1, \dots, x_r, y$. In particular y is a unit and a square modulo every G_i, and Definition 4.5(b) holds for the basis $[x_j]$.

At G_i prescribe

$$c \equiv h\sqrt{y}\,\frac{\epsilon+1}{\epsilon-1} \pmod{G_i},$$

where $\sqrt{y}$ is a square root of y modulo G_i. This is a root modulo G_i of $\delta_+(\epsilon) = c^2(\epsilon-1)^2 - h^2 y(\epsilon+1)^2$, viewed as a polynomial in c, and it is a unit at G_i. The derivative $2c(\epsilon-1)^2$ with respect to c is a unit, so a congruence for c modulo G_i^2 makes $v_{G_i}(\delta_+(\epsilon)) = 1$. Moreover $\delta_- \equiv 4h^2 y\epsilon/(\epsilon-1)^2 \neq 0$ modulo G_i. Once $u \equiv \epsilon \pmod{G_i^2}$ (Step 3), it follows that $v_{G_i}(\delta) = 1$, so G_i ramifies in L (Section 2.3). Add every other prime over each selected rational characteristic to the inert set, using Lemma 4.2. This makes each G_i an isolated ramified prime of L/F: its residue characteristic is unramified in F and F_{G_i} is the field of ℓ-adic numbers for that characteristic ℓ, since G_i has absolute residue degree one. There is at least one such prime, since $[-1] \neq 0$ in $\mathcal{V}(F)$, so that $r \geq 1$.

Step 3: the exponent modulus and the dyadic norms. Choose congruence conditions on c modulo fixed powers of the primes prescribed so far that retain all the local models of Steps 1 and 2. Let $\mathcal{M}$ be divisible by 6 and by the orders of ϵ in the finite unit groups of $\mathcal{O}_F$ modulo these prime powers. Then $k \equiv 1 \pmod{\mathcal{M}}$ gives $u \equiv \epsilon$ modulo each of them, and so retains all the specified congruence classes of u. In addition to the chosen dyadic residues of c, prescribe

$$N_{F_{\mathfrak{e}}/\mathbb{Q}_2} c \equiv 1 \pmod{2^{H-1}}, \qquad N_{F_{\mathfrak{d}}/\mathbb{Q}_2} c \equiv 47(1+2^{H-2}) \pmod{2^{H-1}}. \tag{10}$$

These conditions are compatible with the residues of c, because the norms of principal units in an unramified extension of $\mathbb{Q}_2$ fill $1 + 2\mathbb{Z}_2$. To see this, note that

$$N(1+2^j x) \equiv 1 + 2^j \operatorname{Tr}(x) \pmod{2^{j+1}} \qquad (j \geq 1),$$

and that each residue trace is surjective. So for every $j \geq 1$ the norms of elements of $1 + 2^j\mathcal{O}$ represent both classes of $1 + 2^j\mathbb{Z}_2$ modulo 2^{j+1}, and successive approximation gives every element of $1 + 2\mathbb{Z}_2$ as a norm.

Step 4: realizing the congruences by a ray class. Choose an integral c_{ref} satisfying all these congruences by the Chinese remainder theorem [8, Theorem 1.14]. Adding a large positive rational integer lying in all the moduli of these congruences makes it totally positive. Take a narrow ray modulus $\mathfrak{f}$, containing all real places, whose finite part is supported at the prescribed primes with sufficiently large exponents. The ideal $J = (c_{\mathrm{ref}})A^{-1}$ is prime to this support, since $v_A(c_{\mathrm{ref}}) = 1$ and c_{ref} is a unit at all other prescribed primes. As explained in Section 4.6, a prime R in the ray class of J gives a generator $c = c_{\mathrm{ref}} w$ of AR that is totally positive and satisfies all these congruences. At A the ray condition concerns the ratio c/c_{ref}; increasing the exponent of A in $\mathfrak{f}$ retains the leading coefficient of the nonunit c. Ray class fields exist by [9, Chapter V, Theorems 3.5–3.6], and Chebotarev's theorem gives such primes R (Section 2.3).

Step 5: the residue field of R and the order of ϵ. Let R and c be as in Step 4. Since c is totally positive and A has residue field $\mathbb{F}_{47}$, we have $N_{F/\mathbb{Q}} c = 47NR$. The global norm is the product of the local norms at $\mathfrak{d}$ and $\mathfrak{e}$, the only primes above 2, so (10) implies

$$q = NR \equiv 1 + 2^{H-2} \pmod{2^{H-1}}.$$

For a prime $\mathfrak{p}$ of odd residue characteristic, write $\chi_\mathfrak{p}$ for the residue character of Section 2.3, and write Hilbert symbols additively, as $[x,b]_v \in \mathbb{F}_2$ (Section 2.1). By Section 2.3, $[x,b]_\mathfrak{p} = v_\mathfrak{p}(b)\chi_\mathfrak{p}(x)$ when x is a unit at $\mathfrak{p}$, and the symbol is symmetric [9, Chapter III, Theorem 4.4(b)]. Hilbert reciprocity for (c, ϵ_0^{23}), that is, the product formula [9, Chapter V, Example 5.4], gives

$$\chi_R(\epsilon_0^{23}) = \chi_A(\epsilon_0^{23}) + [c,-1]_\mathfrak{e} = 1 \quad \text{in } \mathbb{F}_2.$$

We check the contributions. The real symbols vanish because c is totally positive. The symbol at $\mathfrak{d}$ vanishes because ϵ_0^{23} is a square in $F_\mathfrak{d}$. At $\mathfrak{e}$ the element ϵ_0^{23} has square class -1, so the symbol there is $[c,-1]_\mathfrak{e}$, and we show that it vanishes. Put $E = F_\mathfrak{e}$, an unramified extension of $\mathbb{Q}_2$; then $\sqrt{-1} \notin E$, and $E(\sqrt{-1})/E$ and $\mathbb{Q}_2(\sqrt{-1})/\mathbb{Q}_2$ are ramified quadratic extensions. By [9, Chapter III, Theorem 4.4(b),(d)], $[c,-1]_\mathfrak{e} = 0$ exactly when c is a norm from $E(\sqrt{-1})$. By [9, Chapter III, Theorem 3.4(b)], the norm groups of the two extensions have index two in $E^\times$ and in $\mathbb{Q}_2^\times$. Both contain $2 = (1+\sqrt{-1})(1-\sqrt{-1})$, a uniformizer, and in a ramified quadratic extension an element whose norm is a unit is itself a unit. So the norms of units form subgroups $\mathcal{N}_E \subset \mathcal{O}_E^\times$ and $\mathcal{N}_{\mathbb{Q}_2} \subset \mathbb{Z}_2^\times$ of index two. Transitivity of norms gives $N_{E/\mathbb{Q}_2}(\mathcal{N}_E) \subseteq \mathcal{N}_{\mathbb{Q}_2}$. Since $N_{E/\mathbb{Q}_2}\colon \mathcal{O}_E^\times \to \mathbb{Z}_2^\times$ is surjective [9, Chapter III, Proposition 1.2], the preimage of $\mathcal{N}_{\mathbb{Q}_2}$ has index two in $\mathcal{O}_E^\times$; it contains $\mathcal{N}_E$, so it equals $\mathcal{N}_E$. By Step 1, c is a unit of E. The norm $N_{E/\mathbb{Q}_2}c$ is congruent to 1 modulo 2^{H-1} by (10), hence modulo 8, so it is a square in $\mathbb{Q}_2$ [8, Theorem 7.32], and in particular a norm from $\mathbb{Q}_2(\sqrt{-1})$. Hence the unit $N_{E/\mathbb{Q}_2}c$ lies in $\mathcal{N}_{\mathbb{Q}_2}$, so $c \in \mathcal{N}_E$, and c is a norm from $E(\sqrt{-1})$. Thus $[c,-1]_\mathfrak{e} = 0$. At odd primes only A and R can contribute, since ϵ_0^{23} is a unit and $(c) = AR$; there the symbols are $\chi_A(\epsilon_0^{23})$ and $\chi_R(\epsilon_0^{23})$. Finally $\chi_A(\epsilon_0^{23}) = 1$, because ϵ_0^{23} has nonsquare residue at A_0 and A lies above A_0 with the same residue field $\mathbb{F}_{47}$. So ϵ_0^{23} is a nonsquare in the cyclic group $(\mathcal{O}_F/R)^\times$ of order $q-1$, and its order has two-adic valuation $v_2(q-1) = H-2$. Since $\epsilon = (\epsilon_0^{23})^{2^{H-3}}$,

$$v_2(\operatorname{ord}_R \epsilon_0^{23}) = H-2, \qquad v_2(\operatorname{ord}_R \epsilon) = 1.$$

Step 6: choice of R and of c. Apply Lemma 4.6 to the ray class field of $\mathfrak{f}$, with γ the image of the class of J under the Artin map, and to the odd prime divisors of $\mathcal{M}$, excluding the primes above $\mathcal{X}$ and above the characteristics of the G_i. By Step 5, infinitely many eligible primes R satisfy

$$\operatorname{ord}_R \epsilon = 2s, \qquad s \text{ odd}, \quad (s, \mathcal{M}) = 1. \tag{11}$$

Before c is fixed, we arrange total positivity of the two discriminant factors. The totally positive units congruent to 1 modulo $\mathfrak{f}$ have logarithms forming a full lattice in the hyperplane of vectors with coordinate sum zero. Indeed, they form a subgroup of finite index in the unit group, since this subgroup contains the squares of the units congruent to 1 modulo $\mathfrak{f}$, and the logarithms of all units form a full lattice there [8, Theorem 5.9]. Multiplying the generator c of AR from Step 4 by one of these units keeps $(c) = AR$, total positivity and all congruences, and makes

$$e^{-\Omega}(47NR)^{1/[F:\mathbb{Q}]} \le \tau(c) \le e^{\Omega}(47NR)^{1/[F:\mathbb{Q}]}$$

at every real embedding τ, with Ω depending only on the fixed modulus. The eligible primes have unbounded norm. Write $\epsilon_\tau = \tau(\epsilon)$; this is positive, since ϵ is a square. Take R with norm large enough that

$$\tau(c)^2 > h^2 \max(\tau(y), 0) \left(\frac{\epsilon_\tau + 1}{\epsilon_\tau - 1}\right)^2 \quad \text{for every } \tau. \tag{12}$$

No conjugate of ϵ equals 1. The absolute value of the fraction is greater than one, and it decreases when ϵ_τ is replaced by ϵ_τ^k with $k \ge 1$. Hence $\delta_- = c^2 - h^2 y$ and every $\delta_+(\epsilon^k) = c^2(\epsilon^k - 1)^2 - h^2 y(\epsilon^k +$

$1)^2$ are totally positive. This balancing preserves every congruence condition and the prime R, and it is done before c and δ_- are fixed.

Step 7: choice of k, integrality, and the field L. Now take

$$k \equiv 1 \pmod{\mathcal{M}}, \qquad k \equiv s \pmod{2s}.$$

These congruences are compatible by (11): $\gcd(\mathcal{M}, 2s) = 2$ divides $s-1$. They define an infinite arithmetic progression of positive integers. Since ϵ^s has order two in the cyclic group $(\mathcal{O}_F/R)^\times$, they give $u = \epsilon^k \equiv \epsilon^s \equiv -1 \pmod{R}$. Since $(c) = AR$, $v_A(y) = 1$ and R divides $u+1$, the element c divides $h^2 y(u+1)$. The expression (8) for t is therefore integral at all odd primes. The original expression $t = (y(1-u^2) - a^2)/(au)$ proves integrality at the dyadic primes, where a is a unit. The same holds for $a = c(u-1)/h$, by (5) at the dyadic primes. The primes G_i ramify, so δ is not a square and L/F is quadratic. By (12), $\delta = \delta_-\delta_+/(chu)^2$ is totally positive, so $L = F(\sqrt{\delta})$ is totally real.

Step 8: norms and ideals. Direct multiplication gives $N\alpha = N\beta = y$: the constant term of $X^2 - tX + y$ is y, and $N(a + u\alpha) = a^2 + aut + u^2 y = y$ by the first expression for t in (4). At A and B the polynomial $X^2 - tX + y$ reduces to $X(X - \bar{t})$ with $\bar{t} \neq 0$, so both primes split. At A, $v_A(a) \geq 2$ puts α and $\beta = a + u\alpha$ at the same prime of L above A. At B, a is a unit, so α and β vanish at the two different primes of L above B. Their norm y has valuation one at A and at B, and no other support. This proves the displayed ideal identities, with A' and B' the primes at which α vanishes. At the other primes P and Q above A and B, the residue of α is the other root $\bar{t}$, and (9) shows that these residues are nonsquares.

Step 9: units and class groups. Let $x \in F^\times$ be a virtual unit that is a norm from L, say $x = N_{L/F} w$. Its unit representative at each ramified G_i has square residue. Indeed, let $\mathfrak{G}$ be the prime of L above G_i. The extension $L_{\mathfrak{G}}/F_{G_i}$ is quadratic and ramified, so the residue fields agree and the nontrivial automorphism acts trivially on the residue field. Since $v_{G_i}(\delta) = 1$, the element $\Pi = \sqrt{\delta}$ is a uniformizer of $L_{\mathfrak{G}}$, and $\delta = \Pi^2$ is a uniformizer at G_i with $N\Pi = -\delta$. Writing $w = \Pi^m w_0$ with w_0 a unit, we get $x = (-\delta)^m N(w_0)$, so $m = v_{G_i}(x)$, which is even because x is a virtual unit, and $x\delta^{-m} = N(w_0)$ has the residue of w_0 squared as its residue. Thus all the characters χ_{G_i} vanish on $[x]$, and by Definition 4.5(b) this virtual unit is a square in F. In particular the norm of a unit of L is a unit of F that is a square in F, hence the square of a unit. So the unit norm image is precisely $\mathcal{O}_F^{\times 2}$, since units of F supply the reverse inclusion. If a fractional ideal I of F capitulates, say $I\mathcal{O}_L = (x)$, then $I^2 = (Nx)$. The assertion on virtual units writes $Nx = b^2$ with $b \in F$, and ideal factorization gives $I = (b)$. Thus the whole ideal class group of F injects into that of L. Finally the ideals $(\alpha), (\beta)$ give $[B'] = -[A']$ and $[Q] = -[A']$, hence $[B\mathcal{O}_L] = -2[A']$; since $(y) = AB$, this gives $2[A'] = [A\mathcal{O}_L]$ and $[P] = [A']$. The class $[A\mathcal{O}_L]$ has order 2^n by injectivity, and $n \geq 1$, so $\operatorname{ord}[A'] = 2^{n+1}$.

The prime G_1 satisfies the last assertion of the proposition, and Steps 1–3 give all prescribed inert and residue conditions for every k in the progression. This completes the finite step. □

4.8 Scaled steps

The later sections use only the following properties of the fields and elements produced by Proposition 4.7. We take them as the definition.

Definition 4.8 (Scaled step)**.** Let $n \geq 1$, and let $F = F_{n-1}$, $A = A_{n-1}$, $B = B_{n-1}$ and $y = y_{n-1}$ satisfy the assumptions stated before (4). Let $r = \dim_{\mathbb{F}_2} \mathcal{V}(F)$. A quadratic extension F_n/F_{n-1} with elements $y_n, \beta_n \in \mathcal{O}_{F_n}$ is a *scaled step* if there are the following data: an integer $H \geq 5$; an inert set, forbidden characteristics and residue prescriptions satisfying conditions (a)–(c) of Proposition 4.7;

an integer k; an element $c \in \mathcal{O}_F$; a prime R of F; and primes $G_1, \dots, G_r$ of F. These data must have the following properties. Form h, ϵ, u, a, t, δ_- and δ_+ as in (4). Then $F_n = L = F(\alpha)$ for a root α of $X^2 - tX + y$, and, with $\beta = a + u\alpha$ and the primes A', P, B', Q of L named in condition (v),

$$y_n = \alpha, \quad \beta_n = \beta, \quad A_n = A', \quad P_n = P, \quad B_n = B', \quad Q_n = Q.$$

Moreover the following conditions hold, with $\mathcal{X}$ the set of Proposition 4.7 for these data.

(i) The exponent k is positive and $k \equiv 1 \pmod 6$. Hence (5) holds, so $u \equiv 1 \pmod 8$ at both dyadic primes. Moreover $u \equiv 1 \pmod A$, $u - 1$ is a unit at B, and u is a square in F.

(ii) The element c is totally positive, $(c) = AR$, and c is a unit at both dyadic primes and at B. The prime R has absolute residue degree one, and its rational characteristic is neither in $\mathcal{X}$ nor the characteristic of any G_i. Moreover

$$N_{F_\mathfrak{e}/\mathbb{Q}_2} c \equiv 1 \pmod{2^{H-1}}, \qquad N_{F_\mathfrak{d}/\mathbb{Q}_2} c \equiv 47(1 + 2^{H-2}) \pmod{2^{H-1}},$$

the order of ϵ modulo R is twice an odd integer, and $u \equiv -1 \pmod R$.

(iii) One has $a, t \in \mathcal{O}_F$; a is a unit at both dyadic primes and at B, and $v_A(a) \geq 2$.

(iv) The elements δ_- and δ_+ are totally positive, and L is totally real.

(v) All displayed conclusions of Proposition 4.7 hold for L, α and β; in them A' and P are the primes of L above A, and B' and Q those above B. Every prescribed inert and residue condition holds; in particular both dyadic primes of F are unramified and inert in L. The residues of α at P and at Q are nonsquares.

(vi) The primes $G_1, \dots, G_r$ are detecting primes for (F, y) in L (Definition 4.5, which applies because $v_A(y) = 1$ is odd, so that $[y] \notin \mathcal{V}(F)$). Here $r \geq 1$, because F is totally real and so $[-1] \neq 0$ in $\mathcal{V}(F)$. Each G_i has absolute residue degree one and is ramified in L; the G_i have distinct rational characteristics, unramified in F and outside the forbidden characteristics, and every other prime of F over such a characteristic is unramified in L.

We then call F_n/F_{n-1} a scaled step *for* H and for this inert set, these forbidden characteristics and these residue prescriptions.

Conditions (v) and (vi) give (T2)–(T5) at level n. The congruences in (i)–(iii) are those used in Sections 7 and 8 and in Appendix B.

Remark 4.9. Every output of the proof of Proposition 4.7 is a scaled step for its input data. Precisely, let H, the inert set, the forbidden characteristics and the residue prescriptions be as in that proposition, let c, R, the primes $G_1, \dots, G_r$ and k be chosen as in its proof, and put $F_n = L$, $y_n = \alpha$, $\beta_n = \beta$, $A_n = A'$, $P_n = P$, $B_n = B'$ and $Q_n = Q$. Then conditions (i)–(vi) of Definition 4.8 hold. For (i), Step 7 takes k positive with $k \equiv 1 \pmod{\mathcal{M}}$, where 6 divides $\mathcal{M}$ by Step 3; the congruences for u at A and B were shown after (5); and $u = \epsilon^k$ is a square because ϵ is. For (ii), Steps 4 and 6 give a totally positive generator c of AR satisfying the congruences of Steps 1–3, among them the unit conditions at the dyadic primes and at B and (10). In Step 6, Lemma 4.6 gives R of absolute residue degree one outside the primes above $\mathcal{X}$ and above the characteristics of the G_i, and (11) gives the order of ϵ modulo R; Step 7 gives $u \equiv -1 \pmod R$. For (iii), Step 7 gives $a, t \in \mathcal{O}_F$, and the valuations of a follow, as shown after (5), from $k \equiv 1 \pmod 6$, $v_A(c) = 1$ and the unit conditions on c. For (iv), see (12) in Step 6 and the end of Step 7. For (v), Step 7

shows that L/F is quadratic, Steps 8 and 9 prove the displayed conclusions and the nonsquare residues of α at P and Q, and Steps 1–3 give every prescribed condition, including the inertness of the dyadic primes (Step 1). For (vi), Step 2 chooses r primes G_i of absolute residue degree one in distinct odd rational characteristics, unramified in F and outside $\mathcal{X}$, which contains the forbidden characteristics; y is a unit and a square modulo each G_i, and Definition 4.5(b) holds for the basis $[x_j]$. Each G_i ramifies in L once $u \equiv \epsilon \pmod{G_i^2}$, which $k \equiv 1 \pmod{\mathcal{M}}$ ensures, and the other primes over its characteristic are added to the inert set in Step 2, so they are inert in L. Thus each G_i is an isolated ramified prime of L/F, and the G_i are detecting primes for (F, y) in L.

For later use, the element $z = \alpha\beta/y$ of a scaled step has a direct description at the dyadic primes:

$$z = -u - \frac{h(u+1)}{c}\alpha, \qquad v_2(-z-1) = H, \qquad \bar{z} = 1 \quad \text{at both dyadic primes.} \tag{13}$$

The first identity follows from $\alpha^2 = t\alpha - y$ and $a + ut = y(1-u^2)/a = -hy(u+1)/c$. For the second, the summand $h(u+1)\alpha/c$ has valuation $H+1$, whereas $v_2(u-1) = H$. Also $\delta_-/c^2 \in 1 + 2^{2H}\mathcal{O}_{F_v}$ is a square at both dyadic places v, and $v_A(\delta_-) = 1$.

4.9 Separating residue offsets

Hypothesis (T6) asks that every maximal ideal $V \neq \mathfrak{m}_A, \mathfrak{m}_B$ of D of odd residue characteristic have a residue offset, and that distinct such V over the same rational prime have distinct residue offsets. This subsection describes a finite procedure, built from the residue prescriptions of Proposition 4.7, that achieves this for all maximal ideals of D over one rational prime. It ends with the explicit residue offsets over 47 and 7. For a rational prime ℓ, the *fibre* over ℓ at a level is the set of primes of that level above ℓ.

Offsets along inert branches. Suppose that a branch $(V_n)_{n \geq m}$ carries an inert prescription at every step after level m, that the residue field of V_m has size $q_m \equiv 1 \pmod 4$, and that the two-primary component of the residue of y_m at V_m has order $2^d > 1$. The identities $N_{F_n/F_{n-1}} y_n = y_{n-1}$ imply that at level $n \geq m$ this order is 2^{d+n-m}. Indeed, the residue field norm of an inert step raises to the power $q_{n-1}+1$, whose two-adic valuation is one, and an element of order 2^e with $e \geq 1$ has a $(q_{n-1}+1)$th power of order 2^{e-1}. The residue offset of the branch is therefore $d - m$. If its component is kept trivial until level $n^* - 1$, and is set to -1 at level n^*, its residue offset is $1 - n^*$.

Splitting a prime with nontrivial component. Consider a finite collection of odd primes other than A, B that are permitted to split, namely the primes of one fibre, and let $\mathcal{F}$ be the set of residue offsets already fixed in this fibre; it is initially empty and stays finite. Prescribe the primes inert until their residue fields satisfy $q \equiv 1 \pmod 4$, $q \geq 81$ and $(q-1)/(q-1)_2 > 4\sqrt{q} + 12$, the residue-field hypotheses of Lemma 4.4. After this growth phase, classify the primes of the current level by their current two-primary components. Throughout the procedure below, whenever a prime with trivial component is kept inert, prescribe its new component to be 1 using part (2) of that lemma. This applies both to the initially trivial components and to those created by splitting. Process the primes whose component is nontrivial one at a time. Split the current prime, with two-primary components 1 and the previous component at the two primes above it. Keep the component at the first of these primes trivial as prescribed above. At the other prime the order of the component is unchanged, whereas its level increases by one. So its *provisional offset*, the value $d-m$ computed at the current level, decreases by one. Repeated splitting makes that offset avoid $\mathcal{F}$ in at most $|\mathcal{F}|+1$ splits. Then prescribe this prime inert at every later step and add its offset to $\mathcal{F}$. Meanwhile the other primes with nontrivial component carry plain inert prescriptions, which leave their provisional

offsets unchanged, since the exponent of the order of the component and the level both increase by one.

Primes with trivial component. When every prime with nontrivial component has been processed, the fibre contains finitely many primes with trivial component: those whose component was trivial after the growth phase, and those created with component 1 by splitting. They need no splitting. Assign them distinct, sufficiently late levels n^*, with $1 - n^* \notin \mathcal{F}$; the resulting offsets $1 - n^*$ are pairwise distinct because the levels are. Keep each component trivial up to level $n^* - 1$, impose in the step to level n^* an inert residue prescription with component -1, and thereafter prescribe the branch inert at every step. Every residue prescription of this procedure is imposed after the growth phase, so the residue-field hypotheses of Lemma 4.4 hold at its prime, since they persist. Every inert residue prescription, with component 1 or -1, is imposed at a prime whose current component is trivial, that is, at which the residue of y has trivial two-primary component. So these prescriptions satisfy condition (c) of Proposition 4.7, which realizes every finite round of them simultaneously. This procedure terminates and gives distinct residue offsets, even for primes ramified in an earlier step, because every step at a treated prime is split or unramified inert.

The fibres over 47 *and* 7. The branches over 47 and 7 formed by the primes P_m, Q_m ($m \geq 1$) and by the conjugates of A_0, B_0 need no such splitting. At the level $m \geq 1$ at which they appear, the primes P_m, Q_m have nonsquare residues of y_m in $\mathbb{F}_{47}$ and $\mathbb{F}_7$, by Definition 4.8(v); their two-primary components have order two. At the first inert step the norm exponents have two-adic valuations $v_2(48) = 4$ and $v_2(8) = 3$, respectively. The new orders are thus 2^5 and 2^4, and subsequent inert steps double them. Their residue offsets are

$$o_{P_m} = 4 - m, \qquad o_{Q_m} = 3 - m.$$

At the conjugates of A_0, B_0 the residues of y_0 are 15 modulo 47 and 3 modulo 7, both nonsquares. Prescribing these primes inert from the start gives residue offsets 4 and 3. The residue offsets within each of these two rational fibres are therefore pairwise distinct.

4.10 The construction schedule

It remains to choose the prescriptions at every step so that all six hypotheses hold in the limit. A construction schedule makes this choice as a function of the tower built so far.

Definition 4.10. *Prescription data* for a step F_n/F_{n-1} consist of an integer $H \geq 5$, an inert set, a finite set of forbidden rational characteristics, and finitely many residue prescriptions (Definition 4.3), as in Proposition 4.7. A *construction schedule* assigns prescription data to every finite tower $F_0 \subset \cdots \subset F_{n-1}$ of scaled steps.

Theorem 4.11 (Existence of the tower)**.** *There exists an infinite tower whose every step is a scaled step in the sense of Definition 4.8, with $H_n \geq 5$ and $H_n \to \infty$, satisfying* (T1)–(T6)*. Thus hypotheses* (T1)–(T6) *can be realized simultaneously.*

Proof. We describe a construction schedule (Definition 4.10), apply Proposition 4.7 at every step, and then check (T1)–(T6) for the resulting tower.

Step 1: fixed prescriptions. Start with F_0 and the data of Section 4.1, and choose, for example, $H_n = n + 5$. Every step below is produced by Proposition 4.7 and is therefore a scaled step for the prescription data of the schedule (Remark 4.9). Prescribe inert at every step both dyadic branches, the conjugates of A_0, B_0, and every P_m, Q_m from level $m+1$ on. Keep every prime over 3 unramified from the start.

Step 2: the list of characteristics. Begin a list with $3, 5, 13$; the characteristics 5 and 13 are ramified in F_0. Subsequently add odd rational characteristics other than $47, 7$. A characteristic remains listed while it is being treated and after its treatment is complete. A listed characteristic is never allowed further ramification: all its primes carry inert prescriptions except during the split or inert operations of the procedure of Section 4.9. After each constructed step, add every newly ramified characteristic not already listed, and also the least odd characteristic other than $47, 7$ not yet listed, the characteristics added after one step being listed in increasing order. Treat the listed characteristics in order, one whole finite fibre at a time. A new entry has only finitely many entries before it, each requiring finitely many steps. The least odd characteristic other than $47, 7$ that is not listed increases strictly at every step, so every such characteristic p is listed after at most p steps. Thus every entry is eventually treated, and all branches of its fibre then carry an inert prescription at every later step. At any step only finitely many fibres are listed, so Proposition 4.7 applies. Selected ramified primes avoid all listed characteristics: the forbidden characteristics passed to Proposition 4.7 at each step are the listed ones. The residue prescriptions passed to it are those of the procedure of Section 4.9, imposed only after the growth phase of their fibre; as shown there, they satisfy its condition (c). In each step, the proof of Proposition 4.7 adds all other primes of the characteristic of a selected detecting prime to the inert set, so they are unramified in that step, and that characteristic is listed after the step. A listed fibre whose treatment has not begun carries only inert prescriptions. During its treatment, classify the primes by their current two-primary components after the initial residue-field growth phase. The treatment of a fibre is complete only after the last level n^* assigned in it; from then on all its branches carry plain inert prescriptions. All remaining choices, namely the integers H_n, the order in which the primes of a fibre are treated, and the late levels n^*, are made by a fixed rule depending only on the tower built so far: for example $H_n = n + 5$, a fixed enumeration of the primes of each fibre, and the least admissible levels n^*. So the prescription data at each step are a function of the tower built so far, as Definition 4.10 requires. They are fixed before the next step is constructed, and every scaled step for these data satisfies them (Definition 4.8(v)); in particular every output of Proposition 4.7 does (Remark 4.9). Hence the procedure of Section 4.9 has the prescribed effect on every fibre whichever output is chosen at each step. The classification of a fibre after its growth phase uses the components in the tower actually constructed, and the resulting treatment is then carried out to completion.

Step 3: verification. New ramification occurs only at previously unramified characteristics. It has relative index two, and no later step ramifies there. The absolute ramification indices at odd primes are therefore at most two. The primes $3, 7, 47$ stay absolutely unramified, and the completions at A_n and B_n remain the prescribed rational local fields. This proves (T1). Conditions (v) and (vi) of Definition 4.8 give (T2)–(T4) by induction, including the order $\operatorname{ord}[A_n] = 2^{n+1}$ needed at the next step; for (T3), $[A_0]$ has order two by Lemma 4.1(b). Conditions (v) and (vi) also give (T5) at each step. Conditions (iv) and (v) also give the assumptions stated before (4) at level n, so the next step is defined. For (T6), let $\ell \neq 47, 7$ be an odd characteristic, and let n_ℓ be a level at which the treatment of its fibre is complete, that is, a level at least every level n^* assigned in the fibre and after which only plain inert prescriptions are imposed there. The fibre over ℓ at level n_ℓ consists of all primes of F_{n_ℓ} above ℓ. By Section 4.9, each of them has been assigned a residue offset, computed as in the paragraph on offsets along inert branches, and these offsets are pairwise distinct. From level n_ℓ on, every prime above ℓ carries an inert prescription at every step, so every prime of the fibre at level n_ℓ has exactly one prime above it at each later level. Hence every maximal ideal V of D above ℓ is determined by its contraction V_{n_ℓ}, and V has the residue offset assigned to V_{n_ℓ}. Distinct such V have distinct contractions to F_{n_ℓ}, so their offsets are distinct. This proves (T6) over ℓ; the explicit residue offsets of the primes P_m, Q_m and of the conjugates of A_0, B_0 prove it over 47 and 7. □

The construction permits primes other than the prescribed ones to ramify in F_n/F_{n-1} at every step, and places no bound on their number. Condition (b) of Definition 4.5 does not give the Selmer condition (S), in which all ramified primes enter, and the local prescriptions above do not give the memberships of unit images in $\mathbb{Z}_p$-modules required by (O). Sections 7 and 8 study these two conditions, which are the remaining hypotheses of Theorem B.

5 Subrings: fraction fields, residue fields and completed orders

Throughout this section the hypotheses (T1)–(T6) of Section 3 hold, and S denotes a subring of D. Section 6 proves that S is atomic when the tower satisfies an approximation condition. This section supplies the structural information about S that the proof needs.

The argument of Section 6 runs as follows. Let $x \in S$ have unbounded factorization lengths. By Proposition 3.6, x has positive valuation at both $\mathfrak{m}_A$ and $\mathfrak{m}_B$, and in D it splits into two one-sided factors (Lemma 6.3; compare Step 1 of the proof of Theorem 3.7). These factors are determined by their divisors only up to units of D, and in general they do not lie in S. To move them into S we must know which elements of D belong to S. Every element of D lies in some field F_N, and, when S is not a BFD, $S \cap F_N$ is an order of F_N (Section 5.2). An order is determined by its completions at the rational primes, and it differs from the maximal order at only finitely many of them. Membership in S is therefore a finite list of local conditions. This section shows that, when S is not a BFD, the conditions at the odd primes are under control. There are none at the odd primes that are unramified in the tower. At each odd prime that ramifies, there is one condition at each maximal ideal above it of ramification index two, measured by a single integer.

In factorization terms the conclusion is that a subring which is not a BFD is forced to be large locally. Precisely, let $S \subseteq D$ be a subring that is not a BFD. Then:

(i) the fraction field of S is K (Proposition 5.2);

(ii) S maps onto the residue field of every maximal ideal of D of odd residue characteristic, and distinct maximal ideals of D over the same odd prime have distinct contractions to S (Proposition 5.9);

(iii) at an odd prime p that is unramified in every F_n, the completion of $S \cap F_n$ at p is the maximal order (Proposition 5.11);

(iv) at an odd prime p that ramifies somewhere in the tower, for $n \geq b_p$ with b_p as in Section 5.6, the completion of $S \cap F_n$ at p is determined by one integer at each maximal ideal of D above p of ramification index two, its conductor exponent, and this integer does not depend on $n \geq b_p$ (Proposition 5.12).

Statement (iv) is what Section 6 needs. There the two factors of x are moved into $S \cap F_N$ by one unit of D whose images at the dyadic places and at the odd places of ramification index two lie in a suitable open subgroup. At an odd prime p this subgroup is read off at one level $n \geq b_p$, where the completed order contains the coefficient rings, and is then transported to the level N.

The proof of (ii) uses the Picard group and the residue offsets of (T6). A long factorization of x in S contains factors supported only on late split-off primes P_j and Q_j (Lemma 5.5). Because their divisors are principal, the classes in (3) force their exponents to balance (Lemma 5.6). Such a factor can then be written in terms of the elements y_n and z_n of (T2), and its residue at a maximal ideal V has two-primary order 2^{o_V+r} for an integer r that does not depend on V and becomes large as the support moves to later primes (Lemma 5.8). Residues of unbounded two-power order

force S to map onto the residue field, and the distinct offsets o_V of (T6) force S to separate the maximal ideals. Statements (iii) and (iv) then follow from (ii) by lifting residues to Teichmüller representatives.

5.1 Subfields of K and subrings with a smaller fraction field

We first dispose of subrings whose fraction field is smaller than K. Hypothesis (T5) gives each step an isolated ramified prime (Definition 3.1). Its role is to control the subfields of the tower. The point is that an automorphism fixing a subfield F' of a step L/F with $F \not\subseteq F'$ moves the isolated ramified prime G to an unramified prime. The proof makes this precise with decomposition groups. For a Galois extension of number fields and a prime of the larger field, the decomposition group of the prime is the Galois group of the corresponding extension of completions [8, Propositions 8.10 and 8.13(a)].

Lemma 5.1. *Let L/F be a quadratic extension of number fields with an isolated ramified prime G over a rational prime ℓ (Definition 3.1). For every subfield $F' \subseteq L$, either $F' \subseteq F$ or $F \subseteq F'$.*

Proof. We show that if $F \not\subseteq F'$, then the prime of F' below the prime of L over G is unramified over ℓ, and that this is incompatible with $F' \not\subseteq F$.

Take a Galois closure L^{gal} of L over $\mathbb{Q}$ and a prime $\mathfrak{G}$ of L^{gal} over G. Set $\Gamma_F = \mathrm{Gal}(L^{\mathrm{gal}}/F)$ and $\Gamma_{F'} = \mathrm{Gal}(L^{\mathrm{gal}}/F')$, and let $\Gamma_{\mathfrak{G}} \subseteq \mathrm{Gal}(L^{\mathrm{gal}}/\mathbb{Q})$ be the decomposition group of $\mathfrak{G}$. The group $\Gamma_{\mathfrak{G}}$ is the Galois group of the completion of L^{gal} at $\mathfrak{G}$ over $\mathbb{Q}_\ell$, and the completion of F at G is the closure of F in that completion [8, Propositions 8.10 and 8.13(a)]. Since $F_G = \mathbb{Q}_\ell$, every element of $\Gamma_{\mathfrak{G}}$ fixes F, that is, $\Gamma_{\mathfrak{G}} \subseteq \Gamma_F$.

Suppose $F \not\subseteq F'$, and choose $\gamma \in \Gamma_{F'} \setminus \Gamma_F$. The prime $\gamma\mathfrak{G}$ lies over a prime of F different from G. Otherwise, since Γ_F acts transitively on the primes of L^{gal} over G [8, proof of Theorem 3.34], we would have $\gamma\mathfrak{G} = \tau\mathfrak{G}$ with $\tau \in \Gamma_F$, hence $\gamma \in \Gamma_F\Gamma_{\mathfrak{G}} = \Gamma_F$. Since γ fixes F', the primes $\gamma\mathfrak{G}$ and $\mathfrak{G}$ lie over the same prime w of F'. The prime of F below $\gamma\mathfrak{G}$ is a prime over ℓ other than G, so it is unramified in L because G is isolated. As ℓ is unramified in F, the prime of L below $\gamma\mathfrak{G}$ is unramified over ℓ. The prime w lies below it, so $e(w/\ell) = 1$.

Now suppose also that $F' \not\subseteq F$. Since L/F is quadratic, $F'F = L$. Inside the completion of L^{gal} at $\mathfrak{G}$, the completion of L at the prime below $\mathfrak{G}$ is therefore the compositum $F'_wF_G = F'_w$, which is unramified over $\mathbb{Q}_\ell$. But G ramifies in L and $F_G = \mathbb{Q}_\ell$, so this completion is totally ramified of degree two over $\mathbb{Q}_\ell$. This contradiction shows that $F' \subseteq F$. □

Proposition 5.2. *Every proper subfield of K is a number field. If $S \subseteq D$ and $\operatorname{Frac} S \neq K$, then S is a BFD.*

Proof. Step 1: the subfields of K. Let $F' \subseteq K$ be a number field. It is finitely generated, so it lies in some F_n; choose the least such n. If $n > 0$, Lemma 5.1 applies to F_n/F_{n-1} by (T5), and minimality of n excludes $F' \subseteq F_{n-1}$. Hence $F_{n-1} \subseteq F' \subseteq F_n$, and $F' = F_n$. Thus every subfield of K of finite degree is either contained in F_0 or equal to one of the F_n. A subfield of infinite degree is the union of its subfields of finite degree, and these have unbounded degree. So it contains F_n for infinitely many n, hence for all n, and it equals K.

Step 2: S is an order. Let $K' = \operatorname{Frac} S \neq K$. By Step 1, K' is a number field. Every element of S is an algebraic integer, so $S \subseteq \mathcal{O}_{K'}$. Let $b \in S \setminus \{0\}$, with monic minimal polynomial $X^d + c_{d-1}X^{d-1} + \cdots + c_0 \in \mathbb{Z}[X]$. Then $c_0 \neq 0$ and

$$b^{-1} = -c_0^{-1}\bigl(b^{d-1} + c_{d-1}b^{d-2} + \cdots + c_1\bigr).$$

So b^{-1} is a polynomial in b with integer coefficients divided by the nonzero integer c_0. Therefore $S \otimes_{\mathbb{Z}} \mathbb{Q} = K'$. As a subgroup of the finitely generated free abelian group $\mathcal{O}_{K'}$, the ring S is free of rank $[K' : \mathbb{Q}]$, so it is a full lattice in K' and hence an order in K'.

Step 3: bounded factorization lengths. Let $b \in S$ be a unit of $\mathcal{O}_{K'}$. The constant term c_0 of its minimal polynomial is $\pm N_{\mathbb{Q}(b)/\mathbb{Q}}(b)$, and the norm of a unit is ± 1. So the display above gives $b^{-1} \in \mathbb{Z}[b] \subseteq S$, and b is a unit of S. Consequently every nonunit of S is a nonunit of $\mathcal{O}_{K'}$, and its norm is an integer of absolute value at least two. If $x \neq 0$ is a product $x = x_1 \cdots x_r$ of nonunits of S, then $2^r \leq |N_{K'/\mathbb{Q}}(x)|$. So the lengths r are bounded, and S is a BFD (Section 2.2). □

5.2 Orders and completed orders

From now on we assume that $S \subseteq D$ is not a BFD. By Proposition 5.2, $\operatorname{Frac} S = K$. Write

$$S_n = S \cap F_n.$$

The argument of Step 2 of the proof of Proposition 5.2 shows that $S \otimes_{\mathbb{Z}} \mathbb{Q} = \operatorname{Frac} S = K$. Hence each element of an integral basis of F_n has a nonzero integer multiple in S, and one nonzero integer multiplies the whole basis into S. So S_n has full rank in $\mathcal{O}_{F_n}$: it is an order in F_n.

An order of a number field is determined by local data at the rational primes. For each rational prime p put

$$\mathcal{A}_{n,p} = S_n \otimes_{\mathbb{Z}} \mathbb{Z}_p, \qquad \mathcal{B}_{n,p} = \mathcal{O}_{F_n} \otimes_{\mathbb{Z}} \mathbb{Z}_p.$$

These are an order and the maximal order, respectively, in the $\mathbb{Q}_p$-algebra $F_n \otimes_{\mathbb{Q}} \mathbb{Q}_p$, which is the product of the completions $(F_n)_v$ over the places v of F_n above p. We call $\mathcal{A}_{n,p}$, the p-adic completion of the order S_n, the *completed order* of S_n at p. The next lemma records the two facts about completed orders that we use: they are compatible with the inclusions $F_n \subseteq F_m$, and they determine S_n.

Lemma 5.3 (Intersection at a fixed level)**.** *Let $S \subseteq D$ be a subring that is not a BFD, and let p be a rational prime. For $m \geq n$,*

$$\mathcal{A}_{m,p} \cap (F_n \otimes_{\mathbb{Q}} \mathbb{Q}_p) = \mathcal{A}_{n,p}. \tag{14}$$

Furthermore

$$S_n = F_n \cap \bigcap_p \mathcal{A}_{n,p}. \tag{15}$$

Proof. The subgroup $S_n = S_m \cap F_n$ is saturated in the free abelian group S_m: if a nonzero integer multiple of an element of S_m belongs to F_n, so does the element. Hence a basis of S_n extends to a basis of S_m. Tensoring with $\mathbb{Z}_p$ gives (14). In coordinates with respect to a basis of S_n, which is also a $\mathbb{Q}$-basis of F_n, the equality (15) becomes $\mathbb{Q} \cap \bigcap_p \mathbb{Z}_p = \mathbb{Z}$. □

Equation (15) reduces membership in S_n to membership in the completed orders. Since S_n has finite index in $\mathcal{O}_{F_n}$, we have $\mathcal{A}_{n,p} = \mathcal{B}_{n,p}$ for all but finitely many p. The rest of this section determines $\mathcal{A}_{n,p}$ at the odd primes p.

5.3 Elements supported on late split-off primes

A subring that is not a BFD contains elements with unbounded factorization lengths. Since one-sided elements are bounded (Proposition 3.6), such an element has positive valuation at both $\mathfrak{m}_A$ and $\mathfrak{m}_B$. We now extract from its long factorizations elements of a very special shape: their divisors involve only the primes P_j and Q_j split off from the two chains at late levels j, with bounded exponents. In this subsection $(f) = fD$ denotes the principal ideal of D generated by f.

Definition 5.4. Let $\kappa \geq 1$ and $j_0 \geq 1$ be integers. A nonzero element $f \in D$ is a *late-supported element at level j_0 with exponent bound κ* if there is an integer $j_1 \geq j_0$ such that

$$(f) = \prod_{j=j_0}^{j_1} P_j^{a_j} Q_j^{b_j}, \qquad 0 \leq a_j, b_j \leq \kappa, \tag{16}$$

and at least one exponent a_j or b_j is positive.

In our applications the exponent bound κ is fixed while the level j_0 increases.

Lemma 5.5. *Let $S \subseteq D$ be a subring that is not a BFD. There is an integer $\kappa \geq 1$ such that, for every sufficiently large j_0, the ring S contains a late-supported element at level j_0 with exponent bound κ. More precisely, if $x \in S$ has unbounded factorization lengths in S, then for every $j_0 \geq 1$ such that (2) holds for x and all $j \geq j_0$, such an element can be chosen as a factor of x in S whose exponents satisfy $a_j \leq v_{\mathfrak{m}_A}(x)$ and $b_j \leq v_{\mathfrak{m}_B}(x)$; in particular $\kappa = \max\{v_{\mathfrak{m}_A}(x), v_{\mathfrak{m}_B}(x)\}$ works.*

Proof. A long factorization of x has more factors than x has prime factors away from the late split-off primes, and the surplus factors are the required elements.

Since S is not a BFD, some $x \in S$ has unbounded factorization lengths in S (Section 2.2), so the first assertion follows from the second. Let $x \in S$ have unbounded factorization lengths. By Proposition 3.6, x is not one-sided, so $v_{\mathfrak{m}_A}(x)$ and $v_{\mathfrak{m}_B}(x)$ are positive; in particular $\kappa = \max\{v_{\mathfrak{m}_A}(x), v_{\mathfrak{m}_B}(x)\} \geq 1$. Let $j_0 \geq 1$ be such that (2) holds for x and all $j \geq j_0$; every sufficiently large j_0 qualifies. By Lemma 3.2, the set $\mathcal{H}$ of maximal ideals of D other than $\mathfrak{m}_A$, $\mathfrak{m}_B$ and the P_j, Q_j with $j \geq j_0$ at which x has positive valuation is finite. It includes those P_j, Q_j with $j < j_0$ at which x has positive valuation.

Consider a factorization $x = x_1 \cdots x_r$ in S. Each x_i is a nonunit of D by Lemma 3.4, and $v_V(x) = \sum_i v_V(x_i)$ at every maximal ideal V. Hence at most $\sum_{V \in \mathcal{H} \cup \{\mathfrak{m}_A, \mathfrak{m}_B\}} v_V(x)$ of the factors have positive valuation at a member of $\mathcal{H} \cup \{\mathfrak{m}_A, \mathfrak{m}_B\}$. Choose a factorization longer than this sum. Then some factor $f = x_i$ has valuation zero at every member of $\mathcal{H} \cup \{\mathfrak{m}_A, \mathfrak{m}_B\}$. Since f divides x in D, its support is contained in the set of primes P_j, Q_j with $j \geq j_0$. This support is finite: its valuations at $\mathfrak{m}_A$ and $\mathfrak{m}_B$ are zero, so by (2) for f it has valuation zero at P_j and Q_j for all large j. It is nonempty because f is not a unit of D, and an element of D whose valuations are all zero is a unit (Lemma 3.2). Finally, since f divides x in D, (2) for x gives $a_j = v_{P_j}(f) \leq v_{P_j}(x) = v_{\mathfrak{m}_A}(x)$ and $b_j = v_{Q_j}(f) \leq v_{Q_j}(x) = v_{\mathfrak{m}_B}(x)$ for $j \geq j_0$. □

The divisor of a late-supported element is principal, so its class in $\mathrm{Pic}(D)$ vanishes. Lemma 3.3 computes this class, and for late support the vanishing forces the exponents a_j and b_j to balance exactly.

Lemma 5.6. *Let f be a late-supported element at level j_0 with exponent bound κ, with divisor (16), and suppose that $\kappa 2^{-j_0} < 1$. Then the following equality holds in $\mathbb{Q}$, and not only modulo $\mathbb{Z}$:*

$$\Sigma := \sum_{j=j_0}^{j_1} a_j 2^{-j} = \sum_{j=j_0}^{j_1} b_j 2^{-j} > 0. \tag{17}$$

Moreover $-v_2(\Sigma) \geq j_0 - 1 - \log_2 \kappa$.

Proof. By (3), the class of (f) in $\mathrm{Pic}(D)$ is $\iota(d + \mathbb{Z})$ with $d = \sum_{j=j_0}^{j_1} (a_j - b_j) 2^{-j-1}$. This class is zero and ι is injective, so d is an integer. But $|d| \leq \kappa \sum_{j=j_0}^{j_1} 2^{-j-1} < \kappa 2^{-j_0} < 1$, so $d = 0$. Since d is

half the difference of the two sums in (17), they are equal. They are positive because all exponents are nonnegative and one of them is positive.

For the last assertion, write $\Sigma = 2^{v_2(\Sigma)}w$ with w a positive rational number whose numerator and denominator are odd. Since the denominator of Σ is a power of two, w is an odd positive integer, and $2^{v_2(\Sigma)} \le \Sigma$. On the other hand $\Sigma \le \kappa \sum_{j \ge j_0} 2^{-j} = \kappa 2^{1-j_0}$. Hence $v_2(\Sigma) \le 1 - j_0 + \log_2 \kappa$. □

In particular, for a fixed exponent bound κ, $-v_2(\Sigma)$ tends to infinity with j_0. This is the quantity that will make residues of late-supported elements large.

5.4 Residues at odd maximal ideals

We now compute the residues of late-supported elements at a maximal ideal of odd residue characteristic. Such an element can be written, up to a unit of D, as a product of the elements y_{j-1}/y_j and z_{j-1}/z_j of (T2). Hypothesis (T6) prescribes the residues of the y_n. The residues of the units of D are not known, but Lemma 5.7 shows that their two-primary components form a finite group, and, for large n, the two-primary components of the residues of the z_n have order at most two. The result is a formula for the two-primary order of the residue of a late-supported element. This argument is what gives both the surjectivity onto residue fields and the separation of maximal ideals in Proposition 5.9; neither is assumed about S.

Setting. Throughout this subsection, V is a maximal ideal of D of odd residue characteristic p, different from $\mathfrak{m}_A$ and $\mathfrak{m}_B$. By (T1), there is a level from which the prime V_n is unramified and inert in F_{n+1} for every n. Every residue degree in the tower is a power of two, because F_0 has degree two over $\mathbb{Q}$ and every step has degree two, and these degrees are unbounded along (V_n) because V_n is inert in every late step. Hence the residue field $k(V) = D/V$ is the union of the finite fields $k(V_n)$ and equals $\bigcup_r \mathbb{F}_{p^{2^r}}$. Let $k(V)^\times_{(2)}$ be the two-primary subgroup of $k(V)^\times$. It is the increasing union of the cyclic groups $k(V_n)^\times_{(2)}$, whose orders are unbounded by (T6). So $k(V)^\times_{(2)}$ is a quasicyclic two-group: each finite subgroup is cyclic, and for each $a \ge 0$ there is exactly one subgroup of order 2^a, consisting of the elements of order at most 2^a.

Let $\overline{E}_V \subseteq k(V)^\times_{(2)}$ be the image of $D^\times$ under reduction at V followed by projection to the two-primary component. It contains -1, the residue of the unit -1, which is the unique element of order two of $k(V)^\times_{(2)}$. For an element g of K that is a unit at V, write $\langle g \rangle_V$ for the class of the two-primary component of its residue in the quotient $k(V)^\times_{(2)}/\overline{E}_V$. We write this quotient additively, so that $\langle gg' \rangle_V = \langle g \rangle_V + \langle g' \rangle_V$.

Two standard facts about one inert step are used repeatedly. Suppose that V_n is unramified and inert in F_{n+1}, and let $q = |k(V_n)|$. The local extension at V is unramified of degree two, its Galois group is generated by σ_{n+1}, and this generator acts on the residue field $k(V_{n+1}) = \mathbb{F}_{q^2}$ as the Frobenius $x \mapsto x^q$ [8, Propositions 7.50(b) and 8.10]. Hence, for $g \in F_{n+1}$ a unit at V, the residue of $N_{F_{n+1}/F_n}(g) = g\,\sigma_{n+1}(g)$ is $\bar{g}^{\,q+1}$, where $\bar{g}$ is the residue of g. If moreover $q \equiv 1 \pmod 4$, then $q+1$ has two-adic valuation one.

Lemma 5.7. *Let V be a maximal ideal of D of odd residue characteristic, different from $\mathfrak{m}_A$ and $\mathfrak{m}_B$. The group $\overline{E}_V$ is finite. Write $|\overline{E}_V| = 2^{a_V}$, and let $\langle y_n \rangle_V$ be the class of the two-primary component of the residue of y_n in $k(V)^\times_{(2)}/\overline{E}_V$, defined when y_n is a unit at V. Then, for n sufficiently large,*

$$2\langle y_{n+1} \rangle_V = \langle y_n \rangle_V, \qquad \operatorname{ord}(\langle y_n \rangle_V) = 2^{n+o_V-a_V}.$$

Proof. Both assertions come from comparing residues across one inert step, where the relative norm becomes the map $x \mapsto x^{q+1}$. In this proof a bar denotes the two-primary component of a residue at V.

Step 1: the starting level. By (T1), the prime V_n is unramified and inert in F_{n+1} for every n from some level on. One inert step later the residue cardinality is a square of an odd number, hence 1 modulo 4, and it remains so. We fix a level n_0 from which both properties hold, and consider $n \geq n_0$ with $q = |k(V_n)|$.

Step 2: $\overline{E}_V$ is finite. Let $\eta \in E_{n+1}$. Hypothesis (T4) gives $N_{F_{n+1}/F_n}\eta = \xi^2$ with $\xi \in E_n$. On two-primary residues this reads $\bar{\eta}^{q+1} = \bar{\xi}^2$. Since $q+1$ has two-adic valuation one, $(q+1)/2$ is odd, and $(\bar{\eta}^{(q+1)/2})^2 = \bar{\xi}^2$. In the cyclic group $k(V_{n+1})^\times_{(2)}$ two elements with the same square differ by an element of order at most two, so $\bar{\eta}^{(q+1)/2} = \pm\bar{\xi}$. As $\bar{\eta}$ has two-power order and $(q+1)/2$ is odd, $\bar{\eta}$ is a power of $\bar{\eta}^{(q+1)/2}$. So $\bar{\eta}$ lies in the subgroup generated by the residues of the units of E_n and -1, which is the image of E_n because $-1 \in E_n$. Thus the image of E_{n+1} is contained in the image of E_n for every $n \geq n_0$. Since $D^\times = \bigcup_n E_n$, the group $\overline{E}_V$ equals the image of E_{n_0}, a subgroup of the finite group $k(V_{n_0})^\times$.

Step 3: the relation $2\langle y_{n+1}\rangle_V = \langle y_n\rangle_V$. Take $n \geq n_0$ so large that y_n and y_{n+1} are units at V. The norm identity $N_{F_{n+1}/F_n} y_{n+1} = y_n$ of (T2) gives $\bar{y}_{n+1}^{q+1} = \bar{y}_n$. The element $\bar{y}_{n+1}^{q-1}$ satisfies $(\bar{y}_{n+1}^{q-1})^{q+1} = \bar{y}_{n+1}^{q^2-1} = 1$, so its order divides $q+1$. Being a two-primary element, it has order at most two, so it lies in $\{\pm 1\} \subseteq \overline{E}_V$. Hence $\bar{y}_{n+1}^{q+1} = \bar{y}_{n+1}^2\,\bar{y}_{n+1}^{q-1}$ has the same class as $\bar{y}_{n+1}^2$, and the norm identity becomes $2\langle y_{n+1}\rangle_V = \langle y_n\rangle_V$.

Step 4: the orders. By (T6), $\bar{y}_n$ has order 2^{n+o_V} for large n. The subgroup $\overline{E}_V$ is the unique subgroup of order 2^{a_V} of the quasicyclic group $k(V)^\times_{(2)}$, that is, the set of elements of order at most 2^{a_V}. For an element g of order 2^e, the multiple $2^k g$ lies in this subgroup exactly when $e - k \leq a_V$. So the class of g has order $2^{\max\{e-a_V,0\}}$. With $e = n + o_V$, which is at least a_V for large n, this gives $\operatorname{ord}(\langle y_n\rangle_V) = 2^{n+o_V-a_V}$. □

The next lemma computes the residue of a late-supported element. The level j_0 at which its support begins must be large compared with several quantities attached to V and κ; we explain the threshold after the statement.

Lemma 5.8. *Fix an integer $\kappa \geq 1$ and a finite set of maximal ideals V of D of odd residue characteristic, all different from $\mathfrak{m}_A$ and $\mathfrak{m}_B$. For each such V, let n_V be a level such that, for every $n \geq n_V$, the prime V_n is unramified and inert in F_{n+1} with residue cardinality congruent to 1 modulo 4, and the conclusions of Lemma 5.7 hold for n. Let $j_V = j$ if V is P_j or Q_j, and $j_V = 0$ otherwise. Put*

$$j_*(V,\kappa) = \max\{n_V + 1,\ j_V + 1,\ a_V - o_V + 2 + \lceil \log_2 \kappa \rceil,\ \lfloor \log_2 \kappa \rfloor + 1\},$$

which depends only on V and κ. Let f be a late-supported element at level j_0 with exponent bound κ, with divisor (16), and suppose $j_0 \geq j_*(V,\kappa)$ for all these V. Then (17) holds, f is a unit at each of these V, and the two-primary component of the residue of f at V has order

$$2^{o_V - v_2(\Sigma)}. \tag{18}$$

In particular this order does not change when f is multiplied by a unit of D.

The four terms in $j_*(V,\kappa)$ have the following roles. The term $n_V + 1$ ensures that every step $F_{j-1} \subset F_j$ with $j \geq j_0$ is one to which Lemma 5.7 applies. The term $j_V + 1$ ensures that V is not one of the primes P_j, Q_j in the support of f, so that f and the auxiliary elements below are units at V. The term $\lfloor \log_2 \kappa \rfloor + 1$ gives $\kappa < 2^{j_0}$, so that Lemma 5.6 applies. The term $a_V - o_V + 2 + \lceil \log_2 \kappa \rceil$ makes the class of f modulo $\overline{E}_V$ nonzero. Residues of units of D are known only modulo $\overline{E}_V$, and a nonzero class modulo $\overline{E}_V$ determines the order of the residue itself.

Proof. We write f as a unit times a product of the elements y_{j-1}/y_j and z_{j-1}/z_j, compute the class of each factor in $k(V)^\times_{(2)}/\overline{E}_V$, and then lift the result from the quotient to $k(V)^\times_{(2)}$.

Step 1: a formula for f. Since $j_0 \geq \lfloor \log_2 \kappa \rfloor + 1 > \log_2 \kappa$, we have $\kappa 2^{-j_0} < 1$, and Lemma 5.6 gives (17). By (T1) and (T2), the elements y_{j-1}/y_j and z_j^{-1} have ideals P_jQ_j and $P_jA_j^{-1}$, and for $j > j_0$ the element z_{j-1}/z_j has ideal $P_j^2P_{j-1}^{-1}$. For $j_0 < j \leq j_1 + 1$ put

$$d_j = -\sum_{i=j_0}^{j-1} (a_i - b_i) 2^{j-1-i},$$

and put $d_{j_0} = 0$. Then $d_{j+1} = 2d_j + b_j - a_j$ for $j_0 \leq j \leq j_1$, and $d_{j_1+1} = -2^{j_1} \sum_i (a_i - b_i) 2^{-i} = 0$ by (17). The element

$$\prod_{j=j_0}^{j_1} \Big(\frac{y_{j-1}}{y_j}\Big)^{b_j} \prod_{j=j_0+1}^{j_1} \Big(\frac{z_{j-1}}{z_j}\Big)^{d_j}$$

has valuation b_j at Q_j and $b_j + 2d_j - d_{j+1} = a_j$ at P_j for $j_0 \leq j \leq j_1$, and valuation zero elsewhere. It has the same divisor as f, so by Lemma 3.2

$$f = \eta \prod_{j=j_0}^{j_1} \Big(\frac{y_{j-1}}{y_j}\Big)^{b_j} \prod_{j=j_0+1}^{j_1} \Big(\frac{z_{j-1}}{z_j}\Big)^{d_j}, \qquad \eta \in D^\times.$$

Step 2: the factors are units at V. Let $j \geq j_0$. As D-ideals, A_jD has valuation one exactly at $\mathfrak{m}_A$ and at the P_i with $i > j$. Since $j \geq j_0 > j_V$ and $V \neq \mathfrak{m}_A, \mathfrak{m}_B$, the ideal V is none of $\mathfrak{m}_A$, P_i, Q_i with $i \geq j_0$. Hence y_{j-1}/y_j and z_j are units at V for $j \geq j_0$, so is z_{j-1}/z_j for $j > j_0$, and V is not in the support of f.

Step 3: the classes of the factors. Let $j \geq j_0$. Since $j - 1 \geq n_V$, the prime V_{j-1} is inert in F_j with residue cardinality $q \equiv 1 \pmod 4$, and $\langle y_{j-1}\rangle_V = 2\langle y_j\rangle_V$. The element z_j has relative norm one by (T2), so its residue $\bar{z}$ satisfies $\bar{z}^{q+1} = 1$. The kernel of the norm map $\mathbb{F}_{q^2}^\times \to \mathbb{F}_q^\times$ has order $q+1$ and $v_2(q+1) = 1$, so the two-primary component of $\bar{z}$ has order at most two. It therefore lies in $\{\pm 1\} \subseteq \overline{E}_V$, and $\langle z_j\rangle_V = 0$. Consequently $\langle z_{j-1}/z_j\rangle_V = 0$ for $j > j_0$. The class of y_{j-1}/y_j is $\langle y_{j-1}\rangle_V - \langle y_j\rangle_V = \langle y_j\rangle_V$, and $\langle \eta\rangle_V = 0$ since $\eta \in D^\times$.

Step 4: the class of f. Iterating $\langle y_{j-1}\rangle_V = 2\langle y_j\rangle_V$ gives $\langle y_j\rangle_V = 2^{j_1-j}\langle y_{j_1}\rangle_V$ for $j_0 \leq j \leq j_1$. By Steps 1 and 3 and (17),

$$\langle f\rangle_V = \sum_{j=j_0}^{j_1} b_j \langle y_j\rangle_V = 2^{j_1}\Sigma\, \langle y_{j_1}\rangle_V.$$

Here $2^{j_1}\Sigma$ is an integer of two-adic valuation $j_1 + v_2(\Sigma)$, and $\langle y_{j_1}\rangle_V$ has order $2^{j_1+o_V-a_V}$ by Lemma 5.7, since $j_1 \geq n_V$. Lemma 5.6 and $j_0 \geq a_V - o_V + 2 + \lceil \log_2 \kappa \rceil$ give

$$o_V - a_V - v_2(\Sigma) \geq o_V - a_V + j_0 - 1 - \log_2 \kappa \geq 1.$$

Hence $j_1 + v_2(\Sigma) < j_1 + o_V - a_V$, and $\langle f\rangle_V$ has order 2^r with $r = o_V - a_V - v_2(\Sigma) \geq 1$.

Step 5: lifting to the residue. In the quasicyclic two-group $k(V)^\times_{(2)}$, let g be an element whose class modulo the subgroup $\overline{E}_V$ of order 2^{a_V} has order $2^r > 1$. Then g has order exactly 2^{r+a_V}: by Step 4 of the proof of Lemma 5.7, the class of an element of order 2^N has order $2^{\max\{N-a_V,0\}}$. Applied to the two-primary component of the residue of f, this gives the order $2^{o_V - v_2(\Sigma)}$ of (18). Multiplying f by a unit of D changes neither its divisor nor, therefore, this computation. □

Proposition 5.9. *Let $S \subseteq D$ be a subring that is not a BFD. For every maximal ideal V of D of odd residue characteristic, the image of S in D/V is the whole residue field. Distinct maximal ideals of D over the same odd rational prime have distinct contractions to S.*

Proof. The late-supported elements of S supplied by Lemma 5.5 have residues of unbounded two-power order by Lemma 5.8; this gives surjectivity, and the offsets of (T6) give separation.

Step 1: the image is a field. Let p be the residue characteristic of V. The image of S in D/V is a subring of an algebraic extension of $\mathbb{F}_p$. For a nonzero element r of this image, $\mathbb{F}_p[r]$ is a finite domain, hence a field, so r^{-1} lies in the image. Thus the image is a field.

Step 2: surjectivity. At $\mathfrak{m}_A$ or $\mathfrak{m}_B$ the residue field is the prime field $\mathbb{F}_{47}$ or $\mathbb{F}_7$, and the image of $\mathbb{Z} \subseteq S$ already fills it. Let V be any other maximal ideal of odd residue characteristic, so that $k(V) = \bigcup_r \mathbb{F}_{p^{2^r}}$. The subfields of $\mathbb{F}_{p^{2^r}}$ are the fields $\mathbb{F}_{p^{2^{r'}}}$ with $r' \leq r$. Hence a subfield of $k(V)$ either contains $\mathbb{F}_{p^{2^r}}$ for every r, and then equals $k(V)$, or it is one of the finite fields $\mathbb{F}_{p^{2^{r'}}}$. By Lemma 5.5 there is a κ such that S contains late-supported elements f with exponent bound κ at arbitrarily large levels j_0. By Lemma 5.8, applied to the set $\{V\}$, and by Lemma 5.6, the two-primary component of the residue of such an f at V has order $2^{o_V - v_2(\Sigma)} \geq 2^{o_V + j_0 - 1 - \log_2 \kappa}$ once $j_0 \geq j_*(V, \kappa)$. These orders are unbounded, whereas the elements of a finite field have bounded orders. So the image of S is not finite, and it is the whole residue field.

Step 3: separation. Let $V \neq V'$ be maximal ideals of D over the same odd rational prime, and suppose that $V \cap S = V' \cap S$. By Steps 1 and 2, the residue maps induce isomorphisms from the field $S/(V \cap S)$ onto D/V and onto D/V'. Composing them gives an isomorphism $D/V \to D/V'$ that sends the residue of each $f \in S$ at V to its residue at V'. The ideals $\mathfrak{m}_A$ and $\mathfrak{m}_B$ have different residue characteristics, so at most one of V, V' is $\mathfrak{m}_A$ or $\mathfrak{m}_B$. If one of them is, its residue field is finite while the other residue field is infinite, and no such isomorphism exists. Suppose that neither is $\mathfrak{m}_A$ or $\mathfrak{m}_B$. The isomorphism preserves the orders of the two-primary components of residues. Take $f \in S$ late-supported with exponent bound κ and $j_0 \geq \max\{j_*(V, \kappa), j_*(V', \kappa)\}$, as in Step 2. Lemma 5.8, applied to the set $\{V, V'\}$, gives orders $2^{o_V - v_2(\Sigma)}$ and $2^{o_{V'} - v_2(\Sigma)}$ for the same Σ. These are different because $o_V \neq o_{V'}$ by (T6). This contradiction shows $V \cap S \neq V' \cap S$. □

5.5 Teichmüller limits and unramified characteristics

Proposition 5.9 is a statement about residues. To turn it into a statement about the completed orders $\mathcal{A}_{n,p}$ we lift residues canonically. The Teichmüller representatives of Section 2.3 [8, Remark 7.36] are such lifts, and they are limits of iterated q-th powers. A completed order is a closed subring, so it contains these limits as soon as it contains elements with the right residues. In an unramified complete discrete valuation ring every element is a p-adic series in Teichmüller representatives, so a closed subring that contains all Teichmüller representatives is the whole ring.

Lemma 5.10 (Teichmüller limits)**.** *Let $\mathcal{O}$ be a complete discrete valuation ring with maximal ideal $\mathfrak{m}$ and finite residue field of characteristic p with q_0 elements, and let $q = q_0^k$ with $k \geq 1$. For every $s \in \mathcal{O}$ the sequence s^{q^r} converges as $r \to \infty$, and its limit is the Teichmüller representative of the residue of s.*

Proof. If $a \equiv b \pmod{\mathfrak{m}^j}$ with $j \geq 1$, then $a^p \equiv b^p \pmod{\mathfrak{m}^{j+1}}$. Indeed, write $a = b + c$ with $c \in \mathfrak{m}^j$; in the binomial expansion of $(b + c)^p$ the terms other than b^p and c^p have coefficients divisible by $p \in \mathfrak{m}$, and $c^p \in \mathfrak{m}^{pj} \subseteq \mathfrak{m}^{j+1}$. Since q is a power of p, it follows that $a^q \equiv b^q \pmod{\mathfrak{m}^{j+1}}$. Every element x of the residue field satisfies $x^q = x$, so $s^q \equiv s \pmod{\mathfrak{m}}$. By induction on r, $s^{q^{r+1}} \equiv s^{q^r} \pmod{\mathfrak{m}^{r+1}}$. So the sequence is Cauchy and converges to some $\lambda \in \mathcal{O}$ with $\lambda \equiv s \pmod{\mathfrak{m}}$. The Teichmüller representative τ of the residue of s satisfies $\tau^{q_0} = \tau$, hence $\tau^{q^r} = \tau$ for all r, and $\tau \equiv s \pmod{\mathfrak{m}}$. By the congruence above, $s^{q^r} \equiv \tau^{q^r} = \tau \pmod{\mathfrak{m}^{r+1}}$ for every r, so $\lambda = \tau$. □

When $\mathcal{O}$ is a finite product of complete discrete valuation rings, as $\mathcal{B}_{n,p}$ is, we apply the lemma in each factor. The *Teichmüller representative* of a tuple of residues is then the tuple of the Teichmüller representatives of its entries.

Proposition 5.11. *Let $S \subseteq D$ be a subring that is not a BFD. If an odd rational prime p is unramified in every F_n, then $\mathcal{A}_{n,p} = \mathcal{B}_{n,p}$ for every n. In particular this holds at 47 and 7.*

Proof. We first show that S maps onto D/pD, using Proposition 5.9 and the idempotents of D/pD; then we lift with Lemma 5.10.

Step 1: the structure of D/pD. Since $p\mathcal{O}_{F_m} \cap \mathcal{O}_{F_n} = p\mathcal{O}_{F_n}$ for $m \geq n$, the ring D/pD is the directed union of the rings $\mathcal{O}_{F_n}/p\mathcal{O}_{F_n}$. These are finite reduced $\mathbb{F}_p$-algebras, because p is unramified in F_n. Hence every finitely generated subalgebra of D/pD is a finite product of finite fields. The same is true of the image $\bar{S}$ of S in D/pD. Such rings are von Neumann regular. Each element r has a *support idempotent* e_r, equal to 1 in the factors where r is nonzero and to 0 elsewhere, and e_r is a power of r: if $\mathbb{F}_p[r] \cong \prod_i \mathbb{F}_{q_i}$, take $e_r = r^M$ with M a common multiple of the $q_i - 1$. In particular $e_r \in \bar{S}$ when $r \in \bar{S}$. Since D has dimension one, the points of $X = \operatorname{Spec}(D/pD)$ are the maximal ideals of D above p. For each r, the set of points where r does not vanish equals the set where $e_r = 1$, and it is open and closed. The space X is compact, and these sets form a basis of its topology. An element of D/pD is zero if it vanishes at every point, because D/pD is reduced.

Step 2: $\bar{S} = D/pD$. By Proposition 5.9, $\bar{S}$ maps onto the residue field at every point of X, and distinct points of X have distinct contractions to S. Let $x \neq y$ be points of X. Some $s \in \bar{S}$ vanishes at exactly one of them, and then e_s or $1 - e_s$ is an idempotent of $\bar{S}$ equal to 1 at x and to 0 at y.

Let $Y \subseteq X$ be open and closed. For $x \in Y$ and each $y \notin Y$, choose an idempotent of $\bar{S}$ equal to 1 at x and 0 at y. The sets where these idempotents vanish cover the compact set $X \setminus Y$, so finitely many suffice. Their product is an idempotent of $\bar{S}$ equal to 1 at x and to 0 on $X \setminus Y$. The sets where these idempotents equal 1, for $x \in Y$, cover the compact set Y. If $e_1, \ldots, e_k$ are finitely many of them that cover Y, then $1 - \prod_i (1 - e_i) \in \bar{S}$ is the characteristic idempotent of Y.

Now let $r \in D/pD$. For each $x \in X$, choose $s_x \in \bar{S}$ with the same residue as r at x. The set where s_x and r agree is the set where $s_x - r$ vanishes, so it is open and closed, and it contains x. Finitely many of these sets $U_1, \ldots, U_k$, for points $x_1, \ldots, x_k$, cover X. The sets $Y_i = U_i \setminus (U_1 \cup \cdots \cup U_{i-1})$ are open, closed, disjoint and cover X, so their characteristic idempotents $e_{Y_i} \in \bar{S}$ are orthogonal and sum to 1. The element $\sum_i e_{Y_i} s_{x_i} \in \bar{S}$ agrees with r at every point of X, so it equals r. This argument applies also to the infinite fibres at 47 and 7.

Step 3: Teichmüller representatives in $\mathcal{A}_{m,p}$. Fix n. The ring $\mathcal{B}_{n,p}$ is the product of the rings of integers $\mathcal{O}_{(F_n)_w}$ over the places w of F_n above p; each is unramified over $\mathbb{Z}_p$, and the residue ring of $\mathcal{B}_{n,p}$ is $\mathcal{O}_{F_n}/p\mathcal{O}_{F_n}$. By Step 2, for each element $\bar{t}$ of the finite ring $\mathcal{O}_{F_n}/p\mathcal{O}_{F_n}$ we may choose $t \in \mathcal{O}_{F_n}$ with residue $\bar{t}$ and $s \in S$ with $s \equiv t \pmod{pD}$. These finitely many representatives s all lie in S_m for some $m \geq n$, and then $s \equiv t \pmod{p\mathcal{O}_{F_m}}$. So, in $\mathcal{B}_{m,p}$, the residues of s agree with those of the corresponding elements t of the embedded ring $\mathcal{B}_{n,p}$. Take $q = p^f$ with $f \geq 1$ divisible by every residue degree of F_m at p; then q is also a power of the residue cardinality of each factor of $\mathcal{B}_{n,p}$. By Lemma 5.10 in each factor, s^{q^r} and t^{q^r} have the same limit, and this limit is the Teichmüller representative of $\bar{t}$ in $\mathcal{B}_{n,p}$. Since $\mathcal{A}_{m,p}$ is a finitely generated $\mathbb{Z}_p$-module, it is closed in $F_m \otimes_{\mathbb{Q}} \mathbb{Q}_p$. It contains the powers s^{q^r}, so it contains the Teichmüller representatives of all residues of $\mathcal{B}_{n,p}$.

Step 4: p-adic expansions. Since p is a uniformizer in each factor of $\mathcal{B}_{n,p}$, every $b \in \mathcal{B}_{n,p}$ is a convergent sum $\sum_{k \geq 0} \omega_k p^k$ with each ω_k a Teichmüller representative: take ω_0 to be the Teichmüller representative of the residue of b, and repeat with $(b - \omega_0)/p$. The partial sums lie in $\mathcal{A}_{m,p}$, which

is closed, so $\mathcal{B}_{n,p} \subseteq \mathcal{A}_{m,p}$. Intersecting with $F_n \otimes_{\mathbb{Q}} \mathbb{Q}_p$ and using (14) gives $\mathcal{B}_{n,p} \subseteq \mathcal{A}_{n,p}$. The reverse inclusion holds because $S_n \subseteq \mathcal{O}_{F_n}$. Finally, 47 and 7 are unramified in every F_n by (T1). □

5.6 Ramified characteristics: coefficient rings and conductor exponents

At a prime p that ramifies in the tower the maximal order is not generated by Teichmüller representatives, and the completed orders can be smaller than the maximal order. The Teichmüller representatives generate the coefficient ring, and at a maximal ideal of ramification index two the local ring of integers is a free module of rank two over it. So the only freedom left for the completed order is in the second coordinate, and we show that it is measured by one integer.

Setting. Let p be an odd rational prime that ramifies in some F_n. By (T1), p is different from 47, 7 and 3, and its absolute ramification indices are at most two. Also by (T1), we may fix a level b_p after which all primes over p are unramified and inert in every subsequent step. For each $n \geq b_p$, the maximal ideals V of D above p then correspond bijectively to the primes of F_n above p, via $V \mapsto V_n$, and

$$F_n \otimes_{\mathbb{Q}} \mathbb{Q}_p = \prod_{V|p} (F_n)_V.$$

The idempotents of the factors $(F_n)_V$ are the primitive idempotents of $F_n \otimes_{\mathbb{Q}} \mathbb{Q}_p$. The ramification index $e(V/p)$ of $(F_n)_V$ over $\mathbb{Q}_p$ is 1 or 2, and it does not depend on $n \geq b_p$.

For a maximal ideal V of D above p and $n \geq b_p$, write $W_{n,V}$ for the ring of integers of the maximal unramified subfield of $(F_n)_V$, the *coefficient ring* of $(F_n)_V$ (Section 2.3; see [8, Proposition 7.50 and Corollary 7.51]). It contains the Teichmüller representatives, it is unramified over $\mathbb{Z}_p$, and it has the same residue field as $(F_n)_V$. If $e(V/p) = 1$, then $W_{n,V} = \mathcal{O}_{(F_n)_V}$.

Suppose that $e(V/p) = 2$. Then $(F_n)_V$ is a ramified quadratic extension of the unramified field $\mathrm{Frac}(W_{n,V})$. For such an extension, with p odd and coefficient ring W, recall that a square uniformizer is a uniformizer π with $\pi^2 \in pW^\times$ (Section 2.3). To obtain one, complete the square in the Eisenstein minimal polynomial, over $\mathrm{Frac}(W_{b_p,V})$, of a uniformizer of $(F_{b_p})_V$; since p is odd, this gives a uniformizer π_V with $\pi_V^2 \in pW_{b_p,V}^\times$. For $n \geq b_p$ the extension $(F_n)_V/(F_{b_p})_V$ is unramified, so π_V is still a uniformizer and $\pi_V^2 \in pW_{n,V}^\times$: the same π_V is a square uniformizer of $(F_n)_V$ for every $n \geq b_p$. It generates the local ring of integers over the coefficient ring [8, Proposition 7.55 and Remark 7.56], so

$$\mathcal{O}_{(F_n)_V} = W_{n,V} \oplus \pi_V W_{n,V}.$$

For an order Λ contained in $\mathcal{O}_{(F_n)_V}$, the *conductor* of Λ is the largest ideal of $\mathcal{O}_{(F_n)_V}$ contained in Λ, namely $\{x \in \mathcal{O}_{(F_n)_V} : x\mathcal{O}_{(F_n)_V} \subseteq \Lambda\}$.

Proposition 5.12. *Let $S \subseteq D$ be a subring that is not a BFD, and let p and b_p be as above. For $n \geq b_p$, $\mathcal{A}_{n,p}$ contains $\prod_{V|p} W_{n,V}$, including the primitive idempotents of $F_n \otimes_{\mathbb{Q}} \mathbb{Q}_p$. Hence $\mathcal{A}_{n,p}$ is the product of its factors in the $(F_n)_V$. Its factor at a maximal ideal V above p with $e(V/p) = 1$ is the full local ring of integers $\mathcal{O}_{(F_n)_V}$. At V with $e(V/p) = 2$ it is*

$$\Lambda_{n,V} = W_{n,V} + \pi_V p^{s_V} W_{n,V} \tag{19}$$

for a nonnegative integer s_V independent of $n \geq b_p$; equivalently, the conductor $\{x \in \mathcal{O}_{(F_n)_V} : x\mathcal{O}_{(F_n)_V} \subseteq \Lambda_{n,V}\}$ equals $p^{s_V}\mathcal{O}_{(F_n)_V}$. We call s_V the conductor exponent *of S at V.*

Proof. We lift residue tuples to Teichmüller representatives, as in the proof of Proposition 5.11, to show that the completed order contains the coefficient rings; what remains is an ideal of the coefficient ring in the second coordinate.

Step 1: the coefficient rings. By Proposition 5.9, the finitely many contractions $V \cap S$ of the maximal ideals V above p are distinct maximal ideals of S, and each residue map $S/(V \cap S) \to D/V$ is onto. The Chinese remainder theorem in S therefore supplies, for every tuple $\rho = (\rho_V)_V$ of residues $\rho_V \in k(V_n)$, an element $s_\rho \in S$ with residue ρ_V at each V. Place these finitely many elements s_ρ in some S_m with $m \geq n$, and take $q = p^f$ with $f \geq 1$ divisible by every residue degree of F_m at p. Lemma 5.10, applied in each factor of $\mathcal{B}_{m,p}$, ramified or not, shows that $s_\rho^{q^r}$ converges to the tuple of Teichmüller representatives of the ρ_V. The Teichmüller representative of ρ_V in $\mathcal{O}_{(F_n)_V}$ is a root of $X^q - X$ with residue ρ_V, so by the last step of the proof of Lemma 5.10 it is this limit. Hence these representatives lie in the coefficient rings $W_{n,V}$ (Section 2.3). As $\mathcal{A}_{m,p}$ is closed, the limit lies in $\mathcal{A}_{m,p}$. Every element of $\prod_V W_{n,V}$ is a convergent sum $\sum_k \omega_k p^k$ of such tuples ω_k (as in Step 4 of the proof of Proposition 5.11), so $\prod_V W_{n,V} \subseteq \mathcal{A}_{m,p}$. This product contains the primitive idempotents: the idempotent of $(F_n)_V$ is the Teichmüller representative of the tuple with entry 1 at V and 0 elsewhere. Equation (14) places $\prod_V W_{n,V}$ already in $\mathcal{A}_{n,p}$. This finite-level intersection is essential: Proposition 5.9 produces elements of S in some S_m, possibly with $m > n$, and surjectivity on residue fields alone does not provide lifts at level n.

Step 2: the shape of each factor. The primitive idempotents lie in $\mathcal{A}_{n,p}$, so $\mathcal{A}_{n,p}$ is the product of its factors $\Lambda_{n,V} \subseteq \mathcal{O}_{(F_n)_V}$, and each $\Lambda_{n,V}$ contains $W_{n,V}$. If $e(V/p) = 1$, then $W_{n,V} = \mathcal{O}_{(F_n)_V}$ and the factor is the full local ring of integers. Let $e(V/p) = 2$. Every element of $\Lambda_{n,V}$ has the form $w_0 + \pi_V w_1$ with $w_0, w_1 \in W_{n,V}$, and subtracting $w_0 \in \Lambda_{n,V}$ shows that $\pi_V w_1 \in \Lambda_{n,V}$. Hence $\Lambda_{n,V} = W_{n,V} + \pi_V J_n$, where $J_n = \{w \in W_{n,V} : \pi_V w \in \Lambda_{n,V}\}$ is an ideal of $W_{n,V}$, because $\Lambda_{n,V}$ is a $W_{n,V}$-module. The ideal J_n is nonzero because $\Lambda_{n,V}$ is a full lattice in $(F_n)_V$. The ring $W_{n,V}$ is a discrete valuation ring with uniformizer p, so $J_n = p^{s_{n,V}} W_{n,V}$ for some integer $s_{n,V} \geq 0$.

Step 3: independence of n. Let $m \geq n \geq b_p$. The primes of F_n and of F_m above p correspond, so $F_n \otimes_{\mathbb{Q}} \mathbb{Q}_p$ and $F_m \otimes_{\mathbb{Q}} \mathbb{Q}_p$ have the same primitive idempotents, and (14) gives $\Lambda_{m,V} \cap (F_n)_V = \Lambda_{n,V}$ for each V. The extension $\mathrm{Frac}(W_{m,V})/\mathrm{Frac}(W_{n,V})$ is unramified, so $p^s W_{m,V} \cap \mathrm{Frac}(W_{n,V}) = p^s W_{n,V}$ for every s. Since $(F_m)_V = \mathrm{Frac}(W_{m,V}) \oplus \pi_V \mathrm{Frac}(W_{m,V})$ with the same π_V as at level n, it follows that $\Lambda_{m,V} \cap (F_n)_V = W_{n,V} + \pi_V p^{s_{m,V}} W_{n,V}$. Comparing with $\Lambda_{n,V}$ gives $s_{m,V} = s_{n,V}$, as required.

Step 4: the conductor. Write $s = s_V$, $W = W_{n,V}$ and $\pi = \pi_V$, with $\pi^2 = p u_0$ for a unit $u_0 \in W^\times$. An element $x = a + \pi b$ of $\mathcal{O}_{(F_n)_V}$, with $a, b \in W$, lies in the conductor exactly when x and $x\pi = p u_0 b + \pi a$ lie in $\Lambda_{n,V}$, because $\Lambda_{n,V}$ is a W-module and $\mathcal{O}_{(F_n)_V} = W + \pi W$; that is, when $b \in p^s W$ and $a \in p^s W$. So the conductor is $p^s W \oplus \pi p^s W = p^s \mathcal{O}_{(F_n)_V}$. Conversely, the conductor determines s, and $\Lambda_{n,V} = W + p^s \mathcal{O}_{(F_n)_V}$. □

Section 6 uses the following description of the units of the local orders $\Lambda_{n,V}$.

Lemma 5.13 (Units of the local orders)**.** *In the situation of Proposition 5.12, let V be a maximal ideal of D above p with $e(V/p) = 2$, and let $n \geq b_p$. Every unit of $\mathcal{O}_{(F_n)_V}$ can be written uniquely as $x_0(1 + \pi_V b)$ with $x_0 \in W_{n,V}^\times$ and $b \in W_{n,V}$. It belongs to $\Lambda_{n,V}^\times$ if and only if $b \in p^{s_V} W_{n,V}$.*

Proof. Write a unit as $x = a + \pi_V a'$ with $a, a' \in W_{n,V}$. It has the same residue as a, so $a \in W_{n,V}^\times$, and $x = a(1 + \pi_V a^{-1} a')$. Conversely $x_0(1 + \pi_V b) = x_0 + \pi_V x_0 b$, so the coordinates of x determine x_0 and then b. If $x \in \Lambda_{n,V}$, then $x_0 b \in p^{s_V} W_{n,V}$, so $b \in p^{s_V} W_{n,V}$. Conversely, if $b \in p^{s_V} W_{n,V}$, then $x \in \Lambda_{n,V}$, and its inverse

$$x^{-1} = x_0^{-1}(1 - \pi_V b)(1 - \pi_V^2 b^2)^{-1}$$

also lies in $\Lambda_{n,V}$, because $1 - \pi_V b \in \Lambda_{n,V}$ and $1 - \pi_V^2 b^2 \in 1 + p W_{n,V}$ is a unit of the coefficient ring. □

We have not assumed that S_n is stable under any automorphism of F_n, and the conductor exponents s_V may differ for different maximal ideals V above the same prime p.

6 Unit approximation and hereditary atomicity

This section proves Theorem 6.1: if the elements z_n of (T2) can be approximated by global units in the sense of condition (JT_A) below, then D is hereditarily atomic. Since D does not satisfy ACCP (Theorem 3.7), this completes the factorization-theoretic part of the proof of Theorem B. Sections 7 and 8 then show that, for a tower as in Theorem B, the conditions (S) and (O) imply (JT_A).

Throughout this section hypotheses (T1)–(T6) hold. For a subring $S \subseteq D$ with $\operatorname{Frac} S = K$, we use the order $S_N = S \cap F_N$, its completion $\mathcal{A}_{N,p} = S_N \otimes_{\mathbb{Z}} \mathbb{Z}_p$ and the maximal order $\mathcal{B}_{N,p} = \mathcal{O}_{F_N} \otimes_{\mathbb{Z}} \mathbb{Z}_p$, as in Section 5. The results of Section 5 about these completed orders are used only when S is not a BFD.

The idea. Let $S \subseteq D$ be a subring and let $x \in S$ be a nonzero nonunit. If $\operatorname{Frac} S \neq K$, or if x has bounded factorization lengths in S, then x is a product of atoms of S. Otherwise x has positive valuation at both $\mathfrak{m}_A$ and $\mathfrak{m}_B$ (Proposition 3.6). In D, such an x is the product of two one-sided factors, as in the proof of Theorem 3.7, and a one-sided element has bounded factorization lengths in every subring that contains it (Proposition 3.6). Hence x is a product of atoms of S as soon as both factors can be chosen in S. A factor of x in D is determined by its divisor only up to a unit of D. The proof has two steps, and the second uses this freedom.

(1) *Split* (Section 6.2). Lemma 6.3 gives a factor x_A of x in D with $v_{\mathfrak{m}_A}(x_A) = v_{\mathfrak{m}_A}(x)$ and $v_{\mathfrak{m}_B}(x_A) = 0$, together with a factor f of x in S, such that the ratio $\rho = f/x_A$ is a product of powers of elements z_n with large n.

(2) *Move* (Section 6.3). We look for a unit η of D such that $f' = \eta^{-1}x_A$ and x/f' both lie in S. Both lie in some field F_N, and by (15) an element of F_N lies in S_N exactly when it lies in the completed order $\mathcal{A}_{N,p}$ for every prime p. Lemma 6.4 produces such an η when the image of ρ lies in the closure of the image of the unit group E_N in a compact group $\mathcal{U}_N$ made of local unit groups.

Why the approximation must be simultaneous. The unit η in step (2) must be one element of E_N that works at all primes p at once. Suppose that for each prime p we only had a unit η_p with $\eta_p\rho \in \mathcal{A}_{N,p}^{\times}$. The candidate factors $\eta_p^{-1}x_A$ would in general be different elements of F_N, each satisfying the local condition at its own prime, and none of them need lie in S_N. Approximating ρ by units at each place separately therefore does not suffice for this argument: we need approximation by one unit in the product of all the local groups. For this reason the closure in (JT_A) is taken in the whole product $\mathcal{U}_N$.

Which places occur. At 47 and 7, the final factors are integral and the completed orders are maximal (Proposition 5.11), so their membership in $\mathcal{A}_{N,p}$ requires no additional approximation condition. For the local units considered in Lemma 6.4, there is also no additional condition at an odd place at which F_N is unramified over p. At an odd place of ramification index two, the completed order contains the coefficient ring (Proposition 5.12 at a level $M \geq \max\{N, b_p\}$, and (14)), so only the class of a unit modulo the units of the coefficient ring matters. At the dyadic places we know only that $\mathcal{A}_{N,2}$ contains $1 + 2^e\mathcal{B}_{N,2}$ for some $e \geq 1$, so the whole dyadic unit group appears. In every case membership is an open condition, so approximation to a finite precision suffices; this is why (JT_A) is stated with a closure.

6.1 The approximation group

Fix a level N. For a place v of F_N above an odd prime p with $e(v/p) = 2$, write $\mathcal{O}_{N,v} = \mathcal{O}_{(F_N)_v}$ for the ring of integers of the completion and $W_{N,v}$ for its coefficient ring (Section 2.3). Define the

compact abelian group

$$\mathcal{U}_N = (\mathcal{O}_{F_N} \otimes \mathbb{Z}_2)^\times \times \prod_{\substack{v \text{ odd} \\ e(v/p)=2}} \mathcal{O}_{N,v}^\times / W_{N,v}^\times. \tag{20}$$

There are finitely many factors, since only finitely many primes of F_N are ramified over $\mathbb{Q}$. The first factor is the product of the unit groups $\mathcal{O}_{(F_N)_v}^\times$ over the dyadic places v. Each factor is compact, because $W_{N,v}^\times$ is a closed subgroup of the compact group $\mathcal{O}_{N,v}^\times$.

Let $z \in F_N^\times$ be a unit at every dyadic prime and at every odd prime of absolute ramification index two. We write $\jmath_N(z)$ for its image in $\mathcal{U}_N$: its image in $(\mathcal{O}_{F_N} \otimes \mathbb{Z}_2)^\times$ in the first factor, and its class in $\mathcal{O}_{N,v}^\times / W_{N,v}^\times$ in the factor of v. On the group of such elements, $\jmath_N$ is a homomorphism. It is defined on all global units E_N, and on z_n and z_n' for $n \le N$. Indeed, the divisors $(z_n) = A_n P_n^{-1}$ and $(z_n') = B_n Q_n^{-1}$ lie above 47 and 7, and by (T1) the primes 47 and 7 are unramified in every F_N. We impose the following *unit approximation condition* on the elements z_n; the subscript A in its name refers to the chain (A_n), since $(z_n) = A_n P_n^{-1}$:

$$\exists n_0 \ \forall n > n_0 \ \exists N \ge n : \quad \jmath_N(z_n) \in \overline{\jmath_N(E_N)}. \tag{JT$_A$}$$

Here $\overline{\jmath_N(E_N)}$ is the closure of $\jmath_N(E_N)$ in $\mathcal{U}_N$. Thus $\jmath_N(z_n)$ lies in it exactly when, for every neighbourhood $\mathcal{N}$ of the identity in $\mathcal{U}_N$, there is a unit $\eta \in E_N$ with $\jmath_N(\eta z_n) \in \mathcal{N}$: one global unit approximates z_n in all factors of $\mathcal{U}_N$ at once.

The closure in the whole product can be strictly smaller than the product of the closures of the projections: the diagonal image of $\mathbb{Z}$ in $\mathbb{Z}_2 \times \mathbb{Z}_2$ has closure the diagonal, while each projection has closure $\mathbb{Z}_2$.

The quantifiers in (JT$_A$) are exactly those that the proof of Theorem 6.1 uses. The condition concerns only z_n with $n > n_0$, because a given element x requires only finitely many z_n, all with n larger than a cutoff n_1 that may be taken as large as we like (Lemma 6.3). The level N may depend on n, because approximation at one level persists at every later level (Lemma 6.2 below), so finitely many z_n can be moved to a common level. In Sections 7 and 8 the condition is verified with $N = n$ (Proposition 8.9).

Theorem 6.1 (Unit approximation criterion)**.** *Assume* (T1)–(T6) *and* (JT$_A$)*. Then D is hereditarily atomic and does not satisfy ACCP. The same conclusion holds if, in* (JT$_A$)*, the elements z_n are replaced by the elements z_n'.*

The theorem is proved in Section 6.4, after the two lemmas that carry out the steps described above. We first compare the groups $\mathcal{U}_N$ at different levels.

Lemma 6.2. *Let $N' \ge N$. There is a continuous homomorphism $\theta_{N,N'} : \mathcal{U}_N \to \mathcal{U}_{N'}$ such that $\theta_{N,N'}(\jmath_N(z)) = \jmath_{N'}(z)$ for every $z \in F_N^\times$ that is a unit at every prime not lying above* 47 *or* 7*. This applies to every global unit $z \in E_N$ and to $z = z_n, z_n'$ with $n \le N$; these z_n, z_n' are the elements of relative norm one of* (T2)*. The homomorphism $\theta_{N,N'}$ maps $\overline{\jmath_N(E_N)}$ into $\overline{\jmath_{N'}(E_{N'})}$.*

Proof. We define $\theta_{N,N'}$ factor by factor, using for each factor of $\mathcal{U}_{N'}$ the factor of $\mathcal{U}_N$ at the prime below it.

Dyadic factor. The inclusion $F_N \otimes \mathbb{Q}_2 \subseteq F_{N'} \otimes \mathbb{Q}_2$ maps $(\mathcal{O}_{F_N} \otimes \mathbb{Z}_2)^\times$ into $(\mathcal{O}_{F_{N'}} \otimes \mathbb{Z}_2)^\times$.

Odd factors. Let w be an odd prime of $F_{N'}$ with $e(w/p) = 2$, and let v be the prime of F_N below it. Then $e(v/p)$ divides $e(w/p)$, so $e(v/p)$ is 1 or 2.

(a) Suppose $e(v/p) = 2$, so that v is already ramified. The completion $(F_N)_v$ embeds in $(F_{N'})_w$. The fraction field of its coefficient ring is unramified over $\mathbb{Q}_p$, so it embeds in the fraction

field of the coefficient ring of $(F_{N'})_w$, and $W_{N,v}^\times \subseteq W_{N',w}^\times$. Hence the inclusion $\mathcal{O}_{N,v}^\times \subseteq \mathcal{O}_{N',w}^\times$ induces a map of quotients $\mathcal{O}_{N,v}^\times/W_{N,v}^\times \to \mathcal{O}_{N',w}^\times/W_{N',w}^\times$. We use this map in the factor of every prime w of $F_{N'}$ above v.

(b) Suppose $e(v/p) = 1$, so that w is newly ramified. Then $(F_N)_v$ is unramified over $\mathbb{Q}_p$, so $\mathcal{O}_{N,v} = W_{N,v} \subseteq W_{N',w}$. Every element of F_N that is a unit at v therefore has trivial class in $\mathcal{O}_{N',w}^\times/W_{N',w}^\times$, and we use the trivial homomorphism in the factor of w.

These maps are continuous homomorphisms, so $\theta_{N,N'}$ is one. Let $z \in F_N^\times$ be a unit at every prime that does not lie above 47 or 7; for instance z is a global unit of F_N, or z_n or z_n' with $n \le N$. By (T1), z is then a unit at every dyadic prime, at every odd prime of absolute ramification index two, and at every prime v in case (b). In case (a) both $\theta_{N,N'}(\jmath_N(z))$ and $\jmath_{N'}(z)$ have the class of z in the factor of w; in case (b) both are trivial there. This proves the compatibility with $\jmath$. Since $E_N \subseteq E_{N'}$, we get $\theta_{N,N'}(\jmath_N(E_N)) \subseteq \jmath_{N'}(E_{N'})$. A continuous map sends the closure of a set into the closure of its image, which gives the statement about closures. □

6.2 Splitting an element into two one-sided factors

Let $x \in S$ have unbounded factorization lengths. Lemma 5.5 provides a factor f of x in S supported on late primes P_j, Q_j. We now construct a factor x_A of x in D that carries the whole valuation of x at $\mathfrak{m}_A$ and none at $\mathfrak{m}_B$. We obtain it from f by multiplying by powers of the elements z_n, with $(z_n) = A_n P_n^{-1}$; each of them moves valuation from a prime P_n split off from the chain (A_n) to $A_n D$, and hence to $\mathfrak{m}_A$. A weighted degree, defined in Step 3 of the proof below, is unchanged by these moves. Together with the prescribed valuations of x_A at the P_j, it determines the valuation that x_A acquires at $\mathfrak{m}_A$. Thus f/x_A is exactly a product of powers of the z_n.

Lemma 6.3. *Let $S \subseteq D$ be a subring and let $x \in S$ have unbounded factorization lengths in S, with $\kappa_A = v_{\mathfrak{m}_A}(x) > 0$ and $\kappa_B = v_{\mathfrak{m}_B}(x) > 0$. For every sufficiently large integer n_1 there are an element $f \in S$ with $x/f \in S$ and an element $x_A \in D$ with $x_A \mid_D x$, such that*

$$\big(v_{\mathfrak{m}_A}(x_A), v_{\mathfrak{m}_B}(x_A)\big) = (\kappa_A, 0), \qquad \big(v_{\mathfrak{m}_A}(x/x_A), v_{\mathfrak{m}_B}(x/x_A)\big) = (0, \kappa_B),$$

and, for some integer n_2,

$$\rho := f/x_A = \prod_{n_1 < n \le n_2} z_n^{m_n}, \qquad m_n \in \mathbb{Z}. \tag{21}$$

Both x_A and ρ are units at every maximal ideal of D that does not lie above 47 or 7.

Proof. We take for f the factor of x given by Lemma 5.5, and obtain x_A by multiplying f by powers of the z_n whose exponents are determined by bounded binary digits.

Step 1: the factor f. Choose n_1 so that $v_{P_j}(x) = \kappa_A$ and $v_{Q_j}(x) = \kappa_B$ for all $j > n_1$ (see (2)) and $2^{n_1+1} > \max\{\kappa_A, \kappa_B\}$. Every larger n_1 has the same two properties. Lemma 5.5, applied with $j_0 = n_1 + 1$, gives a factor $f \in S$ of x with $x/f \in S$ and with finite nonempty support

$$(f) = \prod_{j > n_1} P_j^{a_j} Q_j^{b_j}, \qquad 0 \le a_j \le \kappa_A, \quad 0 \le b_j \le \kappa_B.$$

The element f is late-supported at level $n_1 + 1$ with exponent bound $\max\{\kappa_A, \kappa_B\}$, and $\max\{\kappa_A, \kappa_B\}\, 2^{-n_1-1} < 1$. So Lemma 5.6 gives the balance (17):

$$\Sigma := \sum_{j > n_1} a_j 2^{-j} = \sum_{j > n_1} b_j 2^{-j} > 0.$$

Step 2: bounded binary digits. Choose n_2 at least the largest index in the support of f and so large that $\Sigma' = 2^{n_2}\Sigma$ satisfies $\Sigma' \geq \kappa_A$; this is possible because $\Sigma > 0$ by Step 1. Then Σ' is an integer. Since $a_j \leq \kappa_A$ and $a_j = 0$ unless $n_1 < j \leq n_2$, we have $\Sigma < \kappa_A 2^{-n_1}$, that is, $\Sigma' < \kappa_A 2^{n_2-n_1}$. So the integer $\Sigma' - \kappa_A$ lies in $[0, \kappa_A(2^{n_2-n_1}-1)]$. Every integer in this interval has an expansion in base two with $n_2 - n_1$ digits, each between 0 and κ_A:

$$\Sigma' - \kappa_A = \sum_{n_1<j\leq n_2} a'_j 2^{n_2-j}, \qquad 0 \leq a'_j \leq \kappa_A.$$

We prove this by induction on the number k of digits: for $\kappa \geq 1$ and $k \geq 0$, every integer M with $0 \leq M \leq \kappa(2^k-1)$ has the form $\sum_{i=0}^{k-1} d_i 2^i$ with $0 \leq d_i \leq \kappa$. For $k = 0$ we have $M = 0$. For the inductive step let $0 \leq M \leq \kappa(2^{k+1}-1)$, and let d_0 be whichever of $\min\{\kappa, M\}$ and $\min\{\kappa, M\} - 1$ has the parity of M. If $M \leq \kappa$ then $d_0 = M$. Otherwise $d_0 \geq \kappa - 1$, and the integer $(M - d_0)/2$ is at most $\kappa(2^k-1) + \frac{1}{2}$, hence at most $\kappa(2^k-1)$. In both cases the induction hypothesis applies to $(M - d_0)/2$.

Step 3: the factor x_A. Define integers e_n for $n_1 < n \leq n_2$ recursively by

$$e_n = a_n - a'_n + \sum_{n_1<i<n} e_i,$$

and put

$$x_A = f \prod_{n_1<n\leq n_2} z_n^{e_n}.$$

We write the divisor of an invertible ideal I of D, the family of its valuations $(v_V(I))_V$, additively, and identify an ideal with its divisor. Since $A_n D = A_{n_2} D \prod_{n<j\leq n_2} P_j$ and $(z_n) = A_n P_n^{-1}$,

$$\operatorname{div}(z_n) = A_{n_2} D + \sum_{n<j\leq n_2} P_j - P_n \qquad (n_1 < n \leq n_2).$$

The ideal $A_{n_2}D$ has valuation one at $\mathfrak{m}_A$ and at the P_j with $j > n_2$, and valuation zero elsewhere, so its support is disjoint from $\{P_{n_1+1}, \ldots, P_{n_2}\}$ and from the Q_j. Hence the ideals $A_{n_2}D$, $P_{n_1+1}, \ldots, P_{n_2}$ and the Q_j are multiplicatively independent, and the coefficients of a divisor in the group they generate are well defined. By Step 2 the support of f lies among the P_j, Q_j with $n_1 < j \leq n_2$. Hence, with $E = \sum_n e_n$,

$$\operatorname{div}(x_A) = E\, A_{n_2} D + \sum_{n_1<j\leq n_2} \Big(a_j + \sum_{n_1<n<j} e_n - e_j\Big) P_j + \sum_{j>n_1} b_j Q_j = E\, A_{n_2} D + \sum_{n_1<j\leq n_2} a'_j P_j + \sum_{j>n_1} b_j Q_j,$$

where the second equality is the recursion defining e_j. To compute E, give $A_{n_2}D$ the weight 1 and P_j the weight 2^{n_2-j}, and call the weighted sum of the coefficients of $A_{n_2}D$ and of the P_j with $n_1 < j \leq n_2$ in a divisor its *weighted degree*. The weighted degree of $\operatorname{div}(z_n)$ is $1 + \sum_{n<j\leq n_2} 2^{n_2-j} - 2^{n_2-n} = 0$. So x_A and f have the same weighted degree, namely $\sum_j a_j 2^{n_2-j} = \Sigma'$. Comparing with the displayed divisor of x_A, whose weighted degree is $E + (\Sigma' - \kappa_A)$ by Step 2, gives $E = \kappa_A$.

Hence x_A has valuation κ_A at $\mathfrak{m}_A$ and at P_j for $j > n_2$, valuation $a'_j \leq \kappa_A$ at P_j for $n_1 < j \leq n_2$, valuation $b_j \leq \kappa_B$ at Q_j, and valuation zero elsewhere, in particular at $\mathfrak{m}_B$. All these valuations are nonnegative, so $x_A \in D$, and comparing them with the valuations of x chosen in Step 1 gives $x_A \mid_D x$ (Lemma 3.2), with the stated valuations of x_A and x/x_A at $\mathfrak{m}_A$ and $\mathfrak{m}_B$.

Step 4: the ratio ρ. By construction $f/x_A = \prod_{n_1<n\leq n_2} z_n^{-e_n}$, which is (21) with $m_n = -e_n$. The divisors of x_A and ρ are supported on $\mathfrak{m}_A$, the P_j and the Q_j, which lie above 47 and 7. This proves the last assertion. □

6.3 Moving an element into a fixed order

Let $S \subseteq D$ be a subring that is not a BFD. The next lemma turns approximation in $\mathcal{U}_N$ into membership in the completed orders of the fixed order S_N. Membership of a unit in $\mathcal{A}_{N,p}^{\times}$ is a condition at finitely many primes p, and at each of them it is an open condition on the image of the unit in $\mathcal{U}_N$. At the odd primes these conditions are described by Proposition 5.12, which applies only from the level b_p on. So the proof imposes each condition at a later level M_p, pulls it back to level N along the map θ_{N,M_p} of Lemma 6.2, which allows the unit to be chosen at level N, and then descends the resulting membership from M_p to N by (14).

Let $z \in F_N^{\times}$ be a unit at every prime that does not lie above 47 or 7. By (T1) the primes 47 and 7 are unramified in F_N, so z is a unit at every dyadic prime and at every odd prime of absolute ramification index two, and $\jmath_N(z)$ is defined.

Lemma 6.4 (A unit moving an element into a fixed order)**.** *Let $S \subseteq D$ be a subring that is not a BFD, and let $z \in F_N^{\times}$ be a unit at every prime that does not lie above 47 or 7. If $\jmath_N(z) \in \overline{\jmath_N(E_N)}$, then there exists $\eta \in E_N$ such that*

$$\eta z \in \mathcal{A}_{N,p}^{\times} \qquad \text{for every prime } p \notin \{47, 7\}. \tag{22}$$

Proof. We describe the conditions that define $\mathcal{A}_{N,p}^{\times}$ by an open subgroup $\mathcal{U}_N^S$ of $\mathcal{U}_N$, choose η using the closure hypothesis, and then verify (22) prime by prime.

Step 1: the primes that impose conditions. Since S_N has finite index in $\mathcal{O}_{F_N}$, we can choose $e \geq 1$ with $2^e \mathcal{B}_{N,2} \subseteq \mathcal{A}_{N,2}$. Let $\mathcal{P}$ be the set of odd rational primes p with $\mathcal{A}_{N,p} \neq \mathcal{B}_{N,p}$. It is finite, since $\mathcal{A}_{N,p} = \mathcal{B}_{N,p}$ whenever p does not divide the finite index $[\mathcal{O}_{F_N} : S_N]$. By Proposition 5.11, every $p \in \mathcal{P}$ is ramified at some level of the tower, so Proposition 5.12 applies to it.

Step 2: an open condition at a later level. Fix $p \in \mathcal{P}$, choose a level b_p as in Proposition 5.12, and put $M_p = \max\{N, b_p\}$. By Proposition 5.12, at level M_p the completed order $\mathcal{A}_{M_p,p}$ is the product of its factors at the maximal ideals V of D above p. The factor at V is the full local ring of integers if $e(V/p) = 1$, and the order $\Lambda_{M_p,V}$ of (19) if $e(V/p) = 2$. For V with $e(V/p) = 2$, write $\mathcal{O}_{M_p,V} = \mathcal{O}_{(F_{M_p})_V}$ and $C_V = \mathcal{O}_{M_p,V}^{\times}/W_{M_p,V}^{\times}$ for the factor of $\mathcal{U}_{M_p}$ at the prime of F_{M_p} below V, and let G_V be the image of $\Lambda_{M_p,V}^{\times}$ in C_V. Two properties of G_V are used.

(i) A unit x of $\mathcal{O}_{M_p,V}$ lies in $\Lambda_{M_p,V}^{\times}$ exactly when its class lies in G_V. Indeed, if the class of x is that of $\lambda \in \Lambda_{M_p,V}^{\times}$, then $x = \lambda c$ with $c \in W_{M_p,V}^{\times}$, and $W_{M_p,V}^{\times} \subseteq \Lambda_{M_p,V}^{\times}$ by (19).

(ii) G_V is an open subgroup of C_V. Indeed, (19) gives $p^{s_V}\mathcal{O}_{M_p,V} \subseteq \Lambda_{M_p,V}$, so the open subgroup $1 + p^{s_V+1}\mathcal{O}_{M_p,V}$ of $\mathcal{O}_{M_p,V}^{\times}$ lies in $\Lambda_{M_p,V}$. By Lemma 5.13 it lies in $\Lambda_{M_p,V}^{\times}$: an element $x_0(1 + \pi_V b)$ of $\Lambda_{M_p,V}$ has π_V-coordinate $x_0 b \in p^{s_V} W_{M_p,V}$, so $b \in p^{s_V} W_{M_p,V}$. The quotient map $\mathcal{O}_{M_p,V}^{\times} \to C_V$ is open, so G_V contains an open subgroup of C_V; being a subgroup, G_V is therefore open.

Let $G_p \subseteq \mathcal{U}_{M_p}$ be the subgroup of elements whose coordinate in C_V lies in G_V for every V above p with $e(V/p) = 2$, all other coordinates being unrestricted. By (ii), G_p is an open subgroup of $\mathcal{U}_{M_p}$.

Step 3: an open subgroup at level N and the choice of η. Let $\mathcal{U}_N^S \subseteq \mathcal{U}_N$ be the subgroup of elements whose dyadic coordinate lies in $1 + 2^e\mathcal{B}_{N,2}$ and whose image under θ_{N,M_p} (Lemma 6.2) lies in G_p for every $p \in \mathcal{P}$. Since $e \geq 1$, the set $1 + 2^e\mathcal{B}_{N,2}$ is an open subgroup of $\mathcal{B}_{N,2}^{\times}$. Each θ_{N,M_p} is a continuous homomorphism, so $\theta_{N,M_p}^{-1}(G_p)$ is an open subgroup of $\mathcal{U}_N$. As $\mathcal{P}$ is finite, $\mathcal{U}_N^S$ is an open subgroup of $\mathcal{U}_N$. Because $\jmath_N(z)$ lies in the closure of $\jmath_N(E_N)$, the neighbourhood $\jmath_N(z)\mathcal{U}_N^S$ of $\jmath_N(z)$ contains an element $\jmath_N(\eta^{-1})$ with $\eta \in E_N$. Then $\jmath_N(\eta z) \in \mathcal{U}_N^S$.

Step 4: the dyadic primes and the primes outside $\mathcal{P}$. The element ηz lies in the subgroup $1+2^e\mathcal{B}_{N,2}$, and so does its inverse. Both lie in $\mathcal{A}_{N,2}$, because $1 \in \mathcal{A}_{N,2}$ and $2^e\mathcal{B}_{N,2} \subseteq \mathcal{A}_{N,2}$. At an odd prime $p \notin \mathcal{P} \cup \{47, 7\}$, the element ηz is a unit at every prime above p, so $(\eta z)^{\pm 1}$ already lie in the maximal order $\mathcal{B}_{N,p} = \mathcal{A}_{N,p}$.

Step 5: the primes in $\mathcal{P}$. Fix $p \in \mathcal{P}$ and put $M = M_p$. By Step 1, p ramifies at some level, so $p \neq 47, 7$ by (T1). The element ηz is a unit at every prime that does not lie above 47 or 7, and $\jmath_N(\eta z) \in \mathcal{U}_N^S$ by Step 3, so Lemma 6.2 gives $\jmath_M(\eta z) = \theta_{N,M}(\jmath_N(\eta z)) \in G_p$. We check the factors of $\mathcal{A}_{M,p}$ described in Step 2. At V with $e(V/p) = 2$, the element ηz is a unit of $\mathcal{O}_{M,V}$, since $p \neq 47, 7$, and its class in C_V lies in G_V; by Step 2(i), ηz and its inverse lie in $\Lambda_{M,V}$. At V with $e(V/p) = 1$, the factor is the full local ring of integers, which contains $(\eta z)^{\pm 1}$. Thus $(\eta z)^{\pm 1} \in \mathcal{A}_{M,p}$. Since these elements belong to F_N, (14) returns both to $\mathcal{A}_{N,p}$. This holds for each of the finitely many $p \in \mathcal{P}$ with the same η, which proves (22). □

The primes $p \in \mathcal{P}$, which divide the index $[\mathcal{O}_{F_N} : S_N]$, depend only on the fixed order S_N, and they are chosen before η. Each condition is imposed at its own level M_p and transported to level N by the continuous map θ_{N,M_p}. No comparison of the orders at different levels beyond (14) is required. The levels M_p serve only to describe membership in the completed orders through Proposition 5.12, and (14) intersects this membership back to level N.

6.4 Proof of the unit approximation criterion

Proof of Theorem 6.1. We reduce to an element x with unbounded factorization lengths, split it with Lemma 6.3, use (JT_A) and Lemma 6.2 to approximate the ratio ρ at one common level, and move both factors into S with the unit supplied by Lemma 6.4.

Step 1: reductions. Let $S \subseteq D$ be a subring. If $\operatorname{Frac} S \neq K$, then S is a BFD by Proposition 5.2, and a BFD is atomic. We may therefore assume that S is not a BFD and has fraction field K. An element of S with bounded factorization lengths already has a factorization into atoms: a factorization into nonunits of maximal length has irreducible factors. Take a nonzero nonunit $x \in S$ with unbounded factorization lengths. By Proposition 3.6, x is not one-sided, so $\kappa_A = v_{\mathfrak{m}_A}(x)$ and $\kappa_B = v_{\mathfrak{m}_B}(x)$ are both positive.

Step 2: splitting. Let n_0 be as in (JT_A). Apply Lemma 6.3 with a cutoff $n_1 > n_0$ that is large enough for the lemma. It gives f, x_A, n_2 and the exponents m_n in (21).

Step 3: approximation at one common level. Every z_n that occurs in (21) has $n > n_1 > n_0$, so (JT_A) applies to each of these finitely many z_n and gives a level $N_n \geq n$ with $\jmath_{N_n}(z_n) \in \overline{\jmath_{N_n}(E_{N_n})}$. Let N'' be the largest of the N_n. Lemma 6.2, applied with $\theta_{N_n,N''}$, moves these closure assertions to the common level N''. We then enlarge the level to some $N \geq N''$ such that F_N contains $x, f, x/f, x_A$ and ρ, and apply Lemma 6.2 once more, with $\theta_{N'',N}$. This gives $\jmath_N(z_n) \in \overline{\jmath_N(E_N)}$ for all $n_1 < n \leq n_2$. The closure of a subgroup is a closed subgroup, so it is closed under products and integer powers. Since $\jmath_N$ is a homomorphism, (21) gives

$$\jmath_N(\rho) = \prod_{n_1 < n \leq n_2} \jmath_N(z_n)^{m_n} \in \overline{\jmath_N(E_N)}.$$

Step 4: moving the factors into S. By Lemma 6.3, $\rho \in F_N^\times$ is a unit at every prime that does not lie above 47 or 7. Lemma 6.4, applied to $z = \rho$, supplies $\eta \in E_N$ such that $\eta\rho \in \mathcal{A}_{N,p}^\times$ for every $p \neq 47, 7$. Set

$$f' = \eta^{-1} x_A = f(\eta\rho)^{-1}, \qquad g' = x/f' = (x/f)\eta\rho.$$

Both elements belong to $D \cap F_N = \mathcal{O}_{F_N}$, because $x_A \mid_D x$ and multiplication by a global unit changes no valuation. Since f and x/f lie in $S_N \subseteq \mathcal{A}_{N,p}$, the expressions on the right place f' and

g' in $\mathcal{A}_{N,p}$ for every $p \neq 47, 7$. At 47 and 7 the completed order is maximal by Proposition 5.11, so integrality suffices. Equation (15), applied to the one fixed order S_N, gives $f', g' \in S_N \subseteq S$. Their valuations at $(\mathfrak{m}_A, \mathfrak{m}_B)$ are those of x_A and x/x_A, namely $(\kappa_A, 0)$ and $(0, \kappa_B)$.

Step 5: conclusion. The elements f' and g' are one-sided nonunits of S. By Proposition 3.6 each has bounded factorization lengths in S, so each has a factorization into atoms of S. Their product is x. Thus every subring S of D is atomic, and D is hereditarily atomic. Failure of ACCP was established in Theorem 3.7. Interchanging the roles of $\mathfrak{m}_A$ and $\mathfrak{m}_B$, and accordingly of A_n, P_j, z_n and B_n, Q_j, z'_n, proves the symmetric assertion. □

7 The dyadic places: a Selmer condition

Theorem 6.1 (unit approximation criterion) reduces hereditary atomicity to condition (JT_A): for every sufficiently large n, the element z_n must be approximated by global units in the compact group $\mathcal{U}_N$ of (20) for some $N \geq n$. The group $\mathcal{U}_N$ has a dyadic factor $(\mathcal{O}_{F_N} \otimes \mathbb{Z}_2)^\times$ and factors at the odd places of ramification index two. This section treats the dyadic factor. Its main result, Theorem 7.2, shows that at a scaled step F_n/F_{n-1} the Selmer condition (S) implies that the pro-two part of z_n at the dyadic places is a limit of the pro-two parts of global units of F_n. Section 8 treats the odd places and combines the two results with $N = n$ (Proposition 8.9).

We work at one step, in the notation of Section 2.1: $F = F_{n-1}$, $L = F_n$, $\sigma = \sigma_n$, $A = A_{n-1}$, and $A\mathcal{O}_L = A'P$ with $A' = A_n$ and $P = P_n$. The rest of this introduction explains the strategy; the precise statements begin in Section 7.1.

What must be approximated. The prime 2 is unramified in L, so at each dyadic place v of L the unit group $\mathcal{O}_{L_v}^\times$ is the product of the Teichmüller representatives of $k_v^\times$, a cyclic group of odd order, and the group $1 + 2\mathcal{O}_{L_v}$ of principal units, which is a pro-two group. The odd-order factor concerns residues only, and Section 8 treats it. What remains is the group $U_L = \prod_{v|2}(1 + 2\mathcal{O}_{L_v})$ of dyadic principal units. We must show that the component of z_n in U_L lies in the closure of the components of the global units E_n. A congruence between z_n and a global unit modulo one fixed power of 2 is not enough: we need such a unit for every power of 2.

Why the obstruction is an element of a Galois group. For each m, the ray class group $\mathrm{Cl}_{2^m}(L)$ of modulus $2^m\mathcal{O}_L$ (Section 2.3) contains the group of residues modulo 2^m of elements prime to 2, modulo the residues of global units. An element b prime to 2 is congruent to a global unit modulo 2^m exactly when its ideal (b) is trivial in $\mathrm{Cl}_{2^m}(L)$. Letting m tend to infinity and keeping two-primary parts gives a profinite group X_L. Class field theory identifies X_L with the Galois group of the maximal abelian pro-two extension of L unramified outside 2 and split at the real places, and the class of an ideal with its Artin symbol. In this language (Lemma 7.4), the element z_n is approximated by global units in U_L exactly when the Artin symbol of $(z_n) = A'P^{-1}$ vanishes. Since $P = \sigma A'$, this symbol is the difference between the Artin symbol of A' and its image under σ. So the question is whether σ fixes the Artin symbol of A'.

Why a Selmer group of square classes appears. The group X_L is infinite, and we do not compute it. Instead we use the closed subgroup $\mathcal{W}$ generated by elements that are visibly fixed by σ: the image of the corresponding group X_F of F, and the Frobenius elements of the primes of L that ramify over F. On the quotient $X_L/\mathcal{W}$, the automorphism σ acts as -1. By Kummer theory, the characters of order two of X_L correspond to the quadratic extensions $L(\sqrt{b})$ unramified outside 2 and split at the real places, and hence to square classes $[b]$ of L. Those characters that vanish on $\mathcal{W}$ come exactly from the square classes $[x]$ of F that are totally positive, have even valuation outside the ramified and dyadic primes, and satisfy a local condition at each ramified prime (Lemma 7.5). A group of global square classes cut out by local conditions at every place is called a Selmer group.

Ours is the group $\Delta(F,\delta)$ described in Section 1.1 and defined in Definition 7.1, and modulo $[\delta]$, which becomes a square in L, it is Sel_δ. Thus Sel_δ is the dual of the largest quotient of $X_L/\mathcal{W}$ of exponent two.

Why Hilbert symbols at the dyadic places test it. Duality with Sel_δ sees only the largest quotient of exponent two. Nothing is lost if $X_L/\mathcal{W}$ itself has exponent two. The obstruction to this is a cyclic quotient of $X_L/\mathcal{W}$ of order four. Because σ acts on it by -1, and inertia at a prime of T supplies an involution that lifts σ, such a quotient gives a Galois extension of F with dihedral group of order eight that contains $L(\sqrt{x})$ for a nonzero class $[x]\in\mathrm{Sel}_\delta$. An elementary norm computation (Lemma 7.6) shows that then $x\delta$ is a norm from $F(\sqrt{x})$, so the Hilbert symbol $(x,-\delta)_v$ equals 1 at every place v of F. For classes of $\Delta(F,\delta)$ this symbol equals 1 at every non-dyadic place in any case, so only the dyadic places can detect the obstruction. The map λ_δ records the symbols $(x,-\delta)_v$ at the dyadic places, and condition (S) asks that it be injective. Under (S), therefore, $X_L/\mathcal{W}$ has no cyclic quotient of order four and has exponent two (Proposition 7.7).

Why exponent two suffices. Once $X_L/\mathcal{W}$ has exponent two, its elements are detected by the classes of Sel_δ. The Artin symbol of A' lies in $\mathcal{W}$ exactly when A' splits in $L(\sqrt{x})$ for every $[x]\in\mathrm{Sel}_\delta$, that is, when every class of Sel_δ is a square in the completion $F_A=L_{A'}$. Elements of $\mathcal{W}$ are fixed by σ, so the Artin symbol of $A'P^{-1}$ then vanishes (Corollary 7.8). The local-square condition at A holds at every scaled step. An explicit norm g_A from L has odd valuation at A and at no other prime outside the ramified and dyadic primes, and its quotient by $-\delta$ is a square at the dyadic places. Hilbert reciprocity, applied to x and this quotient, proves the condition for every class of $\Delta(F,\delta)$ (Lemma 7.10).

Organization. Section 7.1 defines $\Delta(F,\delta)$, Sel_δ, λ_δ and (S), states Theorem 7.2, and computes the dimension of Sel_δ. Section 7.2 identifies the obstruction to dyadic approximation with an Artin symbol. Section 7.3 introduces the subgroup $\mathcal{W}$ and identifies the dual of $X_L/(\mathcal{W}+2X_L)$ with Sel_δ. Section 7.4 shows that (S) excludes cyclic quotients of order four. Section 7.5 proves the local-square condition at A and Theorem 7.2.

7.1 The Selmer group and condition (S)

At a scaled step (Definition 4.8) write $F=F_{n-1}$ and $L=F_n=F(\sqrt{\delta})$, with $\delta=t^2-4y$ as in (4). The extension is totally real, 2 is absolutely unramified in L, and the two dyadic primes of F are inert in L. Let T be the set of *all* finite primes of F ramified in L. It contains the detecting primes of Definition 4.8(vi), so it is nonempty, and all its primes are odd. Let $\mathcal{D}$ be the set of dyadic primes of F; thus $|\mathcal{D}|=2$. We keep $|\mathcal{D}|$ in the dimension count below to show the role of the number of dyadic places. Apart from the restatement of (S) after Lemma 7.3, the arguments of this section use only that 2 is unramified in L, and not the inertness of the dyadic primes of F in L.

Definition 7.1 (Selmer group and Selmer map)**.** Let

$$\Delta(F,\delta)=\left\{[x]\in F^\times/F^{\times 2}:\ \begin{array}{l} x \text{ is totally positive,}\\ v_{\mathfrak{p}}(x)\equiv 0 \pmod 2 \quad (\mathfrak{p}\notin T\cup\mathcal{D}),\\ [x]_{\mathfrak{p}}\in\langle[\delta]_{\mathfrak{p}}\rangle \quad (\mathfrak{p}\in T)\end{array}\right\},\qquad \mathrm{Sel}_\delta=\Delta(F,\delta)/\langle[\delta]\rangle,$$

where $[x]_{\mathfrak{p}}$ is the local square class of x in $F_{\mathfrak{p}}^\times/F_{\mathfrak{p}}^{\times 2}$. The *Selmer map* is

$$\lambda_\delta:\mathrm{Sel}_\delta\longrightarrow(\mathbb{F}_2^{\mathcal{D}})^0,\qquad \lambda_\delta([x])=([x,-\delta]_v)_{v\in\mathcal{D}},\qquad (\mathbb{F}_2^{\mathcal{D}})^0=\{(x_v)\in\mathbb{F}_2^{\mathcal{D}}:\textstyle\sum_v x_v=0\},$$

where $[x,-\delta]_v\in\mathbb{F}_2$ is the additive Hilbert symbol of Section 2.1.

The local condition at $\mathfrak{p} \in T$ says that x is a square or δ times a square in $F_{\mathfrak{p}}$. Every prime of T enters this condition, including the primes of T that are not detecting primes.

The Hilbert symbols are those of Section 2.3: the local norm criterion and bimultiplicativity are [9, Chapter III, Theorem 4.4] at the finite places and elementary at the real places, and the global product formula is [9, Chapter V, Example 5.4]. The map λ_δ is $\mathbb{F}_2$-linear by bimultiplicativity, and it descends from $\Delta(F,\delta)$ to Sel_δ because $(\delta,-\delta)_v = 1$. Its values have coordinate sum zero by the product formula applied to x and $-\delta$: at each odd prime of T the symbol is trivial, since the local class of x is 1 or $[\delta]_{\mathfrak{p}}$ and $(\delta,-\delta)_{\mathfrak{p}} = 1$; at every other odd prime both x and $-\delta$ have even valuation, so the symbol is trivial (Section 2.3); and at the real places x is positive. Hence the product of the symbols over the dyadic places is 1.

The *Selmer condition* is

$$\ker \lambda_\delta = 0. \tag{S}$$

For a number field in which 2 is unramified, let pr_2 remove the odd-order Teichmüller factor from each dyadic unit and keep its component in the group of principal units. This component is a pro-two element; it includes torsion, for instance $\mathrm{pr}_2(-1) = -1$. For L this is the projection $\prod_{v|2} \mathcal{O}_{L_v}^\times \to U_L$, defined on every element of L that is a unit at the dyadic primes.

Theorem 7.2 (Dyadic approximation)**.** *Let F_n/F_{n-1} be a scaled step in the sense of Definition 4.8. If this step satisfies condition* (S)*, then*

$$\mathrm{pr}_2(z_n) \in \overline{\mathrm{pr}_2(E_n)} \subseteq \prod_{v|2} (1 + 2\mathcal{O}_{L_v}).$$

The proof is given at the end of Section 7.5. We first show that (S) is a finite test and that, when it holds, the dimension of Sel_δ is forced. For a finite set $\mathcal{S}$ of finite primes of F, let

$$\mathcal{V}_{\mathcal{S}}(F) = \{[x] \in F^\times/F^{\times 2} : v_{\mathfrak{p}}(x) \equiv 0 \pmod 2 \text{ for every prime } \mathfrak{p} \notin \mathcal{S}\},$$

so that $\mathcal{V}_\varnothing(F) = \mathcal{V}(F)$ is the group of square classes of virtual units (Section 2.3). Write $\mathrm{rk}_2 \mathrm{Cl}(F) = \dim_{\mathbb{F}_2} \mathrm{Cl}(F)/2\,\mathrm{Cl}(F)$, which equals $\dim_{\mathbb{F}_2} \mathrm{Cl}(F)[2]$ because $\mathrm{Cl}(F)$ is finite, and let $\langle [\mathfrak{p}] : \mathfrak{p} \in \mathcal{S}\rangle$ be the subspace of $\mathrm{Cl}(F)/2\,\mathrm{Cl}(F)$ spanned by the classes of the primes in $\mathcal{S}$.

Lemma 7.3 (The dimension of the Selmer group)**.** *For every finite set $\mathcal{S}$ of finite primes of F, the group $\mathcal{V}_{\mathcal{S}}(F)$ is finite and*

$$\dim_{\mathbb{F}_2} \mathcal{V}_{\mathcal{S}}(F) = [F:\mathbb{Q}] + \mathrm{rk}_2 \mathrm{Cl}(F) + |\mathcal{S}| - \dim_{\mathbb{F}_2} \langle [\mathfrak{p}] : \mathfrak{p} \in \mathcal{S}\rangle. \tag{23}$$

The group $\Delta(F,\delta)$ is finite, contains the nonzero class $[\delta]$, and satisfies

$$\dim \Delta(F,\delta) \geq |\mathcal{D}|, \qquad \dim \mathrm{Sel}_\delta \geq |\mathcal{D}| - 1. \tag{24}$$

If (S) *holds, then* $\dim \mathrm{Sel}_\delta = |\mathcal{D}| - 1$.

Proof. We compute $\mathcal{V}_{\mathcal{S}}(F)$ through the parity of valuations, and then count the conditions that cut out $\Delta(F,\delta)$.

Step 1: the formula (23). Consider the valuation-parity map $\mathcal{V}_{\mathcal{S}}(F) \to \mathbb{F}_2^{\mathcal{S}}$, $[x] \mapsto (v_{\mathfrak{p}}(x) \bmod 2)_{\mathfrak{p}\in\mathcal{S}}$. Its kernel consists of the classes with even valuation at every prime, that is, $\mathcal{V}(F)$. Since F is totally real, its torsion units are ± 1, and $\dim_{\mathbb{F}_2} \mathcal{V}(F) = [F:\mathbb{Q}] + \mathrm{rk}_2 \mathrm{Cl}(F)$ (Section 2.3); the unit contribution $[F:\mathbb{Q}]$ includes the torsion unit -1. The image of the parity map is the kernel of the map $\mathbb{F}_2^{\mathcal{S}} \to \mathrm{Cl}(F)/2\,\mathrm{Cl}(F)$ that sends $(e_{\mathfrak{p}})$ to the class of $\prod_{\mathfrak{p}\in\mathcal{S}} \mathfrak{p}^{e_{\mathfrak{p}}}$. Indeed, a parity vector lies in

that kernel exactly when the corresponding ideal $\prod_{\mathfrak{p}\in\mathcal{S}}\mathfrak{p}^{e_\mathfrak{p}}$ is a square times a principal ideal (x), and then $[x]\in\mathcal{V}_\mathcal{S}(F)$ has the given parity vector; conversely, if $[x]\in\mathcal{V}_\mathcal{S}(F)$, then (x) is the product of $\prod_{\mathfrak{p}\in\mathcal{S}}\mathfrak{p}^{e_\mathfrak{p}}$, with $e_\mathfrak{p}\equiv v_\mathfrak{p}(x)$, and a square. The image of $\mathbb{F}_2^\mathcal{S}\to\mathrm{Cl}(F)/2\,\mathrm{Cl}(F)$ is $\langle[\mathfrak{p}]:\mathfrak{p}\in\mathcal{S}\rangle$, so the kernel has dimension $|\mathcal{S}|-\dim\langle[\mathfrak{p}]:\mathfrak{p}\in\mathcal{S}\rangle$. Adding the dimensions of kernel and image proves both (23) and the finiteness of $\mathcal{V}_\mathcal{S}(F)$.

Step 2: the lower bounds (24). Take $\mathcal{S}=T\cup\mathcal{D}$. Then $\Delta(F,\delta)$ is the subgroup of $\mathcal{V}_\mathcal{S}(F)$ cut out by positivity and by the local conditions at the primes of T, so it is finite. Positivity imposes at most $[F:\mathbb{Q}]$ linear conditions, one for the sign at each real place. At $\mathfrak{q}\in T$ the group $F_\mathfrak{q}^\times/F_\mathfrak{q}^{\times2}$ has dimension two, because $\mathfrak{q}$ is odd, and the condition $[x]_\mathfrak{q}\in\langle[\delta]_\mathfrak{q}\rangle$ is at most one linear condition. Since $\dim_{\mathbb{F}_2}\langle[\mathfrak{p}]:\mathfrak{p}\in\mathcal{S}\rangle\le\mathrm{rk}_2\,\mathrm{Cl}(F)$, formula (23) gives

$$\dim\Delta(F,\delta)\ge[F:\mathbb{Q}]+|T|+|\mathcal{D}|-[F:\mathbb{Q}]-|T|=|\mathcal{D}|.$$

The class $[\delta]$ is nonzero because $L=F(\sqrt{\delta})$ is a quadratic field. It belongs to $\Delta(F,\delta)$: the element δ is totally positive because L is totally real, its valuation is even at every odd prime unramified in L, and the condition at T holds trivially. Hence $\dim\mathrm{Sel}_\delta=\dim\Delta(F,\delta)-1\ge|\mathcal{D}|-1$.

Step 3: the dimension under (S). The map λ_δ takes values in $(\mathbb{F}_2^\mathcal{D})^0$, which has dimension $|\mathcal{D}|-1$. If λ_δ is injective, then $\dim\mathrm{Sel}_\delta\le|\mathcal{D}|-1$, and with (24) equality holds. □

In the present setting $|\mathcal{D}|=2$, so Sel_δ is never zero and $(\mathbb{F}_2^\mathcal{D})^0$ is the line spanned by $(1,1)$. Condition (S) therefore says that Sel_δ has exactly two elements and that its nonzero class $[x]$ satisfies $\lambda_\delta([x])=(1,1)$, that is, $(x,-\delta)_v=-1$ at both dyadic places v. This is the form of (S) described in Section 1.1.

7.2 Unit approximation as an Artin symbol

Let $\mathrm{Cl}_{2^m}(F)$ be the ray class group of modulus $2^m\mathcal{O}_F$, with *no infinite part*, and define $\mathrm{Cl}_{2^m}(L)$ likewise (Section 2.3). The prime 2 is absolutely unramified in F and in L, so $1+2\mathcal{O}_{F_v}$ is the group of principal units at each dyadic place v of F, and likewise for L. Since the modulus has no infinite part, the principal ideals that are trivial in $\mathrm{Cl}_{2^m}(F)$ are those generated by elements congruent to 1 modulo 2^m at every dyadic place, with no positivity requirement. Put

$$U_F=\prod_{v|2}(1+2\mathcal{O}_{F_v}),\qquad U_F^{(m)}=\prod_{v|2}(1+2^m\mathcal{O}_{F_v}),$$
$$\overline{E}_{F,2}=\overline{\mathrm{pr}_2(\mathcal{O}_F^\times)}\subseteq U_F,\qquad X_F=\varprojlim_m\mathrm{Cl}_{2^m}(F)_{(2)},$$

and define U_L, $U_L^{(m)}$, $\overline{E}_{L,2}$ and X_L in the same way. Recall that the subscript (2) denotes the two-primary subgroup of a finite abelian group. The groups U_F and U_L are written multiplicatively; the ray class groups and their inverse limits are written additively.

Class field theory identifies X_F with the Galois group of the maximal abelian pro-two extension F_2^{ab}/F unramified outside 2 and split at every real place [9, Chapter V, Theorems 3.5–3.6, Corollary 3.7 and Example 3.9]; likewise $X_L=\mathrm{Gal}(L_2^{\mathrm{ab}}/L)$. The splitting at the real places reflects the absence of an infinite part in the moduli. For a fractional ideal I of F prime to 2, the classes of I in the groups $\mathrm{Cl}_{2^m}(F)_{(2)}$ are compatible; we write $\mathrm{Art}_F(I)\in X_F$ for the resulting element, which is the Artin symbol of I under this identification. We define Art_L likewise.

Lemma 7.4 (The ray obstruction to unit approximation)**.** *There is an exact sequence of compact groups*

$$0\longrightarrow U_F/\overline{E}_{F,2}\longrightarrow X_F\longrightarrow\mathrm{Cl}(F)_{(2)}\longrightarrow0.\tag{25}$$

If $b \in F^\times$ is a unit at every dyadic prime, then

$$\mathrm{Art}_F((b)) = 0 \quad \Longleftrightarrow \quad \mathrm{pr}_2(b) \in \overline{E}_{F,2}. \tag{26}$$

The same assertions hold with L in place of F.

Proof. We take two-primary parts of the ray class exact sequence for the modulus 2^m and pass to the limit over m. We give the proof for F; the proof for L is identical.

Step 1: a finite level. The ray class exact sequence [9, Chapter V, Theorem 1.7] for the modulus $2^m\mathcal{O}_F$ with no infinite part reads

$$0 \longrightarrow (\mathcal{O}_F/2^m\mathcal{O}_F)^\times/\operatorname{im}(\mathcal{O}_F^\times) \longrightarrow \mathrm{Cl}_{2^m}(F) \longrightarrow \mathrm{Cl}(F) \longrightarrow 0.$$

By the Chinese remainder theorem, $(\mathcal{O}_F/2^m\mathcal{O}_F)^\times = \prod_{v|2} \mathcal{O}_{F_v}^\times/(1+2^m\mathcal{O}_{F_v})$, and each factor is the product of the Teichmüller representatives of $k_v^\times$, of odd order, and the two-group $(1+2\mathcal{O}_{F_v})/(1+2^m\mathcal{O}_{F_v})$. Taking two-primary parts is exact on finite abelian groups. It removes the odd residue factors and keeps the whole group $U_F/U_F^{(m)}$. For a subgroup H of a finite abelian group G, the two-primary part of H is its image under the projection of G onto $G_{(2)}$, so the two-primary part of the image of $\mathcal{O}_F^\times$ is the image of $\mathrm{pr}_2(\mathcal{O}_F^\times)$. Since $U_F^{(m)}$ is open, $\mathrm{pr}_2(\mathcal{O}_F^\times)U_F^{(m)} = \overline{E}_{F,2}U_F^{(m)}$. We obtain the exact sequence

$$0 \longrightarrow U_F/(\overline{E}_{F,2}U_F^{(m)}) \longrightarrow \mathrm{Cl}_{2^m}(F)_{(2)} \longrightarrow \mathrm{Cl}(F)_{(2)} \longrightarrow 0.$$

Step 2: the limit. These sequences are compatible as m grows. Passing to inverse limits preserves exactness, since all the groups are finite. The groups $U_F^{(m)}$ form a basis of neighbourhoods of 1 in the compact group U_F, and $\overline{E}_{F,2}$ is closed, so $\varprojlim_m U_F/(\overline{E}_{F,2}U_F^{(m)}) = U_F/\overline{E}_{F,2}$. This proves (25).

Step 3: Artin symbols of principal ideals. Let $b \in F^\times$ be a unit at every dyadic prime. The ideal (b) is prime to 2, and the finite ray sequence sends the residue of b modulo 2^m to the class of (b) in $\mathrm{Cl}_{2^m}(F)$, up to an inverse if the reciprocal Artin convention is used, which does not affect vanishing. After projection to two-primary parts, this residue becomes the class of $\mathrm{pr}_2(b)$ in $U_F/(\overline{E}_{F,2}U_F^{(m)})$. Hence $\mathrm{Art}_F((b)) = 0$ exactly when $\mathrm{pr}_2(b)$ lies in $\overline{E}_{F,2}U_F^{(m)}$ for every m, that is, in $\overline{E}_{F,2}$. This proves (26).

Because the modulus has no infinite part, the subgroup in Step 1 is the closure of all global units, not only of the totally positive ones. □

7.3 The action of σ and the Selmer group as a dual

Let $X = X_L$ and let σ be the nontrivial automorphism of L/F. The field L_2^{ab} is defined in terms of L alone, so it is stable under every automorphism of an algebraic closure that fixes F, and L_2^{ab}/F is Galois. Choose an extension $\tilde{\sigma}$ of σ to L_2^{ab}. Then σ acts on X by $g \mapsto \tilde{\sigma}g\tilde{\sigma}^{-1}$. Since X is abelian, this action does not depend on the choice of $\tilde{\sigma}$. We write it additively as $g \mapsto \sigma(g)$. The Artin symbols are compatible with it: $\mathrm{Art}_L(\sigma I) = \sigma(\mathrm{Art}_L(I))$ for every fractional ideal I of L prime to 2. For a prime ideal this is the relation between the Frobenius elements of conjugate primes in the finite subextensions of L_2^{ab}/L that are Galois over F (the proof of [8, 8.14] applies); such subextensions exhaust L_2^{ab}, since a finite subextension M lies in its compositum with $\tilde{\sigma}M$, which is Galois over F. The general case follows by multiplicativity.

Extension and norm of ideals induce continuous homomorphisms

$$\mathrm{ext} : X_F \longrightarrow X, \qquad \nu : X \longrightarrow X_F.$$

They are defined at the finite ray levels, since congruence to 1 modulo 2^m is preserved by extension and by the norm. They satisfy

$$\operatorname{ext}\nu = 1 + \sigma, \qquad \sigma \operatorname{ext} = \operatorname{ext}. \tag{27}$$

The second identity holds because an ideal extended from F is stable under σ. The first follows from ideal factorization: for a prime $\mathfrak{P}$ of L, the ideal $N_{L/F}(\mathfrak{P})\mathcal{O}_L$ equals $\mathfrak{P} \cdot \sigma\mathfrak{P}$, whether the prime of F below $\mathfrak{P}$ splits, is inert or ramifies in L.

For each $\mathfrak{q} \in T$, let $\mathfrak{Q}$ be the unique prime of L above $\mathfrak{q}$. Define the closed subgroup

$$\mathcal{W} = \overline{\langle \operatorname{ext}(X_F), \operatorname{Art}_L(\mathfrak{Q}) : \mathfrak{q} \in T \rangle} \subseteq X. \tag{28}$$

Every generator of $\mathcal{W}$ is fixed by σ: the elements of $\operatorname{ext}(X_F)$ by (27), and the elements $\operatorname{Art}_L(\mathfrak{Q})$ because $\sigma\mathfrak{Q} = \mathfrak{Q}$. For every $g \in X$, the element $g + \sigma(g) = \operatorname{ext}(\nu(g))$ lies in $\mathcal{W}$. Thus

$$\mathcal{W} \subseteq X^\sigma, \qquad \sigma = -1 \text{ on } X/\mathcal{W}. \tag{29}$$

The next lemma identifies the characters of order two of $X/\mathcal{W}$. A continuous character $\psi : X \to \mathbb{F}_2$ corresponds to a quadratic extension of L inside L_2^{ab}, or to L itself if $\psi = 0$, and by Kummer theory (Section 2.3) this extension is $L(\sqrt{b})$ for a square class $[b]$ of L. We write ψ_b for the character, so that $\psi_b(g) = 0$ exactly when g fixes $\sqrt{b}$. In particular, for a prime $\mathfrak{P}$ of L not above 2 at which $L(\sqrt{b})/L$ is unramified, that is, at which b has even valuation (Section 2.3), the value $\psi_b(\operatorname{Art}_L(\mathfrak{P}))$ is 0 exactly when $\mathfrak{P}$ splits in $L(\sqrt{b})$, that is, when b is a square in $L_{\mathfrak{P}}$.

Lemma 7.5 (The quadratic characters of the quotient)**.** *The Kummer pairing identifies*

$$\operatorname{Hom}_{\mathrm{cts}}(X/(\mathcal{W} + 2X), \mathbb{F}_2) \simeq \Delta(F, \delta)/\langle [\delta] \rangle = \operatorname{Sel}_\delta. \tag{30}$$

A class $[x] \in \operatorname{Sel}_\delta$ pairs with a prime of L through the quadratic extension $L(\sqrt{x})/L$.

Proof. We describe all quadratic characters of X by square classes of L, and then translate vanishing on each generator of $\mathcal{W}$ into a condition on the square class. A character with values in $\mathbb{F}_2$ vanishes on $2X$ automatically, so the characters of $X/(\mathcal{W} + 2X)$ are the characters of X that vanish on $\mathcal{W}$, that is, on its generators.

Step 1: characters of X. By Kummer theory [9, Chapter VII, Appendix A, Theorem A.3], the quadratic characters of X correspond to the square classes $[b]$ of L such that $L(\sqrt{b})$ is unramified outside 2 and split at the real places. These are the classes of elements b that are positive at all real places and have even valuation at all odd finite primes (Section 2.3).

Step 2: the transfer identity. For such b, the norm $N_{L/F}b$ is totally positive and has even valuation at every odd prime of F, so it defines a character $\psi_{N_{L/F}b}$ of X_F. We show

$$\psi_b \circ \operatorname{ext} = \psi_{N_{L/F}b}. \tag{31}$$

Exclude the finitely many primes of F that lie above 2, ramify in L/F, or lie below a prime dividing b. At every other prime $\mathfrak{p}$ of F, the value of $\psi_b \circ \operatorname{ext}$ at $\operatorname{Art}_F(\mathfrak{p})$ is the sum of the quadratic residue characters of b at the primes of L above $\mathfrak{p}$. If $\mathfrak{p}$ splits as $\mathfrak{P} \cdot \sigma\mathfrak{P}$, both residue fields equal that of $\mathfrak{p}$, and the residue of $N_{L/F}b$ at $\mathfrak{p}$ is the product of the residues of b at $\mathfrak{P}$ and at $\sigma\mathfrak{P}$; multiplicativity of the quadratic character gives the character of $N_{L/F}b$ at $\mathfrak{p}$. If $\mathfrak{p}$ is inert, with residue field $\mathbb{F}_q$ and $\mathfrak{P} = \mathfrak{p}\mathcal{O}_L$ with residue field $\mathbb{F}_{q^2}$, then the residue of $N_{L/F}b$ is the norm of the residue $\bar{b}$, and the quadratic character of an element of $\mathbb{F}_{q^2}$ equals that of its norm to $\mathbb{F}_q$:

$$(N_{\mathbb{F}_{q^2}/\mathbb{F}_q}\bar{b})^{(q-1)/2} = \bar{b}^{(q^2-1)/2}.$$

Thus the two characters in (31) agree on the Artin symbols of all but finitely many primes. Both factor through a common finite quotient of X_F, and by the Chebotarev density theorem [9, Chapter VIII, Theorem 7.4] every element of a finite quotient is the Frobenius element of infinitely many primes. This proves (31).

Step 3: vanishing on $\mathrm{ext}(X_F)$. By (31), the character ψ_b vanishes on $\mathrm{ext}(X_F)$ exactly when $N_{L/F}b = x'^2$ for some $x' \in F^\times$. In that case b/x' has norm one, and Hilbert's Theorem 90 writes it as $\xi/\sigma(\xi)$ with $\xi \in L^\times$. This instance can be checked directly: for an element $\xi' \neq -1$ of norm one, take $\xi = 1 + \xi'$, and for $\xi' = -1$ take $\xi = \sqrt{\delta}$. It follows that

$$b = \frac{x'}{N_{L/F}\xi}\,\xi^2,$$

so the square class of b comes from $x = x'/N_{L/F}\xi \in F^\times$. Conversely, a class coming from $x \in F^\times$ has square norm x^2, so its character vanishes on $\mathrm{ext}(X_F)$.

Step 4: the local conditions. Let $b = x\xi^2$ with $x \in F^\times$. Since L is totally real, b is positive at all real places of L exactly when x is totally positive. At an odd prime of F outside T, the ramification index in L is one, so the condition that b have even valuation at the primes of L above it says that the valuation of x is even. At $\mathfrak{q} \in T$, the ramification index is two, so every element of F has even valuation in $L_\mathfrak{Q}$. Hence $L(\sqrt{x})$ is unramified or trivial at $\mathfrak{Q}$, and its character vanishes on $\mathrm{Art}_L(\mathfrak{Q})$ exactly when x is a square in $L_\mathfrak{Q} = F_\mathfrak{q}(\sqrt{\delta})$. The kernel of

$$F_\mathfrak{q}^\times/F_\mathfrak{q}^{\times 2} \longrightarrow L_\mathfrak{Q}^\times/L_\mathfrak{Q}^{\times 2}$$

is $\langle[\delta]_\mathfrak{q}\rangle$. Thus the Frobenius conditions at the primes $\mathfrak{Q}$ are exactly the local conditions at T in the definition of $\Delta(F,\delta)$. No condition arises at the dyadic primes, since L_2^{ab}/L may ramify there.

Step 5: the kernel of restriction. By Steps 3 and 4, the characters of X vanishing on $\mathcal{W}$ are the images in $L^\times/L^{\times 2}$ of the classes of $\Delta(F,\delta)$. The kernel of the restriction map $F^\times/F^{\times 2} \to L^\times/L^{\times 2}$ is $\langle[\delta]\rangle$. This proves (30), and the description of the pairing follows from the remark before the lemma. □

7.4 Excluding a cyclic quotient of order four

Lemma 7.5 describes only the largest quotient of $X/\mathcal{W}$ of exponent two. The next elementary norm calculation is the step that upgrades this quadratic information to a statement about the whole pro-two group $X/\mathcal{W}$. It is a form of the classical criterion for embedding a biquadratic field in a dihedral extension of degree eight; we prove the direction we need.

Lemma 7.6 (A dihedral norm)**.** *Suppose that $\widetilde{L}/F$ is a Galois extension with group*

$$\mathrm{Dih}_4 = \langle r, s : r^4 = s^2 = 1,\ srs = r^{-1}\rangle.$$

Assume that the quadratic subfield fixed by the rotation r is $F(\sqrt{\delta})$ and that the biquadratic subfield, fixed by r^2, is $F(\sqrt{\delta},\sqrt{x})$. Then $x\delta$ is a norm from $F(\sqrt{x})$, and

$$(x,-\delta)_v = 1 \quad \textit{at every place } v \textit{ of } F.$$

Proof. We produce an element of $F(\sqrt{x})$ whose norm is $x\delta$ times a square in F. The subgroups of order four of Dih_4 other than $\langle r\rangle$ are $\langle r^2, s\rangle$ and $\langle r^2, rs\rangle$; their fixed fields are $F(\sqrt{x})$ and $F(\sqrt{x\delta})$ in some order, and replacing s by rs exchanges them. Choose the reflection s so that $\widetilde{L}^{\langle r^2, s\rangle} = F(\sqrt{x})$. The element r^2 is central and acts nontrivially on the quadratic extension $\widetilde{L}^{\langle s\rangle}/F(\sqrt{x})$, so we may

choose $0 \neq \xi \in \widetilde{L}^{\langle s \rangle}$ with $r^2\xi = -\xi$. Then ξ^2 is fixed by r^2 and s, so $\xi^2 \in F(\sqrt{x})$. Put $\xi' = \xi\, r(\xi)$. Using $r^2\xi = -\xi$ and $sr = r^{-1}s$,

$$r\xi' = -\xi', \qquad s\xi' = \xi\, r^{-1}(\xi) = -\xi'.$$

Thus ξ' is fixed by r^2 and by rs, so it lies in $F(\sqrt{x\delta})$, and it lies in the nontrivial eigenspace of the automorphism of $F(\sqrt{x\delta})/F$. Write $\xi' = b\sqrt{x\delta}$ with $b \in F^\times$. Since r restricts to the nontrivial automorphism of $F(\sqrt{x})/F$,

$$N_{F(\sqrt{x})/F}(\xi^2) = \xi^2 r(\xi^2) = \xi'^2 = b^2 x\delta.$$

Dividing ξ^2 by b gives an element of $F(\sqrt{x})$ with norm $x\delta$. A global norm is a local norm at every place, so the local norm criterion gives $(x, x\delta)_v = 1$: at the finite places by [9, Chapter III, Theorem 4.4], and at a real place by Section 2.3, since a norm from $\mathbb{R}(\sqrt{x})$ is positive when $x < 0$. Finally $(x, x\delta)_v = (x, -\delta)_v$, since $x\delta = (-x)(-\delta)$ and $(x, -x)_v = 1$. □

Proposition 7.7 (Exponent two under (S))**.** *If* (S) *holds, then* $2X \subseteq \mathcal{W}$*. Consequently the field fixed by* $\mathcal{W}$ *is the compositum of the fields* $L(\sqrt{x})$ *for* $[x] \in \mathrm{Sel}_\delta$*.*

Proof. We show that a failure of $2X \subseteq \mathcal{W}$ produces a dihedral extension of F of degree eight, which by Lemma 7.6 yields a nonzero element of $\ker \lambda_\delta$.

Step 1: a cyclic quotient of order four. Suppose that $X/\mathcal{W}$ is not killed by two. Then some finite quotient of the pro-two group $X/\mathcal{W}$ has exponent at least four, and hence has a cyclic quotient of order four. Let Y be the kernel of the resulting surjection $X \to \mathbb{Z}/4\mathbb{Z}$; it is an open subgroup containing $\mathcal{W}$. By (29), $\mathcal{W} \subseteq X^\sigma$, so $\sigma(\mathcal{W}) = \mathcal{W}$, and $\sigma = -1$ on $X/\mathcal{W}$ maps $Y/\mathcal{W}$ to $-Y/\mathcal{W} = Y/\mathcal{W}$. Hence Y is stable under σ. The fixed field $\widetilde{L}$ of Y is therefore Galois over F, and $\widetilde{L}/L$ is cyclic of degree four.

Step 2: the group is dihedral. Take $\mathfrak{q} \in T$; the set T is nonempty. The extension $\widetilde{L}/L$ is unramified above $\mathfrak{q}$, because $\widetilde{L}$ is contained in L_2^{ab}, which is unramified outside 2, and $\mathfrak{q}$ is odd, since the dyadic primes are unramified in L, whereas L/F has ramification index two there. Thus an inertia group in $\mathrm{Gal}(\widetilde{L}/F)$ above $\mathfrak{q}$ has order two and maps isomorphically onto $\mathrm{Gal}(L/F)$. It supplies an involution outside $\mathrm{Gal}(\widetilde{L}/L)$. This involution acts on the cyclic group $\mathrm{Gal}(\widetilde{L}/L) \simeq X/Y$ as σ does, that is, by inversion. Hence $\mathrm{Gal}(\widetilde{L}/F) \simeq \mathrm{Dih}_4$, with rotation subgroup $\mathrm{Gal}(\widetilde{L}/L)$, whose fixed field is $L = F(\sqrt{\delta})$.

Step 3: a contradiction with (S). The nonzero quadratic character of the cyclic quotient X/Y vanishes on $\mathcal{W} + 2X$. By Lemma 7.5, its field is $L(\sqrt{x})$ for some nonzero $[x] \in \mathrm{Sel}_\delta$. This field is the unique quadratic extension of L inside $\widetilde{L}$, so the biquadratic subfield of $\widetilde{L}/F$ is $F(\sqrt{\delta}, \sqrt{x})$. Lemma 7.6 gives $(x, -\delta)_v = 1$ at every place, so $\lambda_\delta([x]) = 0$, contradicting (S).

Step 4: the fixed field. Hence $2X \subseteq \mathcal{W}$. Now $X/\mathcal{W}$ is an elementary abelian two-group, and its group of continuous characters is Sel_δ by Lemma 7.5. The corresponding quadratic fields $L(\sqrt{x})$ therefore generate exactly the fixed field of $\mathcal{W}$. □

Corollary 7.8 (Obstruction at one split prime)**.** *Suppose that* (S) *holds. Let* A *be an odd prime of* F *that splits in* L *as* $A\mathcal{O}_L = A'P$*, and suppose that* $A'P^{-1} = (z)$ *for some* $z \in L^\times$*. If every class in* Sel_δ *is a square in* F_A*, then* $\mathrm{pr}_2(z) \in \overline{E}_{L,2}$*.*

Proof. We show that $\mathrm{Art}_L(A')$ lies in $\mathcal{W}$, hence is fixed by σ, so that the Artin symbol of (z) vanishes. Since A splits, $L_{A'} = F_A$, and the local-square assumption makes A' split in every quadratic field $L(\sqrt{x})$ with $[x] \in \mathrm{Sel}_\delta$. By Proposition 7.7, $X/\mathcal{W}$ is an elementary abelian two-group whose characters are these quadratic fields. These characters all vanish on $\mathrm{Art}_L(A')$, and the

continuous characters of a profinite elementary abelian two-group separate its points, so $\mathrm{Art}_L(A') \in \mathcal{W}$. This element is fixed by σ, since $\mathcal{W} \subseteq X^\sigma$. As $\sigma A' = P$,

$$\mathrm{Art}_L((z)) = \mathrm{Art}_L(A') - \mathrm{Art}_L(P) = \mathrm{Art}_L(A') - \sigma(\mathrm{Art}_L(A')) = 0.$$

The element z is a unit at the dyadic primes, because its ideal is prime to 2. Lemma 7.4 for L now gives $\mathrm{pr}_2(z) \in \overline{E}_{L,2}$. □

7.5 The local-square condition at A

It remains to verify the hypothesis of Corollary 7.8 at the prime $A = A_{n-1}$ of a scaled step. Return to the elements $\alpha = y_n$ and $\beta = \beta_n$ of a scaled step. The identities below use only the integral formulas (4) over F, with u a global unit of F different from ± 1, $v_A(a) \geq 2$, $v_B(a) = 0$, $u \equiv 1 \pmod{A}$, the splitting of A and B in L, and the dyadic conditions of Definition 4.8(i)–(iii). Put

$$g_+ = a^2 - y(u+1)^2, \qquad g_- = a^2 - y(u-1)^2, \qquad g_A = -ug_+.$$

Lemma 7.9 (An explicit norm with one odd valuation)**.** *At a scaled step, g_+ and g_- are nonzero elements of $\mathcal{O}_F$, and*

$$a^2u^2\delta = g_+g_-, \qquad N_{L/F}(\alpha+\beta) = -g_+/u, \qquad g_A = N_{L/F}\big(u(\alpha+\beta)\big). \tag{32}$$

Moreover $v_A(g_A) = 1$, $v_B(g_A) = 0$, and $v_\mathfrak{p}(g_A)$ is even at every odd prime $\mathfrak{p} \notin T \cup \{A\}$.

Proof. Neither g_+ nor g_- vanishes, since otherwise y would be a square in F, contradicting its odd valuations at A and B. The identities (32) follow by direct calculation from (4).

At A, the element $u+1$ is a unit, since $u \equiv 1 \pmod{A}$ and A is odd, and $v_A(a) \geq 2$, $v_A(y) = 1$; so $v_A(g_+) = 1$. At B, the element a is a unit and y has valuation one, so $v_B(g_+) = 0$. At every other odd prime, $g_+ - g_- = -4yu$ is a unit, so at least one of the integral factors g_+, g_- is a unit. At such a prime outside T, the valuation of δ is even, and (32) then makes both factor valuations even, including at primes dividing a. Since u is a unit, $g_A = -ug_+$ has the same valuations as g_+. Thus the only odd valuation of g_A at a prime outside $T \cup \mathcal{D}$ is at A. □

Lemma 7.10 (The local-square condition at A)**.** *At a scaled step, every class of $\Delta(F,\delta)$ is a square in F_A.*

Proof. We apply Hilbert reciprocity once, to x and the element $t_A = g_A/(-\delta)$, where g_A is the norm of Lemma 7.9; the element t_A is a square at the dyadic places.

Step 1: the dyadic places. At a dyadic place v, the elements a and y are units and $u \equiv 1 \pmod{8}$. The identity

$$\frac{g_A}{-\delta} = \frac{u^3}{1 - y((u-1)/a)^2} \in 1 + 8\mathcal{O}_{F_v} \subseteq F_v^{\times 2} \tag{33}$$

follows from (32) by cancelling g_+. The membership in $1 + 8\mathcal{O}_{F_v}$ holds because $(u-1)/a \in 8\mathcal{O}_{F_v}$ and $u^3 \in 1 + 8\mathcal{O}_{F_v}$. The final inclusion follows from Newton's lemma [8, Theorem 7.32], since 2 is unramified in F. Hence $[x, t_A]_v = 0$ for every $x \in F^\times$ and every $v \in \mathcal{D}$.

Step 2: reciprocity. Let $[x] \in \Delta(F,\delta)$. Since A is odd and splits in L, it lies outside $T \cup \mathcal{D}$, so $v_A(x)$ is even. Write $x = \pi^{2k}w$ with π a uniformizer of F_A and w a unit, so that the residue character χ_A of Section 2.3 satisfies $\chi_A(x) = 0$ exactly when the residue of w is a square. We compute the additive Hilbert symbols $[x, t_A]_v$ at the places $v \neq A$ outside $\mathcal{D}$. At a prime $\mathfrak{q}$ of T, the local class of x is 1 or $[\delta]_\mathfrak{q}$. Here g_A is a norm from L, hence a local norm from $L_\mathfrak{Q} = F_\mathfrak{q}(\sqrt{\delta})$,

so $[\delta, g_A]_{\mathfrak{q}} = 0$; also $[\delta, -\delta]_{\mathfrak{q}} = 0$. Hence $[x, t_A]_{\mathfrak{q}} = 0$. At every other odd prime $\mathfrak{q} \neq A$, the element x has even valuation by Definition 7.1, g_A by Lemma 7.9, and δ because $\mathfrak{q}$ is odd and unramified in $L = F(\sqrt{\delta})$; so x and t_A have even valuation, and the symbol is zero (Section 2.3). At the real places the symbol is zero because x is positive. At A, the element δ is a square in F_A because A splits in L, so $[x, t_A]_A = [x, -g_A]_A = [w, -g_A]_A$, and this is $\chi_A(x)$ because $v_A(-g_A) = 1$ (Section 2.3). Together with Step 1, Hilbert reciprocity [9, Chapter V, Example 5.4] gives $\chi_A(x) = 0$.

Step 3: conclusion. Thus the unit w has a square residue, so it is a square in F_A by Newton's lemma [8, Theorem 7.32], because A is odd. Since $v_A(x)$ is even, x is a square in F_A. □

The local analysis of Step 2, applied to x and g_A instead of t_A, shows that every symbol $[x, g_A]_v$ with $v \neq A$ outside $\mathcal{D}$ vanishes; and $[x, g_A]_A = [w, g_A]_A = \chi_A(x)$ because $v_A(g_A) = 1$. Reciprocity therefore gives

$$\chi_A(x) = \sum_{v \in \mathcal{D}} [x, g_A]_v \qquad ([x] \in \Delta(F, \delta)). \tag{34}$$

By (33), the right side equals $\sum_{v \in \mathcal{D}} [x, -\delta]_v$, which vanishes by the sum-zero property of λ_δ; this is a second route to Lemma 7.10.

The assertion of Lemma 7.10 is well defined on Sel_δ, because δ is a square in F_A at the split prime A. The proof uses every prime of T in the definition of $\Delta(F, \delta)$, but it does not use the injectivity of λ_δ.

Proof of Theorem 7.2. The element z_n satisfies $(z_n) = A_n P_n^{-1} = A' P^{-1}$, and $A = A_{n-1}$ is odd and splits in L. By Lemma 7.10, every class of Sel_δ is a square in F_A; this is the remaining hypothesis of Corollary 7.8. Under (S), that corollary gives $\mathrm{pr}_2(z_n) \in \overline{E}_{L,2} = \overline{\mathrm{pr}_2(E_n)}$, in the whole product of the dyadic principal-unit groups, as asserted. □

Remark 7.11. Lemma 7.10 alone shows only that the image of $\mathrm{Art}_L(A')$ vanishes in the quotient $X/(\mathcal{W} + 2X)$ of exponent two. Condition (S) supplies the inclusion $2X \subseteq \mathcal{W}$, and this is what allows passage to the closure of the global units in the whole group U_L.

We do not prove the corresponding statement that every class of $\Delta(F, \delta)$ is a square at B, and it is not needed: the divisor $A_n P_n^{-1}$ of z_n involves only the primes A_n and P_n, as used in Theorem 6.1. Let χ_B be the residue character at B (Section 2.3). The element $y g_A$ is a norm from L, with $v_A(y g_A) = 2$ and $v_B(y g_A) = 1$ since $(y) = AB$, and applying reciprocity to it as in the derivation of (34) gives, on the same group $\Delta(F, \delta)$,

$$\chi_B(x) = \sum_{v \in \mathcal{D}} [x, y]_v.$$

8 The odd ramified places and the proof of Theorem B

This section completes the proof of Theorem B. By Theorem 6.1 (unit approximation criterion), it suffices to verify condition (JT_A) of Section 6, and we verify it with $N = n$ at every sufficiently late step: the image $\jmath_n(z_n)$ of z_n in the compact group $\mathcal{U}_n$ of (20) must lie in the closure of the image of the unit group E_n. The group $\mathcal{U}_n$ is a finite product of pro-ℓ groups for finitely many primes ℓ, and a closed subgroup of such a product is the product of its ℓ-primary parts (Step 1 of the proof of Proposition 8.9). The approximation problem therefore splits into one problem for each prime ℓ. Theorem 7.2 (dyadic approximation) solves the 2-primary problem under the Selmer condition (S). This section formulates the odd unit condition (O), shows that it solves the p-primary problem for every odd prime p, and combines the two. We first explain the ideas.

The odd places give a problem in $\mathbb{Z}_p$-modules. The odd-primary part of $\mathcal{U}_n$ consists of the quotients $C_v = \mathcal{O}_{(F_n)_v}^{\times}/W_{n,v}^{\times}$ at the places v of F_n of odd residue characteristic p with ramification index $e(v/p) = 2$, together with the multiplicative groups of the residue fields at the dyadic places, which have odd order. The quotients C_v enter because of Proposition 5.12: from the level b_p on, the completed orders that arise in Section 5 have the form $W_{n,v} + \pi_v p^s W_{n,v}$ at such a place, and the proof of Lemma 6.4 transports the resulting conditions to earlier levels. These orders are $W_{n,v}$-modules, so whether a unit of $\mathcal{O}_{(F_n)_v}$ lies in one of them depends only on its class in C_v. Each C_v is a compact abelian pro-p group, a quotient of $1 + \pi_v \mathcal{O}_{(F_n)_v}$ by a closed subgroup (see the paragraph preceding (40)), hence a $\mathbb{Z}_p$-module. So approximating z_n by global units at the places above p means that one vector, the image of z_n in the product $M_{n,p}$ of the groups C_v, lies in the closed $\mathbb{Z}_p$-submodule generated by the images of units. Since z_n has relative norm one, the relative units suffice; their images lie in the submodule $M_{n,p}^{-}$ on which σ_n acts by -1 (Lemma 8.4). Of the structure of the C_v, this reduction and the odd-primary part of the proof of Theorem B use only that they are compact pro-p groups, together with the closure identity (42); they do not use the logarithm. The p-adic logarithm, which for $p \geq 5$ identifies C_v with the additive group of $W_{n,v}$, a free $\mathbb{Z}_p$-module of rank f_v (Lemma 8.1), linearizes the problem further; it is used for the rank computation of Lemma 8.2 and the finite test below.

The dyadic residue restriction. At a dyadic place v of F_n the unit group of the completion is the product of a cyclic group isomorphic to the multiplicative group $k_v^{\times}$ of the residue field, which has odd order, and the pro-2 group $1 + 2\mathcal{O}_{(F_n)_v}$. For an odd prime p, the p-primary part $\mathcal{T}_{n,p}$ of these residue groups lies in the same p-primary factor of $\mathcal{U}_n$ as $M_{n,p}$. The p-primary approximation problem is therefore posed in $\mathcal{T}_{n,p} \times M_{n,p}$. The residue of z_n at both dyadic primes is 1, so its component in $\mathcal{T}_{n,p}$ is zero, and only units whose p-primary dyadic residue vanishes can approximate it. Condition (O) therefore uses the span of the images of those relative units. This restriction is vacuous unless p divides $2^{2^{n-1}} + 1$ (43).

Nakayama's lemma gives a finite sufficient test. The module $M_{n,p}^{-}$ is free of finite rank over $\mathbb{Z}_p$, and the span in (O) is a closed submodule of it. A closed submodule of a finitely generated $\mathbb{Z}_p$-module is the whole module as soon as it is the whole module modulo p. Reduction modulo p is computed from the first two coefficients of a unit in a π_v-adic expansion (36). Hence finitely many relative units whose reductions span an $\mathbb{F}_p$-vector space of the right dimension prove (O) at p (Proposition 8.5).

Why $p = 3$ does not occur. The logarithm argument requires $v_p(\pi_v) > 1/(p-1)$ for a uniformizer π_v of a ramified quadratic extension of an unramified extension of $\mathbb{Q}_p$. Since $v_p(\pi_v) = 1/2$, this means $p \geq 5$. For $p = 3$ the conclusion of Lemma 8.1 can fail. The field $\mathbb{Q}_3(\sqrt{-3})$ is a ramified quadratic extension of $\mathbb{Q}_3$ with coefficient ring $\mathbb{Z}_3$, and it contains a primitive cube root of unity ζ_3. Since $\zeta_3 \notin \mathbb{Z}_3^{\times}$, the class of ζ_3 in $\mathcal{O}_{\mathbb{Q}_3(\sqrt{-3})}^{\times}/\mathbb{Z}_3^{\times}$ has order three, so this quotient is not a free $\mathbb{Z}_3$-module. It is still pro-3, so this failure concerns the logarithm, the ranks and the finite test, and not the group-theoretic step of Lemma 8.4. The description of the completed orders (Proposition 5.12) and the construction assume (T1) throughout, and we do not consider ramification at 3. Hypothesis (T1) keeps 3, 7 and 47 unramified in every F_n. Hence every odd prime p below a place of F_n of ramification index two satisfies $p \geq 5$.

Standing notation. Throughout this section we fix $n \geq 1$ and a scaled step $L = F_n$ over $F = F_{n-1}$ (Definition 4.8) in a tower satisfying (T1)–(T6), and we write $\sigma = \sigma_n$. The letter p

denotes an odd rational prime. By (T1), absolute ramification indices at odd primes are at most two, so the places v of L above p with $e(v/p) = 2$ are exactly those at which $L_v/\mathbb{Q}_p$ is ramified. By the preceding paragraph, $p \geq 5$ whenever such a place exists.

Section 8.1 treats a single place. Section 8.2 collects the places above p into the module $M_{n,p}$, and Section 8.3 defines (O). Section 8.4 shows that (O) is exactly the p-primary part of the approximation problem, Section 8.5 gives the finite test, and Section 8.6 proves Theorem B.

8.1 The local unit quotient at an odd ramified place

Let v be a place of L above p with $e(v/p) = 2$. Write W_v for the coefficient ring of L_v (Section 2.3), k_v for its residue field, which is also the residue field of L_v, and $f_v = [k_v : \mathbb{F}_p]$. Since $e(v/p) = 2$, the field L_v is a ramified quadratic extension of $\mathrm{Frac}(W_v)$. As in Section 2.3, choose a square uniformizer π_v, so that

$$\mathcal{O}_{L_v} = W_v \oplus \pi_v W_v, \qquad \pi_v^2 \in pW_v^\times.$$

Let τ_v be the nontrivial automorphism of L_v over $\mathrm{Frac}(W_v)$. Since $\pi_v^2 \in \mathrm{Frac}(W_v)$ and $\pi_v \notin \mathrm{Frac}(W_v)$, the automorphism τ_v negates π_v. This local involution need not be the restriction of σ: the automorphism σ can exchange v with another place, or fix v and act on L_v as the nontrivial automorphism of an unramified quadratic extension.

The next lemma linearizes C_v by the logarithm, and it describes C_v modulo p through the first two coefficients of a unit. The second description is what makes the test of Section 8.5 finite.

Lemma 8.1 (Local unit quotients at odd ramified places)**.** *For $p \geq 5$ and $e(v/p) = 2$, the quotient $C_v = \mathcal{O}_{L_v}^\times / W_v^\times$ is a free $\mathbb{Z}_p$-module of rank f_v. There is an isomorphism*

$$\mathcal{L}_v : C_v \longrightarrow W_v, \qquad \mathcal{L}_v(x) = \frac{\log(x/\tau_v x)}{2\pi_v}. \tag{35}$$

For a unit x, write $x \equiv x_0 + x_1\pi_v \pmod{\pi_v^2}$ with $x_i \in W_v$. The coefficients are unique modulo pW_v, and x_0 is a unit. The homomorphism

$$\varphi_v(x) = \overline{x_1/x_0} = \frac{1 - x^{p^{f_v}-1}}{\pi_v} \bmod \pi_v \tag{36}$$

identifies C_v/pC_v with the additive group of k_v. Moreover $\mathcal{L}_v(x) \bmod p = \varphi_v(x)$, and τ_v acts as -1 on C_v.

Proof. We remove the Teichmüller factor, apply the logarithm, and read off the π_v-coefficient; the statements modulo p then follow from the first two terms of the expansion.

Step 1: reduction to principal units. The Teichmüller representatives of L_v lie in W_v (Section 2.3). Every unit of $\mathcal{O}_{L_v}$ is the product of a Teichmüller representative and an element of $1 + \pi_v\mathcal{O}_{L_v}$, and every unit of W_v is the product of a Teichmüller representative and an element of $1 + pW_v$. Hence $C_v = (1 + \pi_v\mathcal{O}_{L_v})/(1 + pW_v)$.

Step 2: the logarithm. The logarithm and exponential series are inverse homomorphisms between $1 + \pi_v\mathcal{O}_{L_v}$ and $\pi_v\mathcal{O}_{L_v}$: their convergence and formal composition identities apply because $v_p(\pi_v) = 1/2 > 1/(p-1)$ (Section 2.3). In particular the terms tend to zero, and the nonleading terms have strictly larger valuation. The two series restrict to an isomorphism $1 + pW_v \simeq pW_v$. Since $\mathcal{O}_{L_v} = W_v \oplus \pi_v W_v$ and $\pi_v^2 W_v = pW_v$,

$$\pi_v\mathcal{O}_{L_v} = pW_v \oplus \pi_v W_v.$$

Therefore taking the π_v-coefficient of the logarithm gives an isomorphism $C_v \simeq \pi_v \mathcal{O}_{L_v}/pW_v \simeq W_v$. The ring W_v is the ring of integers of an unramified extension of $\mathbb{Q}_p$ of degree f_v, hence a free $\mathbb{Z}_p$-module of rank f_v, and so is C_v.

Step 3: the formula (35) and the involution. Removing the Teichmüller factor defines the logarithm of an arbitrary unit. This factor lies in W_v, so it is fixed by τ_v, and τ_v commutes with the logarithm. If $\log x = a + b\pi_v$ with $a \in pW_v$ and $b \in W_v$, then $\log(\tau_v x) = a - b\pi_v$. Subtracting the τ_v-conjugate logarithm gives $\log(x/\tau_v x) = 2b\pi_v$, so (35) returns the π_v-coefficient b, and $\mathcal{L}_v$ is the isomorphism of Step 2. Also $x\tau_v x \in W_v^\times$, being the norm of x to $\mathrm{Frac}(W_v)$. Hence the class of $\tau_v x$ in C_v is the inverse of the class of x, which proves the assertion about the involution.

Step 4: reduction modulo p. Modulo $\pi_v^2 \mathcal{O}_{L_v} = p\mathcal{O}_{L_v}$, the direct sum $\mathcal{O}_{L_v} = W_v \oplus \pi_v W_v$ gives the asserted unique coefficients modulo p. Since $x \equiv x_0 \pmod{\pi_v}$, the coefficient x_0 is a unit. From

$$(x_0 + x_1\pi_v)(y_0 + y_1\pi_v) \equiv x_0y_0 + (x_0y_1 + x_1y_0)\pi_v \pmod{\pi_v^2}$$

we see that multiplication adds the ratios x_1/x_0 modulo p. So φ_v is a homomorphism. It vanishes on $W_v^\times$, where $x_1 = 0$, so it is defined on C_v. It is surjective, since $\varphi_v(1 + b\pi_v) = \bar{b}$ for $b \in W_v$. It kills pC_v, because k_v has exponent p. Both C_v/pC_v and k_v have p^{f_v} elements by Step 2, so the kernel of φ_v is exactly pC_v.

Step 5: the other two expressions. The logarithm is congruent to its linear term modulo π_v^2: for $w \in \pi_v\mathcal{O}_{L_v}$ and $k \geq 2$, the term w^k/k has p-adic valuation at least $k/2 - v_p(k) \geq 1$, since $p \geq 5$. Let ω be the Teichmüller representative of the residue of x, and write $x = \omega(1 + w)$ with $w \in \pi_v\mathcal{O}_{L_v}$. Since $\omega \equiv x_0 \pmod{pW_v}$, we have $w \equiv (x_1/x_0)\pi_v \pmod{\pi_v^2}$. Hence the π_v-coefficient of $\log x \equiv w$ is x_1/x_0 modulo p, which proves $\mathcal{L}_v(x) \bmod p = \varphi_v(x)$. Finally, put $q = p^{f_v}$. Since $\omega^{q-1} = 1$ and every binomial term of degree at least two lies in $\pi_v^2\mathcal{O}_{L_v}$,

$$x^{q-1} = (1 + w)^{q-1} \equiv 1 + (q-1)w \equiv 1 - (x_1/x_0)\pi_v \pmod{\pi_v^2}.$$

This proves the second expression in (36). □

8.2 The modules $M_{n,p}$ and their minus parts

We now collect the quotients C_v at all places v above p with $e(v/p) = 2$ into one $\mathbb{Z}_p$-module with an action of σ. Condition (O) concerns the part on which σ acts by -1, because the images of z_n and of the relative units lie there. For each odd rational prime p put

$$M_{n,p} = \prod_{\substack{v \mid p \\ e(v/p)=2}} C_v, \qquad M_{n,p}^- = \ker(1 + \sigma : M_{n,p} \to M_{n,p}). \tag{37}$$

We write these groups additively; an empty product is zero. All places above p with $e(v/p) = 2$ occur, including places above primes already ramified in F_{n-1} and pairs of places exchanged by σ. The automorphism σ acts on $M_{n,p}$: it extends to an isomorphism of completions $L_v \to L_{\sigma v}$, which maps $\mathcal{O}_{L_v}^\times$ onto $\mathcal{O}_{L_{\sigma v}}^\times$ and W_v onto $W_{\sigma v}$, and hence induces an isomorphism $C_v \to C_{\sigma v}$.

Form the reduction map

$$\Phi_p = \prod_v \varphi_v : M_{n,p} \longrightarrow k_{n,p}, \qquad k_{n,p} = \bigoplus_v k_v.$$

We transport the involution to the residue space by this map. The elements $\sigma(\pi_v)$ and $\pi_{\sigma v}$ are uniformizers of $L_{\sigma v}$ whose squares lie in $pW_{\sigma v}^\times$. So $\sigma(\pi_v) = \psi_v \pi_{\sigma v}$ with $\psi_v \in W_{\sigma v}^\times$. Indeed, ψ_v is a unit whose square lies in $\mathrm{Frac}(W_{\sigma v})$. Write $\psi_v = a + b\pi_{\sigma v}$ with $a, b \in \mathrm{Frac}(W_{\sigma v})$. Then

$\psi_v^2 = a^2 + b^2\pi_{\sigma v}^2 + 2ab\pi_{\sigma v}$ forces $ab = 0$. If $a = 0$, then $\psi_v = b\pi_{\sigma v}$ has odd valuation and is not a unit. So $b = 0$ and $\psi_v = a \in W_{\sigma v}^\times$. If $x \equiv x_0 + x_1\pi_v \pmod{\pi_v^2}$, then $\sigma x \equiv \sigma(x_0) + \sigma(x_1)\psi_v\pi_{\sigma v} \pmod{\pi_{\sigma v}^2}$ with $\sigma(x_0), \sigma(x_1) \in W_{\sigma v}$. Hence $\Phi_p \circ \sigma = \sigma_\Phi \circ \Phi_p$ for the $\mathbb{F}_p$-linear involution σ_Φ of $k_{n,p}$ given by

$$(\sigma_\Phi a)_{\sigma v} = \bar{\psi}_v\, \bar{\sigma}(a_v), \tag{38}$$

where $\bar{\sigma} : k_v \to k_{\sigma v}$ is the induced isomorphism of residue fields. This formula includes both the residue-field isomorphism and the change of uniformizer. Let $k_{n,p}^-$ be the minus part of $k_{n,p}$ for this action.

Lemma 8.2. *The group $M_{n,p}^-$ is free over $\mathbb{Z}_p$ of rank*

$$r_{n,p} = \sum_{\substack{v|p \text{ in } L \\ e(v/p)=2}} f_v - \sum_{\substack{w|p \text{ in } F \\ e(w/p)=2}} f_w, \tag{39}$$

and Φ_p induces an isomorphism $M_{n,p}^-/pM_{n,p}^- \simeq k_{n,p}^-$. If p is unramified in F, then $M_{n,p}^- = M_{n,p}$ and $k_{n,p}^- = k_{n,p}$.

Proof. We split $M_{n,p}$ into its ± 1 parts for σ and identify the plus part with the corresponding module for F.

Step 1: reduction modulo p. Lemma 8.1 identifies $M_{n,p}/pM_{n,p}$ with the residue space $k_{n,p}$ through Φ_p, equivariantly for (38). Since two is invertible in $\mathbb{Z}_p$, the projectors $(1 \pm \sigma)/2$ decompose $M_{n,p} = M_{n,p}^+ \oplus M_{n,p}^-$ and commute with reduction modulo p. Hence Φ_p induces $M_{n,p}^-/pM_{n,p}^- \simeq k_{n,p}^-$. As a direct summand of the free module $M_{n,p}$, the module $M_{n,p}^-$ is free.

Step 2: the rank. Let $M_{F,p}$ be the analogous product over the places w of F above p with $e(w/p) = 2$; Lemma 8.1 is a statement about a single local field and applies to these places too. Inclusion of completions gives $\mathrm{ext} : M_{F,p} \to M_{n,p}$, which duplicates coordinates at split places. Products of local norms give $\nu : M_{n,p} \to M_{F,p}$. These maps are defined for the following reasons. At a place w of F ramified over $\mathbb{Q}$, the extension of completions L_v/F_w is split or unramified, because absolute ramification indices are at most two, and norms of coefficient units are coefficient units. At a newly ramified place, F_w is unramified over $\mathbb{Q}_p$, so the quotient C_w for F is zero, and the norm of any unit is a coefficient unit. Consequently

$$\nu \,\mathrm{ext} = 2, \qquad \mathrm{ext}\, \nu = 1 + \sigma.$$

It follows that ext identifies $M_{F,p}$ with $M_{n,p}^+$. It is injective because $\nu\,\mathrm{ext} = 2$ and $M_{F,p}$ is torsion-free, its image is fixed by σ, and for an invariant element x one has $x = \mathrm{ext}(\nu(x)/2)$. By Lemma 8.1 the rank of $M_{F,p}$ is the second sum in (39). Subtracting it from the rank of $M_{n,p}$, the first sum, proves (39).

Step 3: p unramified in F. Let v be a place of L above p with $e(v/p) = 2$, and let w be the place of F below it. Then F_w is unramified over $\mathbb{Q}_p$, so $F_w = \mathrm{Frac}(W_v)$, and L_v is the ramified quadratic extension of $\mathrm{Frac}(W_v)$. Hence σ fixes v and acts on L_v as τ_v, which acts as -1 on C_v by Lemma 8.1. Hence $M_{n,p}^- = M_{n,p}$, and Step 1 gives $k_{n,p}^- = k_{n,p}$. □

8.3 Relative units and the odd unit condition

The primary decomposition of $\mathcal{U}_n$. Let $\mathcal{T}_{n,p}$ be the p-primary part of $\prod_{v|2} k_v^\times$, and put $\mathcal{T}_{n,p}^- = \ker(1 + \sigma : \mathcal{T}_{n,p} \to \mathcal{T}_{n,p})$. Two is unramified in F_0, whose discriminant 65 is odd, and by (T1) the dyadic primes are unramified in every step; so two is unramified in L. The factor $(\mathcal{O}_L \otimes \mathbb{Z}_2)^\times$ of

(20) is $\prod_{v|2}\mathcal{O}_{L_v}^{\times}$, and the unit group of $\mathcal{O}_{L_v}$ at a dyadic place v is the product of the Teichmüller representatives, a cyclic group isomorphic to $k_v^{\times}$ of odd order, and the pro-2 group $1+2\mathcal{O}_{L_v}$. The factors C_v at the places v above p with $e(v/p)=2$ are pro-p. Indeed, $\mathcal{O}_{L_v}^{\times}$ is the product of the nonzero Teichmüller representatives and the pro-p group $1+\pi_v\mathcal{O}_{L_v}$, where π_v is a uniformizer, and the Teichmüller representatives lie in $W_v^{\times}$; so C_v is a quotient of $1+\pi_v\mathcal{O}_{L_v}$ by a closed subgroup. This uses neither the logarithm nor $p\geq 5$. We obtain the primary decomposition

$$\mathcal{U}_n = U_L \times \prod_{p \text{ odd}} (\mathcal{T}_{n,p} \times M_{n,p}), \qquad U_L = \prod_{v|2}(1+2\mathcal{O}_{L_v}). \tag{40}$$

Only finitely many factors are nonzero at this fixed level.

The maps $\mu_{n,p}$ and $\mathrm{res}_{n,p}$. For $z\in F_n^{\times}$ that is a unit at every place v above p with $e(v/p)=2$, let $\mu_{n,p}(z)\in M_{n,p}$ be the tuple of the classes of z in the quotients C_v. This is a homomorphism on the group of such elements, and $\mu_{n,p}(\sigma z)=\sigma\mu_{n,p}(z)$. On z_n and on global units it agrees with the projection of $\jmath_n$ to the factor $M_{n,p}$ of (40). Similarly, for z that is a unit at both dyadic places, let $\mathrm{res}_{n,p}(z)\in\mathcal{T}_{n,p}$ be the p-primary component of the tuple of its residues there.

Put $\mathscr{E}_{n,p}=E_n^1\otimes_{\mathbb{Z}}\mathbb{Z}_p$, where $E_n^1=\ker(N_{L/F}:E_n\to E_{n-1})$ is the group of relative units. Finite generation of units [8, Theorem 5.1] makes this a finitely generated, hence compact, $\mathbb{Z}_p$-module. Since $M_{n,p}$ and $\mathcal{T}_{n,p}$ are $\mathbb{Z}_p$-modules, the restrictions of $\mu_{n,p}$ and $\mathrm{res}_{n,p}$ to E_n^1 extend uniquely to $\mathbb{Z}_p$-linear maps on $\mathscr{E}_{n,p}$, which are continuous and which we denote by the same letters:

$$\mu_{n,p}:\mathscr{E}_{n,p}\to M_{n,p}^{-}, \qquad \mathrm{res}_{n,p}:\mathscr{E}_{n,p}\to\mathcal{T}_{n,p}^{-}.$$

Their images lie in the minus parts because $\sigma\eta=\eta^{-1}$ for a relative unit η. Define

$$\begin{aligned} U_{n,p}^{\mathrm{all}} &= \mu_{n,p}(\mathscr{E}_{n,p}), \qquad & U_{n,p}^{0} &= \mu_{n,p}(\ker \mathrm{res}_{n,p}),\\ E_{n,p}^{1,0} &= \{\eta\in E_n^1 : \mathrm{res}_{n,p}(\eta)=0\}, \qquad & z_{n,p} &= \mu_{n,p}(z_n)\in M_{n,p}^{-}. \end{aligned} \tag{41}$$

The element z_n is a unit at every ramified place, because $(z_n)=A_nP_n^{-1}$ and 47 is unramified in F_n by (T1). Its relative norm is one, since y_n and β_n both have relative norm y_{n-1} by (T2); so $z_{n,p}$ lies in $M_{n,p}^{-}$.

The span $U_{n,p}^0$ as a closure. The groups $U_{n,p}^{\mathrm{all}}$ and $U_{n,p}^0$ are compact, hence closed. The image of E_n^1 is dense in $\mathscr{E}_{n,p}$, since $\mathbb{Z}$ is dense in $\mathbb{Z}_p$. Since the group $\mathcal{T}_{n,p}^{-}$ is finite and discrete, the kernel of $\mathrm{res}_{n,p}$ is open, and the intersection of the image of E_n^1 with this kernel, which is the image of $E_{n,p}^{1,0}$, is dense in that kernel. By continuity and compactness, it follows that

$$U_{n,p}^{0}=\overline{\mu_{n,p}(E_{n,p}^{1,0})}. \tag{42}$$

Thus $U_{n,p}^0$ is the closed $\mathbb{Z}_p$-submodule of $M_{n,p}^{-}$ generated by the images of the relative units whose dyadic residues have trivial p-primary component.

The odd unit condition. The *odd unit condition* is

$$z_{n,p}\in U_{n,p}^{0} \quad \text{for every odd rational } p \text{ ramified in } F_n. \tag{O}$$

Section 8.4 shows that (O) at p is equivalent to the p-primary part of the approximation problem.

When the dyadic restriction is vacuous. The two dyadic primes of $F_0 = \mathbb{Q}(\sqrt{65})$ have residue field $\mathbb{F}_2$, since $65 \equiv 1 \pmod 8$, and by (T1) they are inert in every step. So the two dyadic residue extensions at this step are $\mathbb{F}_{q^2}/\mathbb{F}_q$ with $q = 2^{2^{n-1}}$, and σ acts on them as the Frobenius $x \mapsto x^q$. The kernel of $1+\sigma$ on $\mathbb{F}_{q^2}^\times$ is therefore the group of elements of norm one, which is cyclic of order $q+1$. Thus

$$p \nmid 2^{2^{n-1}} + 1 \quad \Longrightarrow \quad \mathcal{T}_{n,p}^- = 0, \quad E_{n,p}^{1,0} = E_n^1, \quad U_{n,p}^0 = U_{n,p}^{\mathrm{all}}. \tag{43}$$

Even in this case, the $M_{n,p}$-components at places above primes already ramified in F_{n-1} remain part of (O).

Remark 8.3 (The residue obstruction). The difference between $U_{n,p}^0$ and $U_{n,p}^{\mathrm{all}}$ is measured by a finite group. Suppressing n and p, there is an exact sequence

$$0 \longrightarrow U^0 \longrightarrow U^{\mathrm{all}} \longrightarrow \mathrm{res}(\mathscr{E})/\,\mathrm{res}(\ker \mu) \longrightarrow 0, \qquad \mu(e) \longmapsto [\mathrm{res}(e)]. \tag{44}$$

The map is well defined, because two lifts e of the same element of U^{all} differ by an element of $\ker \mu$. It is onto, since the class of $\mathrm{res}(e)$ is the image of $\mu(e)$ for every $e \in \mathscr{E}$, and it vanishes on U^0. Conversely, if $\mathrm{res}(e) = \mathrm{res}(e')$ with $e' \in \ker \mu$, then $e - e'$ lies in $\ker \mathrm{res}$ and has the same image $\mu(e)$; so the kernel consists precisely of the images under μ whose residue can be removed without changing the image, that is, of U^0. The target is finite because $\mathcal{T}_{n,p}^-$ is. Thus (O) first requires $z_{n,p} \in U^{\mathrm{all}}$, and then the vanishing of this finite obstruction for a lift of $z_{n,p}$.

8.4 The odd unit condition as an approximation condition

The next lemma shows that (O) at p says exactly that the p-primary component $(0, z_{n,p})$ of $\jmath_n(z_n)$ can be approximated by global units, simultaneously at the dyadic residues and at all places v above p with $e(v/p) = 2$. It uses arbitrary units, not only relative units, and shows that relative units with trivial p-primary dyadic residue suffice.

Lemma 8.4 (Membership in the closure of the unit image)**.** *Let $\Gamma_{n,p}$ be the closure of the image of all global units E_n in $\mathcal{T}_{n,p} \times M_{n,p}$. For $g \in M_{n,p}^-$,*

$$(0, g) \in \Gamma_{n,p} \quad \Longleftrightarrow \quad g \in U_{n,p}^0.$$

Proof. We reduce to units with trivial residue component and then use $\eta \mapsto \eta/\sigma\eta$ to pass to relative units.

Step 1: units with trivial residue. Let $E' = \ker(E_n \to \mathcal{T}_{n,p})$, and let Γ' be the closure of its image in $M_{n,p}$. The image of E' lies in $\{0\} \times M_{n,p}$. Conversely, suppose that $(0, g)$ is a limit of images of units e_i. Since the residue factor $\mathcal{T}_{n,p}$ is finite and discrete, the residue components of the e_i are eventually 0, so eventually $e_i \in E'$, and $g \in \Gamma'$. Hence $\Gamma_{n,p} \cap (\{0\} \times M_{n,p}) = \{0\} \times \Gamma'$.

Step 2: passage to relative units. The group Γ' is stable under σ. For $\eta \in E'$, the unit $\eta/\sigma\eta$ has relative norm one and trivial residue component, so it belongs to $E_{n,p}^{1,0}$; its image is $(1-\sigma)\mu_{n,p}(\eta)$. Taking closures gives $(1-\sigma)\Gamma' \subseteq U_{n,p}^0$. Now let $g \in \Gamma' \cap M_{n,p}^-$. Then $(1-\sigma)g = 2g \in U_{n,p}^0$. A closed subgroup of a compact abelian pro-p group is a $\mathbb{Z}_p$-submodule, so it is closed under division by two. Hence $g \in U_{n,p}^0$, and $\Gamma' \cap M_{n,p}^- \subseteq U_{n,p}^0$. With Step 1 this proves the implication from left to right.

Step 3: the converse. The units in $E_{n,p}^{1,0}$ lie in E', and their images in $\mathcal{T}_{n,p} \times M_{n,p}$ are $(0, \mu_{n,p}(\eta))$. So the reverse inclusion $U_{n,p}^0 \subseteq \Gamma'$ follows directly from (42). □

As explained in Section 6, approximation at each place separately would not suffice; this is why (O) is stated in the whole product $\mathcal{T}_{n,p} \times M_{n,p}$.

8.5 A finite sufficient test

Condition (O) asks for membership of one vector in a closed submodule. The following proposition gives a condition under which every vector of $M^-_{n,p}$ is a member, and that condition can be checked modulo p with finitely many units.

Proposition 8.5 (A finite sufficient test for (O))**.** *For an odd ramified rational prime p,*

$$U^0_{n,p} = M^-_{n,p} \quad \Longleftrightarrow \quad \operatorname{span}_{\mathbb{F}_p} \Phi_p \mu_{n,p}(E^{1,0}_{n,p}) = k^-_{n,p}.$$

In particular, any finite collection $\eta_1, \dots, \eta_m \in E^{1,0}_{n,p}$ whose images under $\Phi_p\mu_{n,p}$ have rank $r_{n,p}$ proves (O) at p, and also the analogue of (O) with $z_{n,p}$ replaced by any element of $M^-_{n,p}$.

Proof. We translate the right side into a statement modulo p and apply Nakayama's lemma in its form for compact $\mathbb{Z}_p$-modules. Write $M = M^-_{n,p}$ and $U = U^0_{n,p}$.

Step 1: the condition modulo p. By Lemma 8.2, Φ_p induces an isomorphism $M/pM \simeq k^-_{n,p}$. By (42), which rests on density and the finiteness of $\mathcal{T}^-_{n,p}$, U is the closure of $\mu_{n,p}(E^{1,0}_{n,p})$. Since pM is open in M, the sets U and $\mu_{n,p}(E^{1,0}_{n,p})$ have the same image in M/pM. This image is a subgroup, hence equal to its $\mathbb{F}_p$-span, and it lies in $k^-_{n,p}$ after applying Φ_p. So the right side is the assertion $U + pM = M$.

Step 2: Nakayama's lemma. Suppose $M = U + pM$. Given $g \in M$, write $g = g_0 + ph_1$ with $g_0 \in U$ and $h_1 \in M$, then $h_1 = g_1 + ph_2$, and so on. Repeated substitution gives

$$g = g_0 + pg_1 + \cdots + p^{a-1}g_{a-1} + p^a h_a, \qquad g_i \in U, \quad h_a \in M,$$

for every $a \geq 1$. The residual term $p^a h_a$ tends to zero, and U is closed, so $g \in U$. The converse is immediate.

Step 3: the finite collection. By Lemma 8.2, $k^-_{n,p} \simeq M/pM$ has dimension $r_{n,p}$ over $\mathbb{F}_p$. If the images of $\eta_1, \dots, \eta_m$, which lie in $k^-_{n,p}$, have rank $r_{n,p}$, they span $k^-_{n,p}$, so $U = M$ by Steps 1 and 2, and every element of M, in particular $z_{n,p}$, lies in U. □

Remark 8.6 (Computing the test). The units in this test can be produced by a finite computation. Start with any global units $\gamma_1, \dots, \gamma_m$ and put $\eta_i = \gamma_i/\sigma\gamma_i \in E^1_n$. Compute their dyadic p-residues and the integer kernel lattice

$$\mathcal{N}_p = \left\{ (a_i) \in \mathbb{Z}^m : \sum_i a_i \operatorname{res}_{n,p}(\eta_i) = 0 \right\}.$$

It has finite index in $\mathbb{Z}^m$ because the residue group is finite. For an integer basis $a^{(1)}, \dots, a^{(m)}$ of this lattice, put $\xi_j = \prod_i \eta_i^{a_i^{(j)}}$. Then $\xi_j \in E^{1,0}_{n,p}$. Compute $\Phi_p\mu_{n,p}(\xi_j)$ using (36), which requires only reduction modulo π_v^2 at every place v above p with $e(v/p) = 2$. After choosing $\mathbb{F}_p$-bases for the residue fields and for the subspace $k^-_{n,p}$ with the action (38), form one matrix from these images at all places above p together. By Proposition 8.5, rank $r_{n,p}$ certifies the equality $U^0_{n,p} = M^-_{n,p}$, and therefore (O) at p. When (43) applies, $\mathcal{N}_p = \mathbb{Z}^m$.

Remark 8.7 (Isolated ramified primes). For an isolated ramified prime (Definition 3.1) the test is scalar. Suppose that p is unramified in F, that exactly one prime $\mathfrak{p} \mid p$ ramifies in L/F, that $F_\mathfrak{p} = \mathbb{Q}_p$, and that $v_\mathfrak{p}(\delta) = 2m + 1$. Then the place v of L above $\mathfrak{p}$ is the only place of L above p with ramification index two over p, it has $f_v = 1$, and F has no such place above p; so $r_{n,p} = 1$. Moreover $\pi = \sqrt{\delta}/p^m$ is a local uniformizer. If $\eta = b_0 + b_1\sqrt{\delta} \in E^{1,0}_{n,p}$ with $b_0, b_1 \in F$, then b_0 and

$p^m b_1$ are $\mathfrak{p}$-integral and b_0 is a unit. This follows by taking the half-trace and half-difference of η and $\sigma\eta$, and using integrality at the ramified place. Consequently

$$\varphi_v(\eta) = p^m b_1/b_0 \bmod \mathfrak{p}.$$

By Proposition 8.5, one such unit with nonzero value proves (O) at p.

Remark 8.8 (What a deficient rank shows). A deficient matrix from selected units gives no negative conclusion about the full unit image. Even if those units generate $E^{1,0}_{n,p}$ modulo p-th powers, deficient rank proves only $U^0_{n,p} \neq M^-_{n,p}$, and the element $z_{n,p}$ may still belong to that proper submodule. A unit image of finite index does not suffice either: $p\mathbb{Z}_p \subset \mathbb{Z}_p$ has finite index but does not contain 1.

8.6 Proof of Theorem B

We now combine the two halves of the approximation problem. The 2-primary half is Theorem 7.2, which uses (S); the odd-primary half is condition (O), which Lemma 8.4 identifies with approximation of the odd-primary components. The following proposition puts them together at one step.

Proposition 8.9. *Let F_n/F_{n-1} be a scaled step in the sense of Definition 4.8, in a tower satisfying* (T1)–(T6). *If the step F_n/F_{n-1} satisfies both* (S) *and* (O), *then $\jmath_n(z_n) \in \overline{\jmath_n(E_n)} \subseteq \mathcal{U}_n$.*

Proof. We check the membership one primary component at a time, using the decomposition (40).

Step 1: primary components of a closed subgroup. A closed subgroup of a profinite abelian group is stable under all primary projections: in each finite quotient, a primary projection is multiplication by an integer chosen by the Chinese remainder theorem, and passage to the limit uses closedness. Thus $\overline{\jmath_n(E_n)}$ is the product of its primary subgroups, and each projection of $\overline{\jmath_n(E_n)}$ is its intersection with the corresponding primary factor. Since E_n is mapped into a compact group, the projection of $\overline{\jmath_n(E_n)}$ to a factor is the closure of the projection of $\jmath_n(E_n)$. It therefore suffices to show that each primary component of $\jmath_n(z_n)$ lies in the corresponding projection of $\overline{\jmath_n(E_n)}$.

Step 2: the 2-primary component. Theorem 7.2 (dyadic approximation) uses (S) and supplies the U_L component of $\jmath_n(z_n)$ in (40), including its two-primary torsion.

Step 3: the dyadic residues of z_n. The element z_n has residue 1 at both dyadic primes. By Definition 4.8(i) and (iii), $u \equiv 1 \pmod 2$ and a is a dyadic unit, and y is a dyadic unit because $(y) = AB$. Reduce modulo the dyadic primes of L. In (4), $t = (y(1-u^2)-a^2)/(au)$ with $1-u^2 \equiv 0$, so $\bar{t} = -\bar{a} = \bar{a}$. Since $\alpha + \sigma\alpha = t$, $\alpha\sigma\alpha = y$ and $\beta = a + u\alpha$, we obtain

$$\bar{t} = \bar{a}, \qquad \bar{\beta} = \bar{a} + \bar{\alpha} = \overline{\sigma\alpha}, \qquad \bar{z}_n = \overline{\alpha\beta/y} = 1.$$

Hence the $\mathcal{T}_{n,p}$-component of $\jmath_n(z_n)$ is zero for every odd p.

Step 4: the odd components. For every odd ramified p, the p-primary component of $\jmath_n(z_n)$ is therefore $(0, z_{n,p})$. Condition (O) and Lemma 8.4 place it in $\Gamma_{n,p}$, which is the projection of $\overline{\jmath_n(E_n)}$ to that primary factor. If p is unramified in L, the product $M_{n,p}$ is zero, and the p-primary component of $\jmath_n(z_n)$ is already zero.

Step 5: conclusion. By Step 1, the primary components combine to give the asserted membership in $\overline{\jmath_n(E_n)}$. □

Proof of Theorem B. Both (S) and (O) refer to the same step F_n/F_{n-1}. By hypothesis there is n_0 such that for every $n > n_0$ the step F_n/F_{n-1} is a scaled step satisfying (S) and (O). For each such n, Proposition 8.9 gives $\jmath_n(z_n) \in \overline{\jmath_n(E_n)}$, which is condition ($\mathrm{JT}_A$) with $N = n$. Theorem 6.1 then proves that D is hereditarily atomic. The strict principal-ideal chain of Theorem 3.7 already proves the failure of ACCP under the hypotheses (T1)–(T6). □

9 The remaining existence problem

Theorem B applies to one infinite tower of scaled steps that satisfies (T1)–(T6) and in which every sufficiently late step satisfies both the Selmer condition (S) and the odd unit condition (O). Theorem 4.11 supplies infinite towers of scaled steps satisfying (T1)–(T6), but its construction imposes neither (S) nor (O). Each of the two conditions is a statement about a single step; Sections 7 and 8 define them and give sufficient tests. What remains is to make infinitely many such choices compatibly. This section states that problem precisely and explains what it involves.

Section 9.1 formulates a one-step extension statement, Hypothesis 9.2 (which this paper does not prove): every finite extension of a fixed initial tower that is built by the construction schedule and whose added steps satisfy (S) and (O) has a further scaled step, for the prescription data of the schedule, that satisfies both. Proposition 9.3 shows that this hypothesis yields a hereditarily atomic domain without ACCP. Section 9.2 explains why (S) and (O) are conditions on the field F_n itself, which the local prescriptions of Section 4 do not determine. Section 9.3 compares (O) with the coefficient precision that the construction can prescribe (Appendix A). Lemma 9.5 shows that prescribed precision gives (O) at p once the unit index exponent $e_{n,p}$ of F_n is small enough, and an example of lattices shows that precision without such an index bound does not suffice.

9.1 Extending finite towers one step at a time

Fix an initial tower $F_0 \subset \cdots \subset F_{n_0}$ obtained by the construction of Section 4. Together with the fields we retain the state of that construction at level n_0: its list of characteristics, the fibres whose treatment is complete, and the current state of the finite procedure for separating residue offsets. Fix also the construction schedule (Definition 4.10) used in the proof of Theorem 4.11, in which the integers H_n (for example $H_n = n + 5$) and all other choices of that proof (the order of treatment within a fibre and the late levels n^*) are made by one fixed rule depending only on the tower built so far. After each step this schedule adds to its list all newly ramified characteristics and the least odd characteristic other than $47, 7$ not yet listed, and it specifies the prescription data for the next step. The schedule is thus a fixed rule, but the prescription data it assigns at a later level depend on the fields already chosen.

Definition 9.1. A *successful finite continuation* of the initial tower is a finite sequence of scaled steps $F_{n_0} \subset F_{n_0+1} \subset \cdots \subset F_m$ (Definition 4.8) such that each step F_n/F_{n-1} with $n_0 < n \leq m$ is a scaled step for the prescription data that the construction schedule assigns at level n, and satisfies both (S) and (O). The initial tower itself is a successful continuation of length zero; no condition (S) or (O) is imposed at its levels. A *successful infinite sequence* of scaled steps is an infinite sequence $F_{n_0} \subset F_{n_0+1} \subset \cdots$ of scaled steps each of whose finite initial segments is a successful finite continuation.

In this definition all local conditions, sets of ramified primes and unit groups are those of the fields F_n of the sequence.

Hypothesis 9.2 (Simultaneous extension)**.** Let $F_{n_0} \subset \cdots \subset F_m$, with $m \geq n_0$, be a successful finite continuation of the initial tower. Then there is a next step F_{m+1}/F_m that is a scaled step in the sense of Definition 4.8 for $H = H_{m+1}$ and the next prescription data of the construction schedule (its inert set, forbidden characteristics and residue prescriptions), and that satisfies both (S) and (O). The auxiliary prime R, the element c and the positive exponent k of this step may depend on the whole continuation and on the next prescription data.

The hypothesis asks for one scaled step that satisfies (S) and (O) together. It does not ask that every preliminary choice of c admit a suitable exponent k. The case $m = n_0$ is included: the hypothesis asserts that the initial tower itself has a successful extension of length one. So the hypothesis is not satisfied vacuously by the absence of successful continuations of positive length.

Proposition 9.3 (The extension hypothesis suffices)**.** *If some initial tower and the construction schedule above satisfy Hypothesis 9.2, then there exists a hereditarily atomic domain that does not satisfy ACCP.*

Proof. We build a successful infinite sequence of scaled steps by repeated use of the hypothesis, and apply Theorem B to the resulting tower.

Step 1: an infinite tower. The initial tower is a successful continuation of length zero. If $F_{n_0} \subset \cdots \subset F_m$ is a successful continuation, Hypothesis 9.2 gives a scaled step F_{m+1}/F_m, a scaled step for the next prescription data of the schedule, that satisfies (S) and (O); then $F_{n_0} \subset \cdots \subset F_{m+1}$ is again a successful continuation. Choosing such a step at every level, by countable dependent choice, gives an infinite tower $F_0 \subset F_1 \subset \cdots$ every step of which is a scaled step; the steps of the initial tower are scaled steps by Remark 4.9, since the construction of Theorem 4.11 produces them by Proposition 4.7.

Step 2: hypotheses (T1)–(T6). Every step after level n_0 is a scaled step for the prescription data of the schedule, so every local prescription of the schedule is retained. The argument of Section 4.9 that these residue prescriptions satisfy condition (c) of Proposition 4.7 uses only that the prescriptions of the earlier steps hold, so it applies to this tower. The termination argument and the argument that each listed characteristic is treated after finitely many steps, in the proof of Theorem 4.11, therefore apply to this tower: each listed fibre has only finitely many predecessors in the list, and each fibre procedure terminates. Hence Definition 4.8(iv)–(vi) and the proof of Theorem 4.11 give (T1)–(T6) for the tower; that proof uses of each step only conditions (iv)–(vi) and the schedule: the field is totally real, every prescription holds, and newly ramified characteristics are listed after each step.

Step 3: conclusion. Conditions (S) and (O) hold at every level above n_0 by construction. The tower therefore satisfies the hypotheses of Theorem B, and the ring of integers of $\bigcup_n F_n$ is hereditarily atomic and does not satisfy ACCP. □

Hypothesis 9.2 is stronger than what Theorem B needs. One successful infinite sequence of scaled steps suffices, even if some successful finite continuations have no successful extension. Two weaker statements do not suffice by themselves. First, the existence of one step satisfying (S) and of another step satisfying (O) does not give a single step satisfying both. Second, the existence of successful finite continuations of every finite length does not give a successful infinite sequence. The successful continuations form a tree, ordered by extension, and this tree need not be finitely branching. For a finitely branching tree, König's lemma would turn nodes of every finite length into an infinite path; without that property the conclusion can fail. For example, the tree of finite strictly decreasing sequences of positive integers, ordered by extension, has nodes of every finite length but no infinite path. The example is purely combinatorial.

9.2 Conditions (S) and (O) concern the field F_n itself

The construction of Section 4 controls finitely many local prescriptions at each step. Conditions (S) and (O) depend in addition on all the primes that ramify in the step and on the unit group of the new field. We describe both dependencies.

Consider a proposed step L/F. Condition (S) is the injectivity of the map λ_δ of Section 7. Its domain Sel_δ is defined using every odd prime ramified in L/F, including the primes ramified in L/F other than the prescribed ones, and all the global square classes permitted by those primes. The construction allows such additional ramified primes at every step and places no bound on their number. The detecting primes for $\mathcal{V}(F)$ (Definition 4.5) give the conclusions of Proposition 4.7 about ideal classes and norms. They do not replace the computation of the Selmer condition, which involves the whole set T of primes ramified in L/F.

At the same step, (O) at p asks that $z_{n,p}$ lie in the closed $\mathbb{Z}_p$-submodule $U^0_{n,p}$ of $M^-_{n,p}$ (Section 8), and it is imposed at every odd rational prime p ramified in L. These primes include the primes already ramified in F as well as the newly ramified ones, and for each p the module $M_{n,p}$ involves all the ramified places of L above p. The module $U^0_{n,p}$ is the closure of the image of the relative units whose dyadic residues have trivial p-primary component (42), so the dyadic residue restriction is part of (O). The finite test of Proposition 8.5 proves the stronger statement $U^0_{n,p} = M^-_{n,p}$, whereas (O) at p asks only that the single element $z_{n,p}$ lie in $U^0_{n,p}$. So a step at which the test fails may still satisfy (O).

Thus (S) and (O) are conditions on the chosen field F_n and its units, at one level at a time. Sections 7 and 8 give their definitions and sufficient tests. Verifying (S) and (O) at finitely many levels proves neither Hypothesis 9.2 nor the existence of a successful infinite sequence of scaled steps.

The quantifiers over the odd primes also matter. Theorem B needs one level after which every step satisfies (O) at every ramified odd prime. Suppose instead that for each fixed p the condition (O) at p holds at all steps after a level depending on p. This does not supply a common level: a new odd prime at which (O) fails at each level is compatible with eventual success at each fixed prime.

9.3 Coefficient precision and the unit index

Appendix A shows that the construction can prescribe, at finitely many odd primes in each step, how closely z_n is approximated by elements of the local coefficient rings (Proposition A.3). This subsection measures what such a prescription gives towards (O). Throughout, the tower consists of scaled steps and satisfies (T1)–(T6). By (T1), the places $v \mid p$ of F_n that are ramified over $\mathbb{Q}$, for an odd prime p, are exactly the places with $e(v/p) = 2$, which index the product $M_{n,p}$.

Definition 9.4. Let v be a finite place of a number field $\mathcal{E}$, with completion $\mathcal{E}_v$, coefficient ring W_v (the ring of integers of the maximal unramified subfield of $\mathcal{E}_v$; Section 2.3) and maximal ideal $\mathfrak{m}_v$ of $\mathcal{O}_{\mathcal{E}_v}$. A unit $z \in \mathcal{O}^\times_{\mathcal{E}_v}$ has *coefficient precision at least* d at v if $z \in W_v^\times(1 + \mathfrak{m}_v^d)$.

Since the Teichmüller representatives lie in W_v (Section 2.3), every unit has coefficient precision at least 1; larger d means that z is closer to an element of $W_v^\times$. The second ingredient measures how far the unit image $U^0_{n,p}$ is from all of $M^-_{n,p}$. For a prime $p \geq 5$ ramified in F_n, let

$$e_{n,p} = \min\{e \geq 0 : p^e M^-_{n,p} \subseteq U^0_{n,p}\}, \tag{45}$$

with value ∞ if the set is empty. We call $e_{n,p}$ the *unit index exponent* at p. When it is finite, $p^{e_{n,p}}$ is the exponent of the finite quotient $M^-_{n,p}/U^0_{n,p}$, not its order. It is finite exactly when $U^0_{n,p}$ has full $\mathbb{Z}_p$-rank in $M^-_{n,p}$, that is, when $U^0_{n,p} \otimes \mathbb{Q}_p = M^-_{n,p} \otimes \mathbb{Q}_p$. Indeed $M^-_{n,p}$ is free of finite rank (Lemma 8.2), so a submodule of full rank has finite index and the quotient is killed by a power of p; conversely a submodule containing $p^e M^-_{n,p}$ has the same span over $\mathbb{Q}_p$. The definition uses $U^0_{n,p}$, the image of the kernel of the dyadic residue map $\mathrm{res}_{n,p}$, and not the larger module $U^{\mathrm{all}}_{n,p}$.

For $z \in F_n^\times$ that is a unit at the places $v \mid p$ with $e(v/p) = 2$, we write $\mu_{n,p}(z) \in M_{n,p}$ for its image, as in Section 8. At such a place v we use the square uniformizer π_v of Section 8, so that $\mathcal{O}_{(F_n)_v} = W_v \oplus \pi_v W_v$ and $\pi_v^2 \in pW_v^\times$.

Lemma 9.5 (Coefficient precision and the unit index)**.** *Let $d \geq 1$ and let $z \in F_n^\times$ have relative norm one and be a unit at all places $v \mid p$ with $e(v/p) = 2$, where $p \geq 5$. Write $\hat{z} = \mu_{n,p}(z)$ and $\mathfrak{m}_v = \pi_v \mathcal{O}_{(F_n)_v}$. Then*

$$z \in W_v^\times(1 + \mathfrak{m}_v^d) \text{ for every } v \mid p \text{ with } e(v/p) = 2 \quad \Longleftrightarrow \quad \hat{z} \in p^{\lfloor d/2 \rfloor} M_{n,p}^-. \tag{46}$$

If these equivalent conditions hold and moreover $e_{n,p} \leq \lfloor d/2 \rfloor$, then $\hat{z} \in U_{n,p}^0$.

Proof. We compute the image of $W_v^\times(1+\mathfrak{m}_v^d)$ in each quotient C_v by the logarithm, and then pass to the product $M_{n,p}$ and its minus part. Put $a = \lfloor d/2 \rfloor$.

Step 1: one place. Let $v \mid p$ with $e(v/p) = 2$. The logarithm isomorphism of Lemma 8.1 between $1+\pi_v\mathcal{O}_{(F_n)_v}$ and $\pi_v\mathcal{O}_{(F_n)_v}$ restricts to an isomorphism $1+\mathfrak{m}_v^d \simeq \mathfrak{m}_v^d$, because both the logarithm and the exponential preserve the valuation of the leading term. For $d = 2a$ and $d = 2a+1$, respectively, the additive ideals are

$$\mathfrak{m}_v^{2a} = p^a W_v \oplus p^a \pi_v W_v, \qquad \mathfrak{m}_v^{2a+1} = p^{a+1} W_v \oplus p^a \pi_v W_v,$$

since $\pi_v^2 \in pW_v^\times$. In both cases the π_v-coefficients are exactly the elements of $p^a W_v$. The isomorphism $\mathcal{L}_v : C_v \to W_v$ of (35) takes the class of an element of $1 + \pi_v\mathcal{O}_{(F_n)_v}$ to the π_v-coefficient of its logarithm. Hence the image of $W_v^\times(1+\mathfrak{m}_v^d)$ in C_v is $\mathcal{L}_v^{-1}(p^a W_v) = p^a C_v$. Since $W_v^\times$ is the kernel of $\mathcal{O}_{(F_n)_v}^\times \to C_v$, a unit at v lies in $W_v^\times(1+\mathfrak{m}_v^d)$ if and only if its class lies in $p^a C_v$.

Step 2: the product. By Step 1, the left-hand side of (46) holds if and only if every component of $\hat{z}$ lies in $p^a C_v$, that is, if and only if $\hat{z} \in p^a M_{n,p}$. Since z has relative norm one, $z\,\sigma_n(z) = 1$, so $\sigma_n \hat{z} = -\hat{z}$ and $\hat{z} \in M_{n,p}^-$. It remains to show $M_{n,p}^- \cap p^a M_{n,p} = p^a M_{n,p}^-$. The inclusion $\supseteq$ is clear. Since p is odd, $(1-\sigma_n)/2$ is a projector of $M_{n,p}$ onto $M_{n,p}^-$. If $x = p^a y \in M_{n,p}^-$ with $y \in M_{n,p}$, then $x = \frac{1-\sigma_n}{2} x = p^a \cdot \frac{1-\sigma_n}{2} y \in p^a M_{n,p}^-$. This proves (46).

Step 3: membership. If $e_{n,p} \leq a$, then $p^a M_{n,p}^- \subseteq p^{e_{n,p}} M_{n,p}^- \subseteq U_{n,p}^0$, so the right-hand side of (46) gives $\hat{z} \in U_{n,p}^0$. □

The index bound in Lemma 9.5 is also necessary, in the following uniform sense. Every vector of $M_{n,p}^-$ satisfying the right-hand condition of (46) lies in $U_{n,p}^0$ if and only if $e_{n,p} \leq \lfloor d/2 \rfloor$. This is the definition of $e_{n,p}$, together with the inclusion $p^a M_{n,p}^- \subseteq p^e M_{n,p}^-$ for integers $a \geq e \geq 0$. It is not a converse for the particular element z_n, with divisor $A_n P_n^{-1}$. A particular element can belong to $U_{n,p}^0$ even when the whole subgroup $p^{\lfloor d/2 \rfloor} M_{n,p}^-$ does not.

The lemma gives one sufficient way to verify (O) at a chosen step n. Choose integers $d_{n,p} \geq 1$. At every odd characteristic p already ramified in F_{n-1}, verify the left-hand precision condition of (46) for z_n with $d = d_{n,p}$, together with the index bound $e_{n,p} \leq \lfloor d_{n,p}/2 \rfloor$; at every newly ramified characteristic, verify the membership $z_{n,p} \in U_{n,p}^0$ directly. All these conditions must hold at the same step. Precision prescribed at a finite list of previously treated fibres does not cover a fibre outside that list, in particular a ramified fibre whose residue offsets are being separated in the current step. Proposition A.3 prescribes precision only at primes that carry no residue prescription.

Increasing precision alone does not control the unit index, because the field changes from step to step. At the level of abstract lattices, take

$$M_n' = \mathbb{Z}_p, \qquad U_n' = p^{n+1}\mathbb{Z}_p, \qquad g_n' = p^n.$$

Every U'_n has finite index in M'_n, and g'_n satisfies the right-hand condition of (46) with $d = 2n$, that is, $g'_n \in p^n M'_n$; yet $g'_n \notin U'_n$ for every n. This abstract example shows that the two properties together do not imply membership. The exponent $e_{n,p}$ depends on the unit image of the field F_n itself, which is not determined while the parameters of the step are being chosen. Convergence of the successive elements z_n at fixed places and finite index of the unit images do not by themselves bound these exponents in terms of the prescribed precisions.

To complete the construction of a counterexample it therefore suffices to prove a simultaneous extension statement such as Hypothesis 9.2, or to construct one successful infinite sequence of scaled steps by another method.

A Finite local prescriptions and precision

This appendix shows how much the construction of Section 4 can prescribe about the element $z = \alpha\beta/y$ of a single step at finitely many odd primes. Its results are not used in the proofs of Theorems A and B. They are background for Section 9. There, Lemma 9.5 shows that the membership required by condition (O) at an odd prime p ramified in F_n follows from two ingredients: the element z_n has coefficient precision at least d (Definition 9.4) at every place v above p with $e(v/p) = 2$, and the unit index exponent $e_{n,p}$ of (45) is at most $\lfloor d/2 \rfloor$. This appendix shows that the first ingredient can be prescribed by the construction, at finitely many odd primes of the inert set in each step (Proposition A.3), and along every fixed odd rational prime other than 47 and 7 from some level on (Corollary A.4). Section 9 explains why this precision does not by itself give (O).

The argument has three parts. Section A.1 works in one completion at an odd prime. There we choose the parameter c so that the prime is inert in the step and z lies in the coefficient ring of the completion of the next field, and we show that small changes of c and u keep z in the coefficient ring up to any prescribed power of the maximal ideal. The tools are Teichmüller representatives and Newton's lemma, which turns an approximate root of a polynomial into a root [8, Theorem 7.32]; see Section 2.3. Section A.2 inserts these local conditions into the proof of Proposition 4.7. Section A.3 records how closely z is approximated by elements of F at the dyadic places. The residue prescriptions used to separate residue offsets are treated in Lemma 4.4; the primes considered here lie in the inert set and carry no residue prescription.

A.1 Local models with z in the coefficient ring at odd places

Throughout this subsection let $\mathcal{K}/\mathbb{Q}_p$ be a finite extension with p odd, with maximal ideal $\mathfrak{m}_{\mathcal{K}}$ and residue field $k_{\mathcal{K}}$. Let W be its coefficient ring, the ring of integers of its maximal unramified subfield $\operatorname{Frac} W$ (Section 2.3). Let W_2 be the ring of integers of the unramified quadratic extension of $\operatorname{Frac} W$, and set $\mathcal{K}_2 = \mathcal{K} \cdot \operatorname{Frac} W_2$. Since $\mathcal{K}/\operatorname{Frac} W$ is totally ramified, $\mathcal{K} \cap \operatorname{Frac} W_2 = \operatorname{Frac} W$, and $\mathcal{K}_2/\mathcal{K}$ is unramified quadratic. Write σ for the nontrivial automorphism of $\mathcal{K}_2/\mathcal{K}$ and N for the norm from $\mathcal{K}_2$ to $\mathcal{K}$. The automorphism σ restricts to the nontrivial automorphism of $\operatorname{Frac} W_2/\operatorname{Frac} W$, so it preserves W_2. The field $\operatorname{Frac} W_2$ is the maximal unramified subfield of $\mathcal{K}_2$, so W_2 is the coefficient ring of $\mathcal{K}_2$.

The first lemma produces one choice of c for which the polynomial of (4) defines the unramified quadratic extension and z lies in W_2.

Lemma A.1. *Let $y, h, u \in \mathcal{O}_{\mathcal{K}}^{\times}$ with $h \in W$ and $u \neq \pm 1$. Suppose that $u \in W$ or $u \equiv 1 \pmod{\mathfrak{m}_{\mathcal{K}}}$. Then there is $c \in \mathcal{O}_{\mathcal{K}}^{\times}$ with the following properties. Put $a = c(u-1)/h$ and $t = (y(1-u^2)-a^2)/(au)$.*

The polynomial $X^2 - tX + y$ defines $\mathcal{K}_2/\mathcal{K}$; its roots $\alpha, \sigma\alpha$ and the elements $\beta = a + u\alpha$ and $\sigma\beta$ are units; and $z = \alpha\beta/y$ lies in $W_2^\times$.

Proof. We first give a formula that recovers c, α and β from a prescribed value of z. In Cases 2 and 3 we choose z so that this formula has a unit solution. In Case 1 we instead construct c over $\operatorname{Frac} W$ and rescale.

Step 1 (an inverse formula). Let $z \in W_2 \setminus W$ be a unit with $N(z) = 1$, so that $\sigma z = z^{-1}$. Put

$$\mathcal{Q}(z) = 1 + u^2 + u(z + z^{-1}) \in \mathcal{K}.$$

Suppose that the right-hand side of the first equation below is the square of a unit of $\mathcal{K}$, and choose c, α and β by

$$c^2 = \frac{yh^2(u+1)^2}{\mathcal{Q}(z)}, \qquad \alpha = -\frac{c(u+z)}{h(u+1)}, \qquad \beta = -\frac{c(1+uz)}{h(u+1)}. \tag{47}$$

Multiplication gives $N\alpha = N\beta = y$, $\beta = c(u-1)/h + u\alpha$, and $\alpha\beta = yz$. Indeed $(u+z)(u+z^{-1}) = (1+uz)(1+uz^{-1}) = \mathcal{Q}(z)$ and $(u+z)(1+uz) = z\mathcal{Q}(z)$. Put $a = c(u-1)/h$; then $a \neq 0$ because $u \neq 1$. Expanding $N(a + u\alpha) = y$ gives $a^2 + au \operatorname{Tr}\alpha + u^2 y = y$, so the trace of α is the element t of the statement, and $\alpha, \sigma\alpha$ are the roots of $X^2 - tX + y$. The expression for α is linear in z with nonzero coefficient, and $z \notin \mathcal{K}$ because $\mathcal{K} \cap \operatorname{Frac} W_2 = \operatorname{Frac} W$. Hence $\mathcal{K}(\alpha) = \mathcal{K}(z) = \mathcal{K}_2$.

Case 1: $u \in W$. The rings W and $\mathcal{O}_\mathcal{K}$ have the same residue field, so we may choose $y^\circ \in W^\times$ with the same residue as y. Since p is odd and $y/y^\circ \equiv 1 \pmod{\mathfrak{m}_\mathcal{K}}$, Newton's lemma [8, Theorem 7.32] gives a unit $\xi \in \mathcal{K}$ with $\xi^2 = y/y^\circ$. Apply Lemma 4.2 over the field $\operatorname{Frac} W$ to the units y°, h, u. It supplies a unit $c^\circ \in W$ for which the polynomial $X^2 - t^\circ X + y^\circ$ of (4), formed with c°, y° in place of c, y, defines $\operatorname{Frac} W_2/\operatorname{Frac} W$. Its trace t° is integral by the second expression for t in (4), whose denominators $c^\circ h$ and $c^\circ hu$ are units. Its roots $\alpha^\circ, \sigma\alpha^\circ$ are therefore integral, with norm the unit y°, and hence are units. The element $a^\circ = c^\circ(u-1)/h$ is integral. The corresponding elements $\beta^\circ = a^\circ + u\alpha^\circ$ and $\sigma\beta^\circ$ are integral, and their norm is $(a^\circ)^2 + a^\circ u t^\circ + u^2 y^\circ = y^\circ$ by the definition of t°; so they are units. Set

$$c = \xi c^\circ, \quad a = \xi a^\circ, \quad t = \xi t^\circ, \quad \alpha = \xi\alpha^\circ, \quad \beta = \xi\beta^\circ.$$

These are the formulas (4) for y: the polynomial $X^2 - tX + y$ has the root $\alpha = \xi\alpha^\circ$, and $\beta = a + u\alpha$. Since $\alpha^\circ \in \operatorname{Frac} W_2 \setminus \operatorname{Frac} W$ and $\mathcal{K} \cap \operatorname{Frac} W_2 = \operatorname{Frac} W$, the element α generates $\mathcal{K}_2$ over $\mathcal{K}$. The elements $c, \alpha, \sigma\alpha, \beta, \sigma\beta$ are units, and

$$z = \frac{\alpha\beta}{y} = \frac{\alpha^\circ\beta^\circ}{y^\circ} \in W_2^\times.$$

Moreover $z \notin W$. Indeed, from $\beta = a + u\alpha$ and $\alpha^2 = t\alpha - y$ one gets $\alpha\beta = (a + ut)\alpha - uy$ with $a + ut = -yh(u+1)/c$, so that $z = -u - h(u+1)\alpha/c$ (this is the first identity in (13)). Hence $z = -u - h(u+1)\alpha^\circ/c^\circ \notin W$.

Case 2: $u \equiv 1 \pmod{\mathfrak{m}_\mathcal{K}}$ and y is a square. Here u need not lie in W. We choose a norm-one unit $z \in W_2 \setminus W$ arbitrarily close to 1. For example, let $\theta \in W_2$ be nonzero with $\sigma\theta = -\theta$, and put

$$z = \frac{1 + p^r\theta}{1 - p^r\theta}, \qquad r \gg 0.$$

Then z is a unit, $\sigma z = z^{-1}$, and z lies outside W. Indeed, a norm-one element of W satisfies $z = \sigma z = z^{-1}$, so $z = \pm 1$; but $z \equiv 1 \pmod{p}$ excludes -1, and $\theta \neq 0$ excludes 1. Since $u + 1 \equiv 2$ is a unit,

$$\frac{\mathcal{Q}(z)}{(u+1)^2} = 1 + \frac{u(z + z^{-1} - 2)}{(u+1)^2},$$

and $z+z^{-1}-2=(z-1)^2/z$ tends to 0 as $r\to\infty$. For sufficiently large r this quotient is therefore congruent to 1 modulo a high power of $\mathfrak{m}_{\mathcal{K}}$, hence the square of a unit by Newton's lemma. Since y and h^2 are squares of units, the right-hand side of (47) is the square of a unit, and (47) has a unit solution c. The numerators $u+z$ and $1+uz$ have residue $2\neq 0$, and c and $h(u+1)$ are units, so α and β, and hence $\sigma\alpha$ and $\sigma\beta$, are units. By Step 1, c has all the required properties.

Case 3: $u\equiv 1 \pmod{\mathfrak{m}_{\mathcal{K}}}$ and y is not a square. Since p is odd and y is a unit, its residue $\bar{y}$ is not a square. Put $q=|k_{\mathcal{K}}|$, and let χ be the quadratic character of $\mathbb{F}_q^\times$. The norm-one subgroup of $\mathbb{F}_{q^2}^\times$ is cyclic of order $q+1$. Choose an element $\bar{z}\neq -1$ of it that is not a square in this subgroup, that is, $\bar{z}^{(q+1)/2}=-1$. The subgroup has $(q+1)/2\geq 2$ nonsquares, at most one of which is -1, so this choice is possible for every odd q, including $q=3$. The only norm-one elements of $\mathbb{F}_q$ are ± 1, and 1 is a square, so $\bar{z}\notin\mathbb{F}_q$. The norm of $1+\bar{z}$ to $\mathbb{F}_q$ satisfies

$$\chi\big(N_{\mathbb{F}_{q^2}/\mathbb{F}_q}(1+\bar{z})\big)=\bar{z}^{-(q+1)/2}=-1.$$

To check the identity, use $N_{\mathbb{F}_{q^2}/\mathbb{F}_q}(1+\bar{z})=(1+\bar{z})(1+\bar{z}^{-1})=(1+\bar{z})^2/\bar{z}$ and $(1+\bar{z})^{q-1}=\bar{z}^{-1}$. Let $z\in W_2$ be the Teichmüller representative of $\bar{z}$. It has norm one, since the Teichmüller representatives are multiplicative and σ acts on them through $\bar{z}\mapsto\bar{z}^q$; and $z\notin W$ because $\bar{z}\notin\mathbb{F}_q$. Since $\bar{u}=1$, the residue of $\mathcal{Q}(z)$ is $2+\bar{z}+\bar{z}^{-1}=N_{\mathbb{F}_{q^2}/\mathbb{F}_q}(1+\bar{z})$, a nonsquare unit. Hence the right-hand side of (47) has a nonzero square residue, and by Newton's lemma it is the square of a unit of $\mathcal{K}$. The elements α,β are units because their numerators have residue $1+\bar{z}\neq 0$, and so are $\sigma\alpha$ and $\sigma\beta$. By Step 1, c has all the required properties.

The three cases cover all u allowed by the hypotheses. □

The second lemma shows that the conclusion of Lemma A.1 survives small changes of c and u, up to a prescribed precision. In the global construction c and u can be controlled only modulo powers of a prime, so this is the form in which the local model is used.

Lemma A.2 (Stability to finite precision)**.** *Fix y,h,u,c as in Lemma A.1 and an integer $d\geq 1$. For all c,u in sufficiently small neighbourhoods of these values, with y,h unchanged, the polynomial X^2-tX+y of (4) still defines $\mathcal{K}_2/\mathcal{K}$, the elements $\alpha,\sigma\alpha,\beta,\sigma\beta$ are units, and, for either choice of the root α,*

$$z\in W_2^\times(1+\mathfrak{m}_{\mathcal{K}_2}^d).$$

Proof. We follow one root by Newton's lemma and use continuity of the formulas for β and z. Write t°, α° and z° for the values of t, α and z given by Lemma A.1, so that $z^\circ\in W_2^\times$. The derivative of $X^2-t^\circ X+y$ at α° is $2\alpha^\circ-t^\circ=\alpha^\circ-\sigma\alpha^\circ$, which is nonzero, though possibly not a unit. Let e be its valuation in $\mathcal{K}_2$. The coefficient t depends continuously on c,u near the fixed values. Congruence conditions on c,u modulo sufficiently high powers of $\mathfrak{m}_{\mathcal{K}}$ therefore keep the valuation of the derivative at α° equal to e, and make the value of X^2-tX+y at α° have valuation arbitrarily larger than $2e$. Newton's lemma [8, Theorem 7.32] then gives a root $\alpha\in\mathcal{K}_2$ of X^2-tX+y as close to α° as desired. The field $\mathcal{K}$ is closed in $\mathcal{K}_2$ and does not contain α°, so a small enough neighbourhood of α° avoids $\mathcal{K}$. Hence α still generates $\mathcal{K}_2$, and the polynomial defines $\mathcal{K}_2/\mathcal{K}$. Being a unit is an open condition, so $\alpha,\sigma\alpha,\beta,\sigma\beta$ remain units. The rational expressions $\beta=a+u\alpha$ and $z=\alpha\beta/y$ are continuous. Thus $z/z^\circ\in 1+\mathfrak{m}_{\mathcal{K}_2}^d$, so $z\in W_2^\times(1+\mathfrak{m}_{\mathcal{K}_2}^d)$. The other choice of root replaces α,β by $\sigma\alpha,\sigma\beta$, and hence z by $\sigma z=z^{-1}$. Conjugation preserves W_2 and $1+\mathfrak{m}_{\mathcal{K}_2}^d$, so the assertion holds for either root. □

Since W_2 is the coefficient ring of $\mathcal{K}_2$, the conclusion of Lemma A.2 says that z has coefficient precision at least d in the sense of Definition 9.4.

A.2 A finite refinement of the global construction

We now insert the local conditions of Section A.1 into the proof of Proposition 4.7. Since both lemmas there involve only congruence conditions on c and u at finitely many primes, they fit into the ray class and Chebotarev arguments of that proof.

Proposition A.3. *Let the data be as in Proposition 4.7. In addition, let $\mathcal{J}$ be a finite set of primes of F of odd residue characteristic, disjoint from $\{A, B\}$ and from every prime carrying a residue prescription, and assign an integer $d_v \geq 1$ to each $v \in \mathcal{J}$. Add the primes of $\mathcal{J}$ to the inert set. Then the step can be chosen so that all conclusions of Proposition 4.7 hold, the step is a scaled step in the sense of Definition 4.8 for the enlarged inert set, and, at the unique prime $\widetilde{v}$ of L above each $v \in \mathcal{J}$,*

$$z, -z \in W_{\widetilde{v}}^{\times}(1 + \mathfrak{m}_{\widetilde{v}}^{d_v}), \qquad z = \alpha\beta/y, \tag{48}$$

where $W_{\widetilde{v}}$ is the coefficient ring of $L_{\widetilde{v}}$, the ring of integers of its maximal unramified subfield, and $\mathfrak{m}_{\widetilde{v}}$ is the maximal ideal of $\mathcal{O}_{L_{\widetilde{v}}}$.

Proof. We obtain congruence conditions on c and u at the primes of $\mathcal{J}$ from Lemmas A.1 and A.2, and check that the proof of Proposition 4.7 can respect them.

Step 1 (local models at the primes of $\mathcal{J}$). Use the unit $u_0 = \epsilon_H$ as the local starting value of u. Let $v \in \mathcal{J}$ and $\mathcal{K} = F_v$. If the residue characteristic of v is neither 5 nor 13, then the prime of F_0 below v is unramified over $\mathbb{Q}$, since the discriminant of F_0 is 65. So the completion of F_0 at that prime is unramified over $\mathbb{Q}_p$ and lies in $\operatorname{Frac} W$, and $u_0 \in W$. At 5 and 13, the residue of $\epsilon_0 = 8 + \sqrt{65}$ at the prime of F_0 above p is 8, which has order four modulo 5 and modulo 13. Since $H \geq 5$, the exponent $46 \cdot 2^{H-4}$ in (4) is divisible by four, and $u_0 \equiv 1 \pmod{\mathfrak{m}_{\mathcal{K}}}$. Also $u_0 \neq \pm 1$, since ϵ_0 has infinite order. The scale $h = 2^H$ lies in W and is a unit because v is odd, and y is a unit at v because $(y) = AB$ and $v \notin \{A, B\}$. Thus Lemma A.1 applies at every prime of $\mathcal{J}$. In its model the polynomial $X^2 - tX + y$ defines the unramified quadratic extension, so v is inert, as the inert prescription requires. Lemma A.2, with $d = d_v$, supplies congruence conditions on c and u modulo finite powers of v that ensure (48) for z. Here $\mathcal{K}_2 = L_{\widetilde{v}}$, and $W_2 = W_{\widetilde{v}}$ is its coefficient ring.

Step 2 (compatibility with the proof of Proposition 4.7). Insert these congruence conditions into the proof of Proposition 4.7. They are compatible with the other local models of that proof: those models are imposed at other primes, except for the inert prescriptions at the primes of $\mathcal{J}$, which the new conditions refine. Enlarge the ray modulus $\mathfrak{f}$ so that it retains the congruence conditions on c. Replace the integer $\mathcal{M}$ by a multiple that is also divisible by the order of ϵ_H in each finite group $(\mathcal{O}_F/v^{m_v})^{\times}$, $v \in \mathcal{J}$, where v^{m_v} is the modulus of the congruence condition on u at v. Then $k \equiv 1 \pmod{\mathcal{M}}$ gives $u = \epsilon_H^k \equiv u_0 \pmod{v^{m_v}}$, so it retains these conditions. Lemma 4.6 applies to the enlarged finite modulus and integer, so the auxiliary prime R and the condition $u \equiv -1 \pmod{R}$ can still be selected. The argument that makes δ_- and δ_+ totally positive multiplies c by units congruent to 1 modulo the enlarged modulus, and so also preserves these congruence conditions. The rest of the proof of Proposition 4.7 is unchanged. The new congruence conditions are local conditions at primes of $\mathcal{J}$ only, and the enlarged $\mathcal{M}$ is still divisible by 6. Hence conditions (i)–(vi) of Definition 4.8 still hold, and the step is a scaled step for the enlarged inert set.

Step 3 (the element $-z$). Finally, -1 is a unit of the coefficient ring at every place of odd residue characteristic, so the approximation (48) for z gives the same approximation for $-z$. □

Applying the proposition at every step, with growing d_v, gives precision along each fixed odd rational prime other than 47 and 7. We use the construction schedule of the proof of Theorem 4.11,

in which the odd rational characteristics other than 47 and 7 are listed and treated one fibre at a time (fibres as in Section 4.9).

Corollary A.4 (Coefficient precision along a fixed rational prime)**.** *The tower of Theorem 4.11 can be chosen so that the following holds. For every odd rational prime $p \neq 47, 7$ there is a level n_p such that, for all $n > n_p$ and every place $v \mid p$ of F_n,*

$$z_n \in W_{n,v}^{\times}(1 + \mathfrak{m}_{n,v}^{n}),$$

where $W_{n,v}$ is the coefficient ring of $(F_n)_v$, the ring of integers of its maximal unramified subfield, and $\mathfrak{m}_{n,v}$ is the maximal ideal of $\mathcal{O}_{(F_n)_v}$.

Proof. We run the construction schedule of Theorem 4.11 and add, at each step, the precision conditions of Proposition A.3 at all fibres whose treatment is complete. At the step that constructs F_n, apply Proposition A.3 to all primes of F_{n-1} in the finitely many odd fibres whose treatment is already complete, with $d_v = n$. Such primes carry an inert prescription at every later step and have no further residue prescriptions, so they are admissible as $\mathcal{J}$. The fibre currently being treated is excluded until its treatment is complete. The construction schedule still completes the treatment of each fixed fibre in finitely many steps; the level at which the treatment of the fibre over p is complete gives n_p. Each step is a scaled step for the prescription data of the schedule, by Proposition A.3, since the primes of $\mathcal{J}$ already carry inert prescriptions. So the verification in the proof of Theorem 4.11 applies. □

A.3 Dyadic approximation of z by elements of F

At the dyadic places the situation is different: z is close to an element of F only up to a bounded depth. For a scaled step, (13) gives

$$v_2(-z - 1) = H, \qquad v_2\big(z/(-u) - 1\big) = H + 1 \tag{49}$$

at both dyadic places. The second identity follows from $z + u = -h(u + 1)\alpha/c$, since $v_2(h) = H$, $v_2(u + 1) = 1$, and α, c, u are units at the dyadic places.

The valuation $H + 1$ is the largest possible among elements of the dyadic completion F_v. The residue polynomial of $X^2 - tX + y$ at v is separable, so $\alpha - \sigma\alpha$ is a unit. Applying σ to the expression for $z + u$ gives $z - \sigma z = -h(u + 1)(\alpha - \sigma\alpha)/c$, and hence

$$v_2(z - \sigma z) = H + 1.$$

For an element b of F_v, the elements $z - b$ and $\sigma z - b$ are conjugate and so have the same valuation. If this valuation were greater than $H + 1$, so would be the valuation of their difference $z - \sigma z$, a contradiction. The element $b = -u$ attains $H + 1$.

Remark A.5. The global unit -1 gives the congruence $z \equiv -1 \pmod{2^H}$ at the dyadic places. By Proposition A.3 it also preserves every odd approximation in (48), since $-z$ satisfies the same condition as z.

The integers d_v in Proposition A.3 are finite inputs to the construction of one step, and changing them can change c and the field L. Likewise, Corollary A.4 concerns the varying fields F_n at each fixed rational prime. Neither result gives the membership required by (O) for a fixed step, or a bound on the unit index exponent $e_{n,p}$ of (45). Neither result gives coefficient precision at a fibre while its residues are being prescribed, whether by a split residue prescription or by an inert residue prescription (with component 1 or -1). Section 9 discusses what more is needed.

B Comparison fields and valuation parities

Condition (S) of Section 7 asks that the map λ_δ on the Selmer group Sel_δ of a scaled step $L = F(\sqrt{\delta})$ be injective. Among the conditions defining Sel_δ is a local condition, stated in terms of δ, at every prime of F that ramifies in L. The discriminant factors as $\delta = \delta_-\delta_+/(chu)^2$, and the ramified primes fall into two disjoint sets: the set T_- of those at which δ_- has odd valuation and the set T_+ of those at which δ_+ has odd valuation (Lemma B.1). The factor $\delta_- = c^2 - h^2 y$ does not involve the unit u, so T_- is determined once c is chosen.

This appendix expresses (S) through the auxiliary quadratic extension $L' = F(\sqrt{\delta_+})$, which we call the *comparison field*, and through the parities of valuations at the primes of T_-. The odd primes ramified in L' are those of T_+ together with A. In the reformulation, the classes of Sel_δ are sorted by the parities of their valuations at the primes of T_-, and those with even valuation at all these primes are described through a Selmer group attached to L', in which the primes of T_- enter only through quadratic residue symbols. The argument has three parts. First, the classes of $\Delta(F, \delta)$ with even valuation at every prime of T_- form a subgroup Δ_0, which injects into Sel_δ with quotient the group of realized valuation-parity vectors on T_- modulo the all-one vector (Lemma B.3). Second, Δ_0 is identified with the subgroup of the Selmer group of L' on which the residue symbols at T_- vanish, compatibly with the dyadic Hilbert symbols (Lemma B.4). Third, the two pieces combine into a short exact sequence of kernels (Proposition B.5). Because this sequence is exact, the reformulation is an equivalence: (S) holds for the step exactly when two explicit maps, one on the Selmer group of L' and one on the parity vectors, are injective. When $|T_-| = 1$, the Selmer data of L and of L' coincide. The equivalence concerns a given step; it does not by itself produce a step that satisfies (S) (Remark B.6).

Setting. Throughout this appendix we fix a scaled step L/F (Definition 4.8) of a tower satisfying (T1)–(T6), with the notation of (4) and of Section 2.1. In particular $(y) = AB$, $(c) = AR$, the unit u is a square in F, the elements δ_- and δ_+ are totally positive, and

$$\delta_- = c^2 - h^2 y, \qquad \delta_+ = c^2(u-1)^2 - h^2 y(u+1)^2, \qquad \delta = \frac{\delta_-\delta_+}{(chu)^2}.$$

The subscripts L and L' distinguish the Selmer group of the constructed step L from the analogous group of the comparison field L'. The odd unit condition (O) is not affected: it always concerns L.

B.1 Ramification sets of L and L'

Define finite sets of odd primes of F by

$$T_- = \{\mathfrak{p} \neq A, B : v_\mathfrak{p}(\delta_-) \equiv 1 \pmod 2\}, \qquad T_+ = \{\mathfrak{p} \neq A, B : v_\mathfrak{p}(\delta_+) \equiv 1 \pmod 2\}.$$

These sets contain every prime with the indicated valuation, whether it was prescribed in the construction or arose from the choice of the step.

Lemma B.1. *The sets T_- and T_+ are disjoint. The sets of all odd primes of F ramified in L/F and in L'/F are, respectively,*

$$T_L = T_- \sqcup T_+, \qquad T_{L'} = \{A\} \sqcup T_+.$$

The element δ_- is a square in F_v at every dyadic place v and in $F_\mathfrak{p}$ at every prime $\mathfrak{p} \in T_+$. At A one has $v_A(\delta_-) = v_A(\delta_+) = 1$ and $[\delta_-]_A = [\delta_+]_A$. Moreover T_- is nonempty.

Proof. We first compute valuations prime by prime and read off the ramification sets, then show that δ_- is a local square at the dyadic places and at T_+ using that it is a norm from L, and finally deduce $T_- \neq \varnothing$ from the class of A in $\mathrm{Cl}(F)/2\,\mathrm{Cl}(F)$.

Step 1: odd primes other than A and B. Expanding the definitions gives the identity

$$\delta_+ = (u-1)^2\delta_- - 4h^2yu. \tag{50}$$

At an odd prime $\mathfrak{p} \neq A, B$, the last term $4h^2yu$ is a unit, because h is a power of 2, $(y) = AB$ and u is a unit. The elements δ_- and δ_+ are integral, so they cannot both lie in $\mathfrak{p}$: if one of them lies in $\mathfrak{p}$, the identity shows that the other is congruent to a unit modulo $\mathfrak{p}$. An element with odd valuation at $\mathfrak{p}$ lies in $\mathfrak{p}$, so $T_- \cap T_+ = \varnothing$.

Step 2: the primes A and B. At A we have $v_A(c) = v_A(y) = 1$ and $u \equiv 1 \pmod{A}$. In $\delta_- = c^2 - h^2y$ the first term has valuation 2 and the second valuation 1, so $v_A(\delta_-) = 1$. In δ_+ the first term has valuation at least 2 and the second valuation 1, since $u+1 \equiv 2 \pmod{A}$ is a unit; so $v_A(\delta_+) = 1$. The prime A splits in L, so δ is a square in F_A, and $[\delta] = [\delta_-\delta_+]$ gives $[\delta_-]_A = [\delta_+]_A$. At B the elements c, h and $u-1$ are units and $v_B(y) = 1$, so δ_- and δ_+ are units at B.

Step 3: the ramification sets. At an odd prime, a quadratic extension $F(\sqrt{x})$ is ramified exactly when $v_\mathfrak{p}(x)$ is odd (Section 2.3, quadratic Kummer theory). The denominator $(chu)^2$ changes no valuation parity, so $v_\mathfrak{p}(\delta) \equiv v_\mathfrak{p}(\delta_-) + v_\mathfrak{p}(\delta_+) \pmod 2$ at every prime; this includes the auxiliary prime R dividing c, which belongs to T_+ exactly when $v_R(\delta_+)$ is odd. By Steps 1 and 2, $v_\mathfrak{p}(\delta)$ is odd exactly at the primes of $T_- \sqcup T_+$, and $v_\mathfrak{p}(\delta_+)$ is odd exactly at A and at the primes of T_+. This proves the two ramification formulas.

Step 4: the dyadic places. At a dyadic place v we have $\delta_-/c^2 \in 1 + 2^{2H}\mathcal{O}_{F_v} \subseteq F_v^{\times 2}$, as noted after Definition 4.8. Hence δ_- is a square in F_v, and $[\delta_+]_v = [\delta]_v$. Thus L' has the same unramified quadratic completion at v as L.

Step 5: the primes of T_+. A direct calculation from the identities for α and β in (4), using $N_{L/F}(\alpha) = y$, $\mathrm{Tr}_{L/F}(\alpha) = t$ and $a = c(u-1)/h$, gives

$$N_{L/F}(\alpha - \beta) = \frac{a^2 - y(u-1)^2}{u} = \frac{(u-1)^2\delta_-}{h^2u}. \tag{51}$$

The factor $(u-1)^2/(h^2u)$ is a square in $F^\times$, because u is a square, and a square s^2 of $F^\times$ is the norm $N_{L/F}(s)$. Consequently δ_- is a norm from L. Let $\mathfrak{p} \in T_+$. By Step 1, δ_- is a unit at $\mathfrak{p}$, and $L_\mathfrak{p}/F_\mathfrak{p}$ is a ramified quadratic extension of odd residue characteristic. Since $\mathfrak{p}$ has a single prime of L above it, δ_- is also a norm from the completion $L_\mathfrak{p}$; write $\delta_- = N_{L_\mathfrak{p}/F_\mathfrak{p}}(w)$ with $w \in L_\mathfrak{p}$. Then w has valuation zero. Since $L_\mathfrak{p}/F_\mathfrak{p}$ is totally ramified, the two fields have the same residue field and the nontrivial automorphism acts trivially on it, so w and its conjugate have the same residue. Hence the residue of δ_- is the square of the residue of w. A unit with square residue at an odd prime is a square, by Newton's lemma [8, Theorem 7.32]. So δ_- is a square in $F_\mathfrak{p}$. The argument does not divide by c, so it applies also when $\mathfrak{p}$ is the auxiliary prime R.

Step 6: T_- is nonempty. Lemma 4.1(b) shows that $\mathrm{Cl}(F_0) \simeq \mathbb{Z}/2\mathbb{Z}$ with $[A_0] \neq 0$, so $[A_0] \notin 2\,\mathrm{Cl}(F_0)$. For $j \geq 1$ the prime A_{j-1} splits in F_j and A_j has residue degree one over it, so the norm map on ideal classes satisfies

$$N_{F_j/F_{j-1}}[A_j] = [A_{j-1}].$$

If $[A_j] = 2x$ with $x \in \mathrm{Cl}(F_j)$, then $[A_{j-1}] = 2N_{F_j/F_{j-1}}(x) \in 2\,\mathrm{Cl}(F_{j-1})$. By induction on j, therefore, $[A_j] \notin 2\,\mathrm{Cl}(F_j)$ for every j; in particular $[A] \notin 2\,\mathrm{Cl}(F)$. By Step 2 and the definition of T_-, the only primes at which δ_- has odd valuation are A and the primes of T_-. Reducing the principal ideal (δ_-) modulo squares of ideals and taking classes in $\mathrm{Cl}(F)/2\,\mathrm{Cl}(F)$ gives

$$[A] + \sum_{\mathfrak{p} \in T_-} [\mathfrak{p}] = 0 \quad \text{in } \mathrm{Cl}(F)/2\,\mathrm{Cl}(F). \tag{52}$$

Since $[A] \neq 0$ in $\mathrm{Cl}(F)/2\,\mathrm{Cl}(F)$, the set T_- is nonempty. □

We keep the set $\mathcal{D}$ of the two dyadic primes of F. By Lemma B.1, T_L is the set T of all finite primes ramified in L used in Section 7, since no dyadic prime ramifies in L. Both L and L' are totally real, because δ and δ_+ are totally positive. The comparison field is ramified at A, whereas the constructed step splits at A, so L' cannot replace L as a step of the tower; it serves only to reformulate (S).

B.2 The Selmer group of the comparison field

We now set up the objects on the side of L': its Selmer group with the dyadic Hilbert-symbol map, a canonical representative of each Selmer class, and the residue symbols at T_-. For the constructed step put

$$\Delta_L = \Delta(F,\delta), \qquad \mathrm{Sel}_L = \Delta_L/\langle[\delta]\rangle = \mathrm{Sel}_\delta, \qquad \lambda_L = \lambda_\delta.$$

Define $\Delta_{L'}$ by the conditions of Section 7, with δ_+ in place of δ and the set $T_{L'}$ of all odd primes ramified in L'/F in place of T. Thus $\Delta_{L'}$ consists of the square classes $[x] \in F^\times/F^{\times 2}$ of totally positive elements x with even valuation at every prime outside $\mathcal{D} \cup T_{L'}$ and with $[x]_\mathfrak{p} \in \langle[\delta_+]_\mathfrak{p}\rangle$ at every $\mathfrak{p} \in T_{L'}$. Let

$$\mathrm{Sel}_{L'} = \Delta_{L'}/\langle[\delta_+]\rangle, \qquad \lambda_{L'} : \mathrm{Sel}_{L'} \longrightarrow (\mathbb{F}_2^{\mathcal{D}})^0, \qquad \lambda_{L'}([x]) = ([x,-\delta_+]_v)_{v\in\mathcal{D}}.$$

Here and below, a Hilbert-symbol formula on a quotient is evaluated on any representative. The map $\lambda_{L'}$ is well defined because $(\delta_+,-\delta_+)_v = 1$. Its values lie in $(\mathbb{F}_2^{\mathcal{D}})^0$ by the argument used for λ_δ: for $[x] \in \Delta_{L'}$ the symbol $(x,-\delta_+)_v$ is trivial at every $\mathfrak{p} \in T_{L'}$ because $[x]_\mathfrak{p}$ is 1 or $[\delta_+]_\mathfrak{p}$; it is trivial at every other odd prime because x and δ_+ both have even valuation there; and it is trivial at the real places because x is totally positive. The product formula for Hilbert symbols [9, Chapter V, Example 5.4] then gives $\sum_{v\in\mathcal{D}}[x,-\delta_+]_v = 0$.

The A-normalized representative. At A the two local classes allowed in $\Delta_{L'}$ are 1 and $[\delta_+]_A$, and the second has odd valuation by Lemma B.1. A class r of $\mathrm{Sel}_{L'}$ has exactly two representatives in $\Delta_{L'}$, namely $[x]$ and $[x\delta_+]$, and exactly one of them is a square in F_A: the one with even valuation at A. We call it the *A-normalized representative* $\hat{r}$ of r. Explicitly,

$$\mathrm{Sel}_{L'} \longrightarrow \Delta_{L'}, \qquad r = [x] \longmapsto \hat{r} = [x\delta_+^{\,v_A(x) \bmod 2}]. \tag{53}$$

Replacing x by $x\delta_+$ changes the parity of $v_A(x)$ and leaves the right side unchanged. The classes of $\Delta_{L'}$ that are squares in F_A form a subgroup that meets $\langle[\delta_+]\rangle$ trivially and maps onto $\mathrm{Sel}_{L'}$. So (53) is the inverse of the resulting isomorphism, and it is a linear section of the quotient map $\Delta_{L'} \to \mathrm{Sel}_{L'}$. The uniqueness here concerns square classes, not elements of $F^\times$.

Residue symbols at T_-. Let $\mathfrak{p}$ be an odd prime and let $[x]$ be a square class with $v_\mathfrak{p}(x)$ even. The residue character $\chi_\mathfrak{p}$ of Section 2.3 is defined on such classes, and $[x]$ is a square in $F_\mathfrak{p}$ exactly when $\chi_\mathfrak{p}(x) = 0$. The primes of T_- lie outside $\mathcal{D} \cup T_{L'}$ by Lemma B.1, so every class of $\Delta_{L'}$ has even valuation at them. Define

$$\Theta : \mathrm{Sel}_{L'} \longrightarrow \mathbb{F}_2^{T_-}, \qquad \Theta(r) = \big(\chi_\mathfrak{p}(\hat{r})\big)_{\mathfrak{p}\in T_-}. \tag{54}$$

This map is linear, because $r \mapsto \hat{r}$ is linear and $\chi_\mathfrak{p}$ is a homomorphism on the classes of even valuation at $\mathfrak{p}$. We also write

$$(\Theta,\lambda_{L'}) : \mathrm{Sel}_{L'} \longrightarrow \mathbb{F}_2^{T_-} \times (\mathbb{F}_2^{\mathcal{D}})^0, \qquad r \longmapsto (\Theta(r),\lambda_{L'}(r)).$$

Lemma B.2. *The image of Θ lies in the hyperplane of vectors with coordinate sum zero. In particular* $\operatorname{rank}\Theta \le |T_-| - 1$.

Proof. We apply the product formula to the pair (x, δ_-), where x is the A-normalized representative of $r \in \mathrm{Sel}_{L'}$, and check that only the symbols at T_- can be nontrivial. The symbol $(x, \delta_-)_v$ is trivial at A, because x is a square in F_A. It is trivial at the dyadic places and at the primes of T_+, because δ_- is a square there by Lemma B.1. At an odd prime outside $\{A\} \cup T_+ \cup T_-$, both x and δ_- have even valuation, so the symbol is trivial. The real symbols are trivial because x is totally positive. At $\mathfrak{p} \in T_-$, the valuation $v_\mathfrak{p}(\delta_-)$ is odd and $v_\mathfrak{p}(x)$ is even, so the additive symbol $[x, \delta_-]_\mathfrak{p}$ equals $\chi_\mathfrak{p}(x)$ (Section 2.3, Hilbert symbols). The product formula [9, Chapter V, Example 5.4] yields $\sum_{\mathfrak{p}\in T_-} \chi_\mathfrak{p}(x) = 0$. The hyperplane of vectors with coordinate sum zero in $\mathbb{F}_2^{T_-}$ has dimension $|T_-| - 1$. □

B.3 Vanishing valuation parities and the comparison

We now compare the two Selmer groups. Define the *valuation-parity map* on T_- and its image by

$$\mathrm{par} : \Delta_L \longrightarrow \mathbb{F}_2^{T_-}, \qquad \mathrm{par}([x]) = (v_\mathfrak{p}(x) \bmod 2)_{\mathfrak{p}\in T_-}, \qquad \mathrm{Par} = \operatorname{im}\mathrm{par},$$

and let

$$\Delta_0 = \ker \mathrm{par} = \{[x] \in \Delta_L : v_\mathfrak{p}(x) \equiv 0 \pmod 2 \text{ for all } \mathfrak{p} \in T_-\}.$$

Let $\mathbf{1} \in \mathbb{F}_2^{T_-}$ denote the all-one vector, and set $\overline{\mathrm{Par}} = \mathrm{Par}/\langle \mathbf{1}\rangle$. At every $\mathfrak{p} \in T_-$ the valuation $v_\mathfrak{p}(\delta)$ is odd, because δ_+ is a unit there by Lemma B.1; hence $\mathrm{par}([\delta]) = \mathbf{1}$, and $\mathbf{1} \in \mathrm{Par}$. Since T_- is nonempty, $\mathbf{1} \neq 0$.

Lemma B.3. *The composite $\Delta_0 \to \Delta_L \to \mathrm{Sel}_L$ is injective, and the map* par *induces an exact sequence*

$$0 \longrightarrow \Delta_0 \longrightarrow \mathrm{Sel}_L \longrightarrow \overline{\mathrm{Par}} \longrightarrow 0, \qquad \dim \overline{\mathrm{Par}} \le |T_-| - 1. \tag{55}$$

Proof. The kernel of $\Delta_L \to \mathrm{Sel}_L$ is $\langle[\delta]\rangle$, and $\mathrm{par}([\delta]) = \mathbf{1} \neq 0$, so it meets Δ_0 trivially. Since $\mathrm{par}([\delta]) = \mathbf{1}$, the map par descends to a map $\mathrm{Sel}_L \to \overline{\mathrm{Par}}$, which is surjective by the definition of Par. Let a class of Sel_L, represented by $[x]$, lie in its kernel. Then $\mathrm{par}([x])$ is 0 or $\mathbf{1}$. In the second case we replace x by $x\delta$, whose parity vector is 0. So the class has a representative in Δ_0, and this representative is the unique one of $[x]$ and $[x\delta]$ with parity vector 0. This proves exactness. Finally $\mathrm{Par} \subseteq \mathbb{F}_2^{T_-}$ and $\mathbf{1} \neq 0$, so $\dim \overline{\mathrm{Par}} \le |T_-| - 1$. □

Through Lemma B.3 we regard Δ_0 as a subgroup of Sel_L, and we write $\lambda_0 = \lambda_L|_{\Delta_0}$, that is, $\lambda_0([x]) = ([x, -\delta]_v)_{v\in\mathcal{D}}$ for $[x] \in \Delta_0$.

Lemma B.4. *Reduction modulo $\langle[\delta_+]\rangle$ defines a canonical isomorphism*

$$\Delta_0 \xrightarrow{\sim} \ker\Theta \subseteq \mathrm{Sel}_{L'}, \qquad [x] \longmapsto [x] \bmod \langle[\delta_+]\rangle, \tag{56}$$

whose inverse sends $r \in \ker\Theta$ to its A-normalized representative $\hat{r}$. Under this isomorphism λ_0 corresponds to $\lambda_{L'}|_{\ker\Theta}$, and therefore

$$\ker\lambda_0 \simeq \ker(\Theta, \lambda_{L'}). \tag{57}$$

Proof. We compare the local conditions defining Δ_0 and $\Delta_{L'}$ prime by prime, then check that the two maps are mutually inverse and compare the dyadic symbols.

Step 1: local behaviour of Δ_0. Let $[x] \in \Delta_0$. At a prime $\mathfrak{p} \in T_-$, the two local classes allowed in Δ_L are 1 and $[\delta]_\mathfrak{p}$, and the second has odd valuation. Since $v_\mathfrak{p}(x)$ is even, x is a square in $F_\mathfrak{p}$. Also x is a square in F_A by Lemma 7.10, which holds for every class of $\Delta(F, \delta)$ without assuming (S) (its proof uses (33) and Hilbert reciprocity, but not the injectivity of λ_δ). At a prime $\mathfrak{p} \in T_+$ the local conditions of Δ_L and of $\Delta_{L'}$ agree: δ_- is a square in $F_\mathfrak{p}$ by Lemma B.1, so $[\delta]_\mathfrak{p} = [\delta_+]_\mathfrak{p}$.

Step 2: the forward map. Let $[x] \in \Delta_0$. Then x is totally positive and has even valuation outside $\mathcal{D} \cup T_+$, since $T_L = T_- \sqcup T_+$ and the valuations at T_- are even; in particular it has even valuation outside $\mathcal{D} \cup T_{L'}$. At A the class $[x]_A = 1$ is allowed, and at T_+ the local conditions agree by Step 1. Hence $[x] \in \Delta_{L'}$. Let r be its class in $\mathrm{Sel}_{L'}$. Since x is a square in F_A, we have $\hat{r} = [x]$. Since x is a square at every prime of T_-, $\Theta(r) = 0$. The forward map is injective: if $[x] \in \langle [\delta_+] \rangle$ then $[x] = 1$, because $[x] \neq [\delta_+]$, as $v_A(x)$ is even while $v_A(\delta_+) = 1$.

Step 3: the inverse map. Let $r \in \ker \Theta$. Its A-normalized representative $\hat{r}$ is totally positive, has even valuation outside $\mathcal{D} \cup \{A\} \cup T_+$, and is a square in F_A; so it has even valuation outside $\mathcal{D} \cup T_+$, and in particular outside $\mathcal{D} \cup T_L$. At T_+ its local class is allowed in Δ_L by Step 1. At $\mathfrak{p} \in T_-$ it has even valuation and $\chi_\mathfrak{p}(\hat{r}) = 0$, so it is a square in $F_\mathfrak{p}$, which is allowed in Δ_L. Hence $\hat{r} \in \Delta_L$, and $\hat{r} \in \Delta_0$. The two maps are mutually inverse: the class of $\hat{r}$ modulo $\langle [\delta_+] \rangle$ is r, and by Step 2 the A-normalized representative of the image of $[x] \in \Delta_0$ is $[x]$.

Step 4: the dyadic symbols. At a dyadic place v, δ_- is a square in F_v by Lemma B.1, so $[\delta]_v = [\delta_+]_v$ and $[x, -\delta]_v = [x, -\delta_+]_v$. Hence $\lambda_0([x]) = \lambda_{L'}(r)$ when $[x] \in \Delta_0$ corresponds to $r \in \ker \Theta$. Thus $\ker \lambda_0$ corresponds to the set of $r \in \ker \Theta$ with $\lambda_{L'}(r) = 0$, which is $\ker(\Theta, \lambda_{L'})$. □

The map λ_L induces a map

$$\bar{\lambda} : \overline{\mathrm{Par}} \longrightarrow (\mathbb{F}_2^{\mathcal{D}})^0 / \lambda_0(\Delta_0)$$

by sending a class of $\overline{\mathrm{Par}}$ to the image of $\lambda_L(s)$, where $s \in \mathrm{Sel}_L$ is any lift under (55). Two lifts differ by an element of Δ_0, so their values under λ_L differ by an element of $\lambda_0(\Delta_0)$, and $\bar{\lambda}$ is well defined.

Proposition B.5 (Comparison formulation of (S))**.** *There is an exact sequence*

$$0 \longrightarrow \ker(\Theta, \lambda_{L'}) \longrightarrow \ker \lambda_L \longrightarrow \ker \bar{\lambda} \longrightarrow 0. \tag{58}$$

Consequently (S) is equivalent to the conjunction of $\ker(\Theta, \lambda_{L'}) = 0$ *and* $\ker \bar{\lambda} = 0$. *If* $|T_-| = 1$, *there is a canonical isomorphism*

$$(\mathrm{Sel}_L, \lambda_L) \simeq (\mathrm{Sel}_{L'}, \lambda_{L'}).$$

Proof. We restrict the exact sequence (55) to the kernels of λ_0, λ_L and $\bar{\lambda}$.

Step 1: exactness. The intersection of $\ker \lambda_L$ with the subgroup Δ_0 is $\ker \lambda_0$, which is isomorphic to $\ker(\Theta, \lambda_{L'})$ by (57); this gives the injective map on the left. If $s \in \ker \lambda_L$, then the image of s in $\overline{\mathrm{Par}}$ lies in $\ker \bar{\lambda}$, by the definition of $\bar{\lambda}$; this gives the map on the right. An element $s \in \ker \lambda_L$ maps to zero in $\overline{\mathrm{Par}}$ exactly when $s \in \Delta_0$, by (55), that is, exactly when $s \in \ker \lambda_0$. This is exactness in the middle. For surjectivity on the right, let a class of $\overline{\mathrm{Par}}$ lie in $\ker \bar{\lambda}$, and choose a lift $s \in \mathrm{Sel}_L$. Then $\lambda_L(s) \in \lambda_0(\Delta_0)$, say $\lambda_L(s) = \lambda_0(d)$ with $d \in \Delta_0$. The element $s - d$ lies in $\ker \lambda_L$ and has the same image in $\overline{\mathrm{Par}}$.

Step 2: the equivalence. Condition (S) states that $\ker \lambda_L = 0$. By the exact sequence (58), this holds exactly when both outer terms vanish.

Step 3: the case $|T_-| = 1$. The hyperplane of vectors with coordinate sum zero in $\mathbb{F}_2^{T_-} = \mathbb{F}_2$ is zero, so Lemma B.2 forces $\Theta = 0$ and $\ker \Theta = \mathrm{Sel}_{L'}$. Also $\mathrm{Par} = \langle \mathbf{1} \rangle$, because Par contains the nonzero vector $\mathbf{1}$ of the one-dimensional space $\mathbb{F}_2^{T_-}$; hence $\overline{\mathrm{Par}} = 0$. By Lemmas B.3 and B.4,

$$\mathrm{Sel}_L \simeq \Delta_0 \simeq \mathrm{Sel}_{L'},$$

and these isomorphisms carry λ_L to λ_0 and λ_0 to $\lambda_{L'}$. □

In this tower $|\mathcal{D}| = 2$, so $(\mathbb{F}_2^{\mathcal{D}})^0$ is one dimensional, and Proposition B.5 takes the following form. If $\lambda_0 \neq 0$, its image is the whole codomain, so the target of $\bar{\lambda}$ is zero and $\bar{\lambda} = 0$. Condition (S) is then equivalent to

$$\dim \Delta_0 = 1, \qquad \mathrm{Par} = \langle \mathbf{1} \rangle.$$

Indeed $\ker \lambda_0 = 0$ means that Δ_0 embeds in the one-dimensional space $(\mathbb{F}_2^{\mathcal{D}})^0$, while $\lambda_0 \neq 0$ forces $\Delta_0 \neq 0$; and $\ker \bar{\lambda} = \overline{\mathrm{Par}}$ vanishes exactly when $\mathrm{Par} = \langle \mathbf{1} \rangle$. If $\lambda_0 = 0$, then $\ker \lambda_0 = \Delta_0$, and (S) is equivalent to $\Delta_0 = 0$ together with the injectivity of $\bar{\lambda}$.

Remark B.6. The conditions above are equivalent forms of (S) for the chosen step; they do not produce a step with $|T_-| = 1$. Although δ_- is fixed once c has been chosen, changing c to meet another finite prescription can change T_-, and the realized image Par also depends on δ_+. The reformulation therefore leaves the simultaneous extension hypothesis of Section 9 unchanged.

C Proof dependencies

This appendix lists, for each main result, the results and facts that its proof uses; Section 1.2 describes how the arguments fit together. Theorem A rests on the construction of Theorem 4.11 and on the factorization results of Section 3. The proof of Theorem B has two halves. Theorem 6.1 deduces hereditary atomicity from (T1)–(T6) and the unit approximation condition (JT_A). Proposition 8.9 shows that (S) and (O) at one scaled step (Definition 4.8) of a tower satisfying (T1)–(T6) give the unit approximation of z_n in $\mathcal{U}_n$ at that step; if this holds at every sufficiently late step, it gives (JT_A) with approximation level $N = n$. All results in the table are proved in this paper. The standard facts from algebraic number theory that the proofs use most often are collected, with references, in Section 2.3; the others, such as Minkowski's bound and the Chinese remainder theorem, are cited where they are used. A few elementary facts, such as the criterion that $\mathcal{E}(\sqrt{x})$ is split at a real place of $\mathcal{E}$ exactly when x is positive there, are used without citation.

Result	Inputs used	Role
Theorem 4.11	The base field F_0, Proposition 4.7 (whose outputs are scaled steps by Remark 4.9), and the construction schedule (Definition 4.10) of Section 4, and the procedure of Section 4.9.	Realizes (T1)–(T6), without (S) or (O).
Lemma 3.3, Proposition 3.6, and Theorem 3.7	(T1)–(T3), descent of principality to a finite level, the injection ι of Lemma 3.3, integer valuations, Lemma 3.2, and Lemma 3.4.	One-sided boundedness; an atomic almost Dedekind domain without ACCP.
Theorem A	Theorem 4.11, Lemma 3.2, and Theorem 3.7.	An explicit atomic almost Dedekind domain without ACCP.

Result	Inputs used	Role
Proposition 5.2	The isolated ramified primes of (T5) with Lemma 5.1, integrality, and Lemma 3.4 ($S^\times = S \cap D^\times$).	Reduces a subring that is not a BFD to fraction field K.
Proposition 5.9	A subring that is not a BFD, Lemmas 5.5, 5.6, 5.7 and 5.8, the classes of Lemma 3.3, (T4), the identities $N_{F_n/F_{n-1}} y_n = y_{n-1}$, and the distinct residue offsets of (T6).	Recovers all odd residue fields and separates the maximal ideals over each odd prime.
Propositions 5.11 and 5.12	(T1), Proposition 5.9, Lemma 5.10, compactness of the infinite fibres, saturated intersection, and eventual unramified extension.	Describes the completed orders and fixes their conductor exponents.
Lemma 6.4 and Theorem 6.1	Theorem 3.7 (failure of ACCP), Proposition 5.2, Propositions 5.11 and 5.12 with Lemma 5.13 (the completed orders at one level $\geq b_p$), Lemma 5.3, closure in the whole product $\mathcal{U}_N$ and its propagation (Lemma 6.2), Lemma 6.3, and one-sided boundedness (Proposition 3.6).	Deduces hereditary atomicity from (JT_A).
Theorem 7.2	The Selmer condition (S), the ray class and Kummer descriptions of Lemmas 7.4 and 7.5, the dihedral norm of Lemma 7.6, the presence of odd ramified primes in the step, Proposition 7.7 and Corollary 7.8, Lemma 7.9, and Lemma 7.10.	Gives the dyadic principal-unit component of the approximation.
Lemma 8.4 and Proposition 8.5	Absolute odd ramification index at most two, the pro-p structure of the quotients C_v, and the image of the relative units in the whole product $M_{n,p}$, with its dyadic residue restriction; for Proposition 8.5 also 3 unramified and Lemmas 8.1 and 8.2.	Identifies (O); supplies a finite sufficient test.
Proposition 8.9 and Theorem B	(S) and (O) on the same scaled step, Theorem 7.2, Lemma 8.4, residue 1 of z_n at the dyadic primes, primary decomposition, Theorems 6.1 and 3.7.	Proves Theorem B.
Proposition 9.3	Hypothesis 9.2, dependent choice, the construction schedule, Definition 4.8(iv)–(vi) and the verification of (T1)–(T6) in the proof of Theorem 4.11, and Theorem B.	Reduces the hypotheses of Theorem B to a one-step extension statement.
Lemma 9.5 and Appendix A	The logarithm on the quotients C_v (Lemma 8.1), the unit index exponent $e_{n,p}$, local models with z in the coefficient ring, congruence conditions modulo prime powers, and the construction schedule.	Relates prescribed coefficient precision to membership in the unit image.

Result	Inputs used	Role
Proposition B.5	The totally positive elements δ_-, δ_+, the ramification sets of L and L', Lemma 7.10, and Hilbert reciprocity.	Reformulates (S) by the comparison field and valuation parities.

References

[1] D. D. Anderson, D. F. Anderson, and M. Zafrullah, *Factorization in integral domains*, J. Pure Appl. Algebra **69** (1990), 1–19. doi:10.1016/0022-4049(90)90074-R.

[2] J. Coykendall, F. Gotti, and R. E. Hasenauer, *Hereditary atomicity in integral domains*, J. Pure Appl. Algebra **227** (2023), 107249. doi:10.1016/j.jpaa.2022.107249. arXiv:2112.00264v1. The result numbers cited here refer to this arXiv version.

[3] M. Fontana, E. Houston, and T. Lucas, *Factoring Ideals in Integral Domains*, Lecture Notes of the Unione Matematica Italiana **14**, Springer, Heidelberg, 2013. doi:10.1007/978-3-642-31712-5.

[4] A. Geroldinger and F. Halter-Koch, *Non-Unique Factorizations: Algebraic, Combinatorial and Analytic Theory*, Pure and Applied Mathematics **278**, Chapman & Hall/CRC, 2006. doi:10.1201/9781420003208.

[5] R. Gilmer, *Multiplicative Ideal Theory*, corrected reprint, Queen's Papers in Pure and Applied Mathematics **90**, Queen's University, 1992.

[6] F. Gotti and B. Li, *Arithmetic properties encoded in undermonoids*, Semigroup Forum **112** (2026), 427–449. doi:10.1007/s00233-025-10578-3. arXiv:2412.11199v1. The result numbers cited here refer to this arXiv version.

[7] A. Grams, *Atomic rings and the ascending chain condition for principal ideals*, Proc. Cambridge Philos. Soc. **75** (1974), 321–329. doi:10.1017/S0305004100048532.

[8] J. S. Milne, *Algebraic Number Theory*, version 3.08 (2020). https://www.jmilne.org/math/CourseNotes/ANT.pdf.

[9] J. S. Milne, *Class Field Theory*, version 4.03 (2020). https://www.jmilne.org/math/CourseNotes/CFT.pdf.

[10] J. S. Milne, *The Riemann Hypothesis over Finite Fields: From Weil to the Present Day*, in *The Legacy of Bernhard Riemann after One Hundred and Fifty Years*, Vol. II, Adv. Lect. Math. (ALM) **35.2**, International Press, 2016, 487–565. Reprinted in ICCM Notices **4** (2016), no. 2, 14–52. arXiv:1509.00797v1 (2015). The Section 1 pinpoint refers to the arXiv version.